// a calculus book, from zero
// read this if class made no sense

You Think You're F**ked at Calculus.

You're Not.

Calculus 2 Didn't Have to Be Like That

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WHY THIS BOOK EXISTS

I was the one sitting there not getting it.

I took Calculus 2 with a professor who could not explain things. I would sit in that room, watch the board fill up, and have no idea what was happening. Asking questions didn't help either. It had a way of turning into an argument instead of an answer, so after a while I stopped asking.

Most of that class didn't make it. People failed, people dropped, and by the end only a handful of us were still showing up. I spent a lot of that semester feeling like a dumbass, because I would look at a problem, know I was supposed to understand it, and just not.

Here's the part that matters. I wanted an A in that class. Not a pass, an A. And I already knew the only move I had, because it's the same move that made me a developer. I'm self taught. Nobody handed me that one either. I sat down with it for eight months, nine months, a year, and I kept sitting down until I could actually do it.

So I did the same thing with math. Sit down. Read the problem. Get stuck. Stay stuck for a while on purpose, before going to look for the easy answer. Then go find the right resource and work it all the way through. That's the whole method. It's not clever and it's not fast, but it works.

And this is the thing I actually want you to take out of this book. I don't remember most of the formulas in here. I'm not going to pretend I do. But I know that if I sit down with any of them, give it my time, and go find what I need, I will get it back. I have no doubt about that at all. That certainty is worth more than the formulas ever were, and it's the one thing I'm genuinely able to hand you.

Your professor might be great. Mine wasn't. Either way, this book starts from zero and assumes you know nothing. And it makes you one promise, in writing, chapter by chapter: before any formula shows up, you will always be told *why the idea exists*, what question forced someone to invent it, and *what it's actually for*, where you'd use it in the real world or in the very next chapter. If a section ever hands you a rule without paying that debt first, that section failed, not you.

Get stuck on purpose. Turn the page.

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Learn Calculus 2 From Zero

Start here • read this before anything else

A self-learner's field guide to the most beautiful part of math

❧ Hey. Before any math, let's talk about why you're actually here.

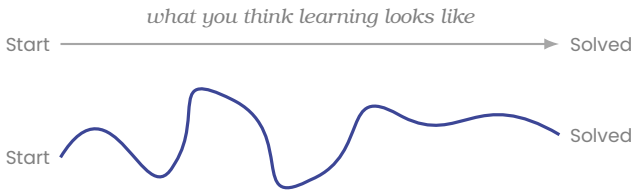
Real talk first, because it matters more than any formula in this whole series.

You will probably forget most of the actual content in two years. The chain rule, the ratio test, the exact formula for arc length. Gone. And that is completely fine, because **the content was never the point.**

Here is the point. The thing you truly keep from learning something hard is the *proof you gave yourself that you can do it.* You sat down in front of something that looked impossible, you got stuck, you did not go hunting for the easy way out, and you cracked it anyway. Do that enough times and something quietly rewires in you. It stops being about math and becomes a plain fact about who you are: **"I don't know this yet, but if I sit down and give it my time, I can learn anything."**

That sentence is the real prize. It works on a new language. It works on a terrifying codebase. It works on a job you are not qualified for yet, and a life you have not built yet. Everyone's timeline is different, mine included, and none of that matters. What matters is knowing that the door opens if you are willing to push on it.

So one more thing, and I mean this: math is not boring. Math is unreasonably beautiful. A good problem is a locked room, and the click when it opens is one of the cleanest feelings there is. You are allowed to enjoy it. In fact that is the whole trick, the people who get good are the ones who chase that click for fun, not because a deadline is chasing them.



Nobody solves a real problem in a straight line. Getting stuck, doubling back, trying the dumb idea: that is not you failing, that is the work. Expect it. Enjoy it.

★ The promise this whole book runs on

Every chapter owes you two things *before* any formula shows up: **why the idea exists**, meaning the question that forced someone to invent it, and **what it's for**, meaning where you would actually use it, in the real world or in the very next chapter. Formulas come and go. If you know where an idea comes from and can picture it, you can rebuild the formula on demand. Any section that hands you a rule without paying that debt first has failed you, and you should treat it that way.

How these guides work

Every chapter is built to make you *do* the math, not just watch it go by. You will keep bumping into colored boxes. Here is each one, being itself, so you know it on sight:

★ Key idea

The one thing from a section worth burning into memory. If you remember nothing else, remember what is in here.

✎ Try it yourself (get stuck on purpose)

Stop reading, close the guide, and actually attempt it. **Get stuck on purpose.** Being stuck is your brain mid-rewire, so sit in it a minute before you peek at the answer.

☆ See it

A picture to study. Cover the words and ask "what is this telling me?" first. The picture is the real thing, a formula is just its shadow, and if you lose the formula you can always redraw it from the picture.

☞ Watch out

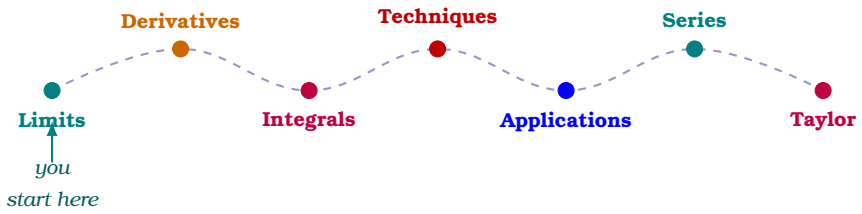
A trap people fall into right at this spot. Read these ones twice, they are usually the difference between a right answer and a confident wrong one.

➜ Want it on video?

A link to Professor Leonard, whose full-length lectures pair perfectly with these guides. Different voice, same ideas, and sometimes a second voice is exactly what makes it click.

The road ahead

Here is the whole journey. It genuinely builds in order, each stop leaning on the one before it, so start at the left and walk right.



Limits is the seed everything grows from. Derivatives and integrals are both just limits in disguise, and series are limits of running totals. Learn to ask "what is this heading toward?" and the whole map opens up.

★ The only rule that matters

Do the problems. Get stuck. Do not look up the answer the second it gets hard. Sit in the stuck, because that exact discomfort is the thing building the version of you that can learn anything. Everything else in these guides is just here to make that process more fun. Now turn the page and go meet *Limits*.

Limits

The very first idea in calculus • Chapter 1

How to pin down a number you never actually reach

❧ Welcome to Chapter 1. Before anything else, a promise.

Calculus has a terrible reputation, and I want to be careful about why. It is not that the ideas are beyond you. It is that calculus is usually taught in an order that makes it nearly impossible: symbols first, meaning later, maybe, if there's time. If that order wrecked you once, the problem was the order, not your brain.

This chapter runs in the opposite order, on purpose. You will see where the idea of a limit *comes from*, you will compute your first one yourself with a calculator before I ever show you the official notation, and every single symbol will be built in front of you before it gets used. Nothing arrives by decree.

One more thing, and keep these in your pocket because I'm going to cash every one of them in later: you already run limits in your head all day. When you walk toward a wall and stop a hair away, you know exactly where you were headed even though you never touched it. When your GPS says "arriving" as the dot slides into the pin, that's one. When a movie fades to black, the brightness is sliding toward zero and your eyes know it. Hold onto the wall especially. In a few pages I'll show you precisely which part of the math is the wall, which part is you walking, and which part is the spot you never quite touch.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- explain why calculus needed a brand-new idea at all (there's a 2,000-year-old paradox behind it),
- read $f(x)$, lim, and the little arrow like plain English, because we build each one from scratch,
- find your first limit *yourself*, with a calculator, before the official definition shows up,
- say what it means when a limit "does not exist," and why that's a finding, not a mistake,
- un-jam a $\frac{0}{0}$ with factoring or a conjugate, and know exactly why those moves are legal,
- pass the pen test: what "continuous" really means,
- call the winner when functions race off to infinity, and explain why.

Why calculus needed a brand-new idea

Here's a question that sounds stupid and is actually one of the deepest questions humans ever asked.

You're in a car. The speedometer says 43. *What does that mean?*

Speed is distance divided by time. Thirty miles in half an hour, 60 miles per hour, fine. But the speedometer isn't claiming anything about half an hour. It's claiming you are going 43 *right now*, at this exact instant. And at an exact instant, how much time passes? Zero. How much distance do you cover? Zero. So "speed right now" is zero divided by zero, which is, as your math teachers have been telling you your whole life, forbidden. The number on your dashboard is a number your own arithmetic says cannot exist.

Sit with that for a second, because people sat with it for two thousand years. The Greek philosopher Zeno poked at this exact wound with his arrow paradox: at any single frozen instant, a flying arrow occupies one fixed spot, not moving. Stack up

a pile of frozen, motionless instants and ask: when does the arrow actually *move*? Nobody had a clean answer.

In the 1600s, Newton and Leibniz got tired of waiting and built calculus anyway, on top of the shaky idea of quantities "infinitely small but not quite zero." It worked spectacularly and it drove people crazy, because nobody could say what that meant. A philosopher named Bishop Berkeley mocked these not-quite-zero quantities as "ghosts of departed quantities," which remains one of the all-time great math insults. It took about 150 more years before Cauchy and Weierstrass finally found the airtight way to say it, and the thing they found is the subject of this chapter.

The fix is called a **limit**, and the move is sneaky: instead of demanding to know what happens *at* the forbidden instant, you watch what happens as you creep *toward* it, and you ask where the creeping is headed. You never divide zero by zero. You walk right up to the edge of it and read off the destination.

★ Why you should care, up front

Everything hard in calculus is this one idea in a different costume. The derivative (Chapter 2) is a limit of slopes. The integral (Chapter 4) is a limit of sums. Infinite series (Calc 2) are limits of running totals. Learn to ask "what is this heading toward?" and you have already met the main character of the entire story. This chapter is the whole game.

First: what a function actually is

I promised to assume nothing, so let's spend two minutes on the thing every calculus book assumes you know. If you're solid on functions, skim this, but don't skip the part about the dots at the end, it's new even for people who've seen functions before.

A **function** is a machine. You feed it a number, it follows a fixed rule, and it hands you back a number. Same input, same output, every time, no moods.



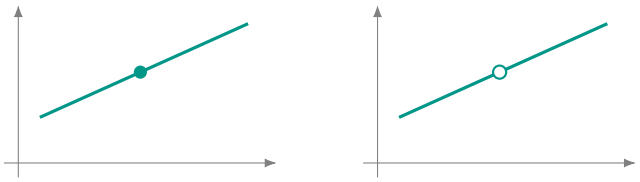
We name the machine with a letter, usually f or g, and we write the rule like this:

$$g(x) = 2x + 1.$$

Read $g(x)$ out loud as "g *of* x," never "g times x." This matters: $g(x)$ is **not multiplication**. It means "what the machine g spits out when you feed it x." So $g(3) = 2(3)+1 = 7$, and $g(10) = 21$, and $g(0) = 1$. The letter x is just a placeholder for "whatever you feed in."

The **graph** of a function is a picture of every input-output pair at once: the input x runs along the horizontal axis, and the output, the *height* of the curve above that input, is the function's answer. When I say "the function's height at $x = 3$," I mean $g(3)$, the output. Input across, output up. That's the whole dictionary.

One last piece of visual grammar, and it's the one that trips people up later, so look at it now while nothing is at stake. Graphs use two kinds of dots:



A **filled dot** means the function really does have that value there. An **open circle** means a hole: at that one input, the function has no output at all, or its output lives somewhere else. Keep both in mind. The open circle is about to become the most important symbol in this chapter.

Your first limit, found by you, not by me

No definition yet. First, a crime scene.

Here is a perfectly innocent-looking function:

$$f(x) = \frac{x^2 - 4}{x - 2}.$$

Feed it 3: $f(3) = \frac{9-4}{3-2} = 5$. Feed it 0: $f(0) = \frac{-4}{-2} = 2$. The machine hums along. Now feed it 2:

$$f(2) = \frac{4 - 4}{2 - 2} = \frac{0}{0}.$$

The machine jams. Not "the answer is zero," not "the answer is undefined so give up." The machine has *no output* at $x = 2$. There is a genuine gap in this function's life at exactly one input.

Here's where you come in. We can't ask what f does *at* 2. Fine. Let's be nosy and ask what it does *near* 2. Grab a calculator, actually do this, and fill in the blanks:

x	1.9	1.99	1.999	$\rightarrow$	$\leftarrow$	2.001	2.01	2.1
$f(x)$	?	?	?			?	?	?

Seriously, compute at least two of them by hand before reading on. The whole point of this book is that you do the thing.

Done? Here's what the machine says:

x	1.9	1.99	1.999	$\rightarrow$	$\leftarrow$	2.001	2.01	2.1
$f(x)$	3.9	3.99	3.999			4.001	4.01	4.1

Look at what's happening. From the left side, the outputs march 3.9, 3.99, 3.999, … From the right side they march down 4.1, 4.01, 4.001, … Both columns are being herded, relentlessly, toward one number. Say it out loud. It's 4.

Now pause and appreciate what you just did, because it's the entire subject in one move. **You found out where a function was headed at a point where the function does not exist.** f has no value at 2. And yet "4" is obviously, undeniably the right answer to *some* question about f at 2. The question it answers is: *where was it heading?*

That is a limit. You just computed one. Everything else is naming.

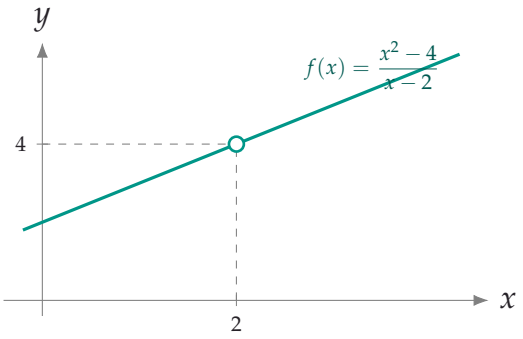
★ The name and the notation, now that you've earned it

What you found is written

$$\lim_{x \to 2} f(x) = 4,$$

and every symbol is a name for something you just did. lim is the question-asker: it takes a function and a target and asks "where is this headed?" The arrow $\rightarrow$ is read "creeps toward": $x \rightarrow 2$ means x takes values like 1.9, 1.99, 1.999, … without ever landing on 2, exactly like your table. And the 4 is the destination your two columns were herded toward. Read the whole line out loud: "the limit of $f(x)$ as x goes to 2 is 4." In general we write $\lim_{x \to a} f(x) = L$, where a is whatever input you're sneaking up on and L is whatever destination the outputs are heading for.

Before I show you the picture, predict it. You have the table. The function is $\frac{x^2-4}{x-2}$, it behaves like a perfectly tame function everywhere, except something must go wrong at $x = 2$. Sketch in your head what that graph has to look like. Then look:



The graph of f: a straight line with exactly one pixel punched out.

Walk your eye along it. Coming in from the left, the curve climbs smoothly toward height 4. Coming in from the right, it slides smoothly down toward height 4. And at the one spot where everything was pointing, there's an open circle: the hole, the jammed machine, the missing output. Your table *was* this picture, written in numbers. The limit is 4 because both approaches aim at height 4; the open circle says $f(2)$ itself doesn't exist; and the limit does not care even a little bit. The limit only ever watches the approach, never the landing.

And now let me pay off the promise from the pep talk. The wall you walk toward? The wall is $x = 2$, the input you never land on. You, walking, are the x values in the table: 1.9, then 1.99, then 1.999, each step covering most of the remaining distance. "Where you were headed" is $L = 4$. The limit is the math-precise version of you standing a hair from the wall saying: I never touched it, but there is zero doubt about where I was going.

★ The whole idea in one line

A limit answers "where is this function *headed*?", not "where does it *land*?" Those can be different, and the gap between them is where every interesting thing in calculus lives. You've now seen the gap with your own eyes: headed to 4, lands nowhere.

When "where is it headed?" has no answer

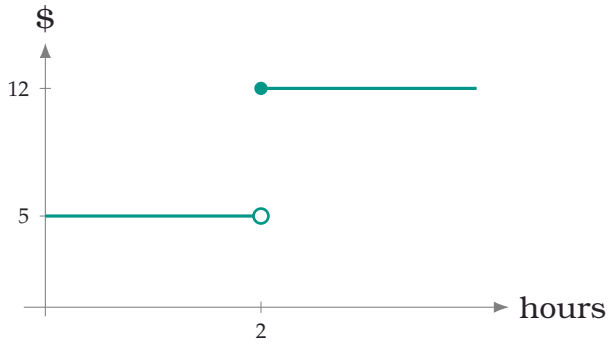
A fair question to ask right now: can you *always* answer "where is it headed"? Is the destination always there?

Try this one yourself before any math shows up. A parking garage charges a flat \$5 if you stay up to two hours, and a flat \$12 if you stay any longer. Picture the graph of cost versus time: a flat shelf at 5, then at the two-hour mark it leaps to a flat shelf at 12. Now the mean question: as your parking time creeps toward exactly two hours, *what price is the function heading toward?*

Think about it. If you creep toward two hours from below, 1.9 hours, 1.99 hours, the price is 5, 5, 5, … headed for 5. If you creep down from above, 2.1 hours, 2.01 hours, it's 12, 12, 12, … headed for 12. You want to give two answers. That itch you're feeling, that "well, it depends which side," is a real mathematical object, and it has notation:

$$\lim_{x \to 2^-} f(x) = 5 \qquad \text{and} \qquad \lim_{x \to 2^+} f(x) = 12.$$

These are **one-sided limits**, and read that superscript carefully, because everyone misreads it the first time. The little minus in $x \to 2^-$ does *not* mean negative two. It means you're sneaking up from the minus side, the side where numbers are a bit *smaller* than 2 (and smaller numbers live to the left on the axis, so this is the from-the-left limit). The little plus means approaching from the side of slightly *bigger* numbers, from the right.



The parking garage. Open circle at $(2, 5)$: as you cross two hours, the \$5 price is no longer yours. Filled dot at $(2, 12)$: at exactly two hours and beyond, you pay \$12.

Trace it with your eye. Left shelf aims at height 5. Right shelf aims at height 12. The two sides are each perfectly confident, and they flatly disagree. So what is *the* limit at 2?

Here's the honest answer, and I want to frame it correctly because this is where students panic for no reason. A limit, by its job description, is *one single number*: the destination. If the left side names 5 and the right side names 12, then no single number does the job, and we report:

$$\lim_{x \to 2} f(x) \text{ does not exist (written DNE).}$$

> ☞ **Watch out**
>
> DNE is a **finding**, not an error message. It does not mean you made a mistake, and it does not mean the universe is broken. It means you investigated, and the honest result of the investigation is "this function has no single destination there." On an exam, DNE is a full-credit answer when it's the true one. The rule in one line: *the two-sided limit exists only when both one-sided limits exist and agree.*

Since one-sided limits are things you compute, not just look at, here's one computed. Take

$$f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ 6 - x & \text{if } x \geq 2 \end{cases}$$

(a *piecewise* function: one machine, two rules, and which rule fires depends on the input). Creeping toward 2 from the left, the inputs are all less than 2, so the top rule is in charge: $x + 1$ heads toward $2 + 1 = 3$. Creeping from the right, the bottom rule is in charge: $6 - x$ heads toward $6 - 2 = 4$. Left says 3, right says 4, so $\lim_{x \to 2} f(x)$ **DNE**. And notice something sneaky: $f(2)$ itself exists just fine, it's $6 - 2 = 4$ by the bottom rule. The function *lands* somewhere, but it wasn't *heading* anywhere in particular. Landing and heading are independent. That's the theme of this whole chapter.

There's one more way a limit can fail, and it fails upward. Try the function $\frac{1}{x^2}$ near $x = 0$, by table, because tables are now officially our move:

x	0.1	0.01	0.001	-0.001	-0.01	-0.1
$\frac{1}{x^2}$	100	10,000	1,000,000	1,000,000	10,000	100

Both sides agree violently: the outputs aren't settling toward any number, they're exploding. We write

$$\lim_{x \to 0} \frac{1}{x^2} = \infty,$$

and here I owe you a straight explanation of that symbol, because this is its first appearance and most books never come clean about it. ∞ is **not a number**. You can't add it, you can't divide by it, it doesn't live on the number line. Writing "$= \infty$" is shorthand for a *description*: "the outputs grow without any ceiling." It names a direction, not a destination. So, slightly maddeningly, both of these statements are true at once: the limit *equals infinity* (that's the description of the behavior) and the limit *does not exist as a number* (because no number is the destination). On a graph, this behavior is a **vertical asymptote**: an invisible vertical wall at $x = 0$ that the curve hugs while rocketing upward.

The plot twist: plugging in usually just works

Now I have to deal with something that should be bothering you. If it wasn't bothering you, it will in ten seconds, so let's just have it out.

I've spent this whole chapter insisting that the limit ignores the value at the target. The limit watches the approach; the landing is irrelevant; headed-to and lands-at are different things. Fine. So compute this one with a table: $\lim_{x \to 3} (x^2 - 2x)$.

x	2.9	2.99	2.999	3.001	3.01	3.1
$x^2 - 2x$	2.61	2.9601	2.996	3.004	3.0401	3.41

Headed for 3, clearly. But wait: just plugging in gives $f(3) = 9 - 6 = 3$. The same answer. The boring way worked. *So was all that hole business a waste of your time?*

No, and the resolution is one of the nicest ideas in the chapter. For most functions you've ever met, the destination and the landing spot **happen to be the same place**. The function heads toward a height and then actually shows up there. Functions with that politeness have a name, **continuous** (the pen-test section later in this chapter is all about them), and for continuous functions, computing a limit really is as easy as plugging in. **Polynomials**, meaning functions built only from powers of x with ordinary coefficients, like $x^2 - 2x$ or $5x^3 - x + 7$, are continuous everywhere, bless them.

So why does the subject of limits exist at all? Because of a spectacular coincidence: *every genuinely interesting object in calculus breaks exactly at the point you care about.* The speedometer from the opening section is a $\frac{0}{0}$ at the instant you care about. The slope of a curve (next chapter) is a $\frac{0}{0}$ at the point you care about. Our friend $\frac{x^2-4}{x-2}$ jams

precisely at 2. Limits weren't invented for the polite functions. They were invented because the important ones are rude, always, at the worst possible spot.

★ Your compute-a-limit instinct, version 1

Plug in first, every single time. If you get an ordinary number and the function is built from polite pieces, you're done, that's the limit. If you get $\frac{0}{0}$, the machine jammed and the next section is your toolkit. If you get $\frac{\text{nonzero}}{0}$, the outputs are exploding near a vertical asymptote, think $\pm\infty$, not algebra.

When plugging in jams: how to read $\frac{0}{0}$

First, what $\frac{0}{0}$ actually is, because "indeterminate" deserves to be a lesson instead of a vocabulary word.

$\frac{0}{0}$ is what you get when the top and the bottom of a fraction are *racing to zero at the same time*, and here's the thing: the result of that race depends entirely on who's shrinking faster. Don't take my word, watch three races as $x \to 0$, all of which jam into $\frac{0}{0}$ if you plug in:

$$\frac{x}{x} \to 1 \qquad \frac{x^2}{x} = x \to 0 \qquad \frac{5x}{x} \to 5$$

Same $\frac{0}{0}$ costume, three different endings. (And $\frac{x}{x^2} = \frac{1}{x}$ wears the costume too, and it explodes.) So $\frac{0}{0}$ carries *no information at all.* It isn't an answer, it isn't an error, it's the problem clearing its throat and saying: *you can't judge me in this form, rewrite me first.* That's exactly what **indeterminate** means, and now you know it rather than memorize it.

So the game is: rewrite the fraction into a form that doesn't jam. Two rewrites cover most of what you'll meet.

Rewrite 1: factor and cancel. Let's finally close the case on $\frac{x^2-4}{x-2}$, properly, with reasons. The top factors: $x^2 - 4 = (x-2)(x+2)$. So

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x\to 2} (x+2) = 4.$$

Now, the step where we canceled $(x-2)$: why was that legal? Because $\frac{(x-2)(x+2)}{x-2}$ and $x+2$ are *the same function at every single input except* $x=2$, where the first one jams and the second one doesn't. And the limit, remember, only watches the approach, inputs *near* 2, never 2 itself. On every input the limit actually looks at, the two functions agree perfectly. So they must be heading to the same place. Canceling isn't a trick here. It's a *discovery*: our jammed function was secretly the line $x+2$ wearing a disguise, with one pixel missing. That's why its graph back in the crime-scene section was a straight line with a hole. The algebra just told us which line.

Rewrite 2: the conjugate, which you are about to invent. Try plugging into

$$\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x}.$$

You get $\frac{1-1}{0} = \frac{0}{0}$. Jammed. Factoring doesn't obviously help, the square root is the thing in the way. So think it through with me, because you have everything you need. What's the one algebra move you know that destroys a square root? *Squaring it.* But you can't just square the top of a fraction, that changes the problem. What *are* you allowed to do to a fraction without changing it? Multiply it by 1, in disguise: anything over itself, $\frac{\text{thing}}{\text{thing}}$. So the question becomes: what "thing" could you multiply $\sqrt{x+1}-1$ by, so that a square appears? You've seen the pattern $(A-B)(A+B) = A^2 - B^2$ before, and that right side is exactly "things getting squared." So multiply by the same expression with the sign flipped:

$$\frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1} = \frac{(x+1)-1}{x\,(\sqrt{x+1}+1)} = \frac{x}{x\,(\sqrt{x+1}+1)} = \frac{1}{\sqrt{x+1}+1},$$

and now plugging in $x=0$ is safe: $\frac{1}{\sqrt{1}+1} = \frac{1}{2}$.

That sign-flipped partner is called the **conjugate**: for $A-B$ it's $A+B$, and its whole purpose in life is to make A^2-B^2 happen, which executes the square root. You didn't memorize a trick just now. You rebuilt it from "what kills roots?" plus "what am I allowed to do?", and that means you can rebuild it in an exam when the memory fails.

And because I don't want you to take $\frac{1}{2}$ on faith, check it the way this chapter checks everything: $x = 0.001$ gives $\frac{\sqrt{1.001}-1}{0.001} = 0.49988\ldots$ The algebra and the creeping numbers are telling the same story. They always will; that's what makes the algebra trustworthy.

☞ Watch out

The wrong ideas almost everyone carries out of a bad calculus class, named and retired:

- *"The limit is just $f(a)$."* Only for continuous functions. The interesting cases are exactly the ones where that fails.
- *"I got DNE, so I messed up."* DNE is a legitimate verdict. You investigated; the sides disagreed; case closed.
- *"$\frac{0}{0}$ means no answer."* $\frac{0}{0}$ means *no answer yet*. Rewrite and re-ask.
- *"$x \to 2^-$ has something to do with negative numbers."* It's the side where numbers are slightly smaller than 2. That's all.
- *"$\frac{5}{0}$ is also indeterminate."* No. Nonzero over zero isn't a race, it's an explosion: vertical asymptote, infinite behavior. Only $\frac{0}{0}$ (and $\frac{\infty}{\infty}$, coming in the L'Hôpital section) are genuine races with unknown winners.

✎ Quick practice, before you move on

Practice 1. $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$.

Practice 2. $\lim_{x \to 0} \dfrac{\sqrt{x+9} - 3}{x}$.

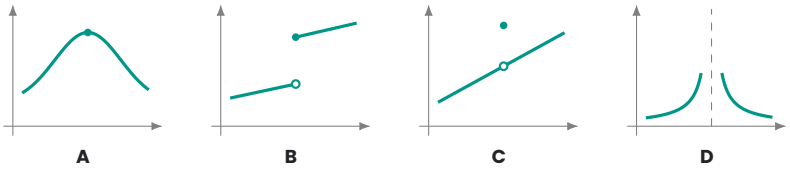
How to think about it: plug in, get $\frac{0}{0}$, notice the square root. What kills a root? What are you allowed to multiply by? Build the conjugate, watch the top collapse, cancel, then plug in.

Practice 3 (in words, one sentence). Your friend computes a limit by plugging in and announces, "limits are pointless, it's just $f(a)$." For which functions are they accidentally right, and where does their method fall apart?

Answers at the end of the chapter. Wrestle first, that's the deal.

Continuity: the pen test

Grab a pencil, this section is an activity. Below are four graphs. For each one, imagine tracing it left to right without lifting your pencil. Two questions as you go: *which ones force you to lift the pencil*, and, the sharper question, *at the exact instant your hand leaves the paper, what exactly went wrong?* Put your answer into words before reading past the figure. Your words, not mine.



Here's what most people write down, in their own words. A: pen never lifts. B: the graph *jumps*, my pen was at one height and the curve teleported to another. C: there's a *hole*, and the function's actual value is a lonely dot floating off the curve, so my pen has to hop off and back. D: the curve *blows up*, runs off the top of the page, and comes back from nowhere.

Congratulations: you just wrote the formal definition of continuity. Mathematicians compress your three failure modes into three conditions. A function is **continuous at** a when:

$$(1)\ f(a) \text{ exists}, \qquad (2)\ \lim_{x\to a} f(x) \text{ exists}, \qquad (3)\ \lim_{x\to a} f(x) = f(a).$$

In plain words, and this is the line to remember: **a continuous function lands exactly where it was heading.** There's a destination (2), there's a landing (1), and they're the same spot (3).

Now match them yourself before reading on: which condition does B break? C? D?

B breaks (2): the sides disagree, no single destination, the limit DNE. C breaks (3): the function heads toward the height of the hole, but it *lands* on the floating dot, destination and landing disagree. (C's hole is called **removable**, and here's the lovely connection: when we factored $\frac{x^2-4}{x-2}$ into $x+2$ back in the $\frac{0}{0}$ section, that canceling literally *removed* a hole just like this one. Removable holes are the gentle failures, algebra can patch them.) And D breaks (2) in the most dramatic way available, by exploding: that's our vertical asymptote from the no-single-answer section, and as a bonus $f(a)$ usually doesn't exist there either, so it breaks (1) too.

Why should you care about continuity beyond the vocabulary? Two reasons. The practical one: for continuous functions, limits are free, just plug in, which is why the plot-twist section worked. The structural one: continuity is the exact dividing line between the functions where calculus is easy and the points where it gets interesting, and later chapters will lean on it constantly (one gift for the road: with continuous functions you may carry a limit inside the function, a move we'll cash in during the series unit of Calc 2).

✎ **Quick practice, before you move on**

Practice 4. Find the value of k that makes

$$f(x) = \begin{cases} x^2 + k & \text{if } x < 1 \\ 4x & \text{if } x \geq 1 \end{cases}$$

continuous at $x = 1$.

How to think about it: continuity at 1 means "lands where it was heading," and here that means the left piece and the right piece must be heading to the same height at $x = 1$. Compute what each piece heads toward, set them equal, solve for k.

Answer at the end of the chapter.

Way out there: limits at infinity

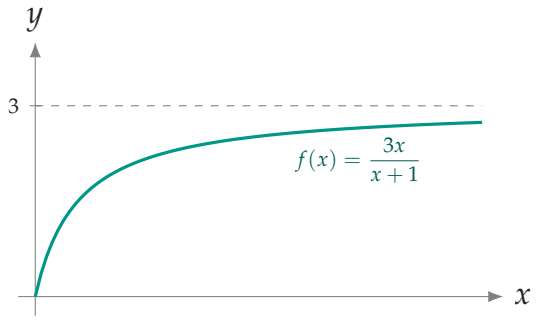
So far the input has crept toward a finite spot. The other question worth asking flips that: what does $f(x)$ do as x just keeps *growing*, past 100, past a million, forever? This isn't idle. It's the question behind "where does this population level off?", "what speed does a falling object settle into?", and, for the programmers, "how does my algorithm behave as the input gets huge?" The long run is a thing people pay money to know.

Notation first, built honestly: we write $x \to \infty$, and you already know to read ∞ as a direction, not a place. There's no wall to creep up on this time, the instruction is just "keep going and watch."

So let's watch. Take $f(x) = \frac{3x}{x+1}$ and do the thing we always do first:

x	10	100	1,000	10,000
$\frac{3x}{x+1}$	2.727...	2.970...	2.997...	2.9997...

No suspense about the destination. Before the picture: predict what a graph does when its outputs behave like this as x grows. It has to flatten out, hugging some height, right? Look:



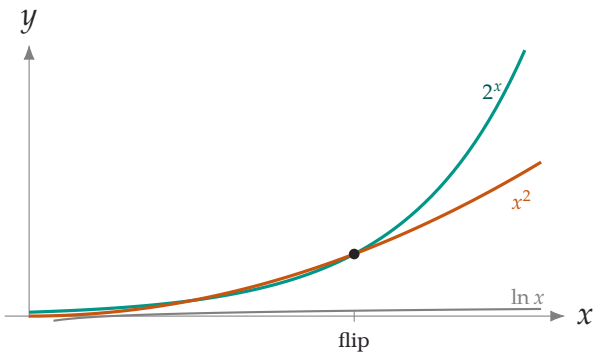
*Flattening toward the dashed line $y = 3$ without ever arriving. That line is a **horizontal asymptote**, the long-run cruising altitude, and we write $\lim_{x\to\infty} \frac{3x}{x+1} = 3$.*

Why 3, from the formula itself? Here's the honest reason, and once you see it you can skip the table forever. When x is huge, say $x = 1{,}000$, compare the sizes of the pieces. The bottom is $x + 1 = 1{,}001$, and that $+1$ is contributing one part in a thousand, pure noise. Out in the far distance, every expression is *dominated by its biggest piece*: the fraction behaves like $\frac{3x}{x} = 3$. That's the whole theory. Watch it work on something meaner: in $5x^3 - x$ at $x = 1{,}000$, the $5x^3$ part is five billion while the $-x$ part is a measly thousand, an adjustment of 0.00002 percent. The big power swamps everything.

That swamping argument is worth having as reflexes for **polynomials** (sums of powers of x; the **degree** is the biggest power, and the **leading coefficient** is the number attached to it). For a ratio of polynomials as $x \to \infty$, only the two leading pieces matter, so: same degree top and bottom, the limit is the ratio of leading coefficients. Bottom degree bigger, the bottom swamps, limit is 0. Top degree bigger, the top swamps, the thing blows up, $\pm\infty$. Not three rules to memorize: one swamping idea wearing three outfits.

Now for the best bet in mathematics. **Which is bigger: 2^x or x^{100}?** Write down your gut answer. Then check small cases, like we always do: at $x = 10$, 2^{10} is about a thousand, while 10^{100} is a googol, a one with a hundred zeros. It's not close, x^{100} is ahead by 97 zeros. At $x = 100$: $2^{100} \approx 10^{30}$ versus $100^{100} = 10^{200}$. Still a massacre. Any sane person declares x^{100} the winner and goes home.

And yet. At $x = 1{,}000$: $2^{1000} \approx 10^{301}$, while $1000^{100} = 10^{300}$. *The exponential just pulled ahead.* At $x = 2{,}000$ it's 10^{602} against 10^{330}, a lead of 272 zeros and growing. The race flips near $x \approx 996$, and after the flip it isn't a race anymore. Exponentials always, always beat powers in the end, because a power multiplies its lead by a fixed amount per lap while an exponential doubles, and doubling compounds.



The same drama in miniature, 2^x versus x^2: the power is winning at $x = 3$ (nine beats eight), they tie at the marked flip ($x = 4$: sixteen all), and after that the exponential is gone. The gray crawler is $\ln x$. Every race has its flip; before it, the eventual loser can look like a champion.

With the flip understood, here's the full pecking order. Two quick introductions so no symbol goes unexplained: $\ln x$ is the *natural logarithm*, the undo-button of the exponential, and all you need here is its personality: it grows achingly slowly, to nudge $\ln x$ up by just 1 you must multiply x by about 2.7. And $x!$ ("x factorial") means $1 \cdot 2 \cdot 3 \cdots x$, which grows even more absurdly than exponentials because its multiplier gets bigger every step. The order, slowest to fastest:

$$\ln x \ \ll \ x^p \ \ll \ a^x \ (a > 1) \ \ll \ x!$$

where $\ll$ means "is eventually left hopelessly behind by," no matter how the early laps looked. In any ratio, whoever is further right wins the race to infinity: that's why $\frac{\ln x}{x} \to 0$ and $\frac{x^{100}}{2^x} \to 0$, no computation required.

A spoiler called L'Hôpital

I'll end with a tool you can't fully use yet, and I'm going to be honest about that instead of pretending. It's here because you'll want to know it exists, and because the idea behind it is a perfect send-off into Chapter 2.

Think back to the $\frac{0}{0}$ races: top and bottom both rushing to zero, and the answer decided by *who gets there faster*. So here's a question worth sitting with: if everything hinges on how fast each part is shrinking, what would you need to be able to measure? Some gadget that takes a function and reports its speed of change. That gadget exists. It's called the **derivative**, it is the entire subject of the next chapter, and the moment you own it, the following becomes available:

★ L'Hôpital's Rule (a preview, redeemable after Chapter 2)

If plugging in gives exactly $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and only then, you may replace top and bottom by their *rates of change* (their derivatives, written f' and g') and ask again:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}.$$

Why that's reasonable: when both runners hit the tape at zero together, the photo finish is decided by their speeds, and derivatives are precisely speed. One taste: this rule turns $\frac{\ln x}{x}$ into $\frac{1/x}{1}$, which clearly heads to 0, confirming the growth race's verdict that ln loses. The two rewrites in that sentence ($\ln x$ becomes $\frac{1}{x}$, and x becomes 1) are derivative facts you'll be able to verify yourself about twenty pages from now. Trust, then verify.

Two footnotes you'll enjoy. First, the rule is named after the Marquis de L'Hôpital, who published it in 1696 in the world's first calculus textbook, but he didn't discover it: he paid Johann Bernoulli a retainer for his mathematical discoveries, and this was one of them. The best-named theorem in calculus is named for the buyer. Second, a warning that will save you real exam points someday:

☞ Watch out

L'Hôpital is *only* for a true $\frac{0}{0}$ or $\frac{\infty}{\infty}$. If plugging in already gives an honest number, the limit is done, and "one more L'Hôpital for luck" will cheerfully hand you a confident wrong answer. Check the form every single time before you reach for it.

Carrying this forward

★ Leaving Chapter 1 with this

A limit asks "where is this heading?" and does not care where, or whether, it lands. You can read one off a table, off a graph, or by algebra: plug in first; a $\frac{0}{0}$ means rewrite (factor, or conjugate) and re-ask; disagreeing sides mean DNE, which is a finding, not a failure; exploding outputs mean ∞, a direction rather than a number. Continuous functions are the polite ones that land where they were heading, and the far horizon belongs to whoever swamps whom. And remember why any of this exists: the speedometer. Every deep idea coming, slopes, areas, infinite sums, is a limit in a costume. You now speak the language the rest of the story is written in.

✎ The chapter workout (get stuck on purpose)

Mixed, like real life. Cover the answers, wrestle honestly, then check.

F1. $\lim_{x\to 3} \dfrac{x^2-9}{x-3}$ **F2.** $\lim_{x\to\infty} \dfrac{5x^3-x}{2x^3+7}$

F3. $\lim_{x\to 1} g(x)$ where $g(x) = \begin{cases} 2x & x < 1 \\ x+3 & x \geq 1 \end{cases}$

F4. $\lim_{x\to 5} \dfrac{1}{(x-5)^2}$ **F5.** $\lim_{x\to\infty} \dfrac{\ln x}{x}$

F6. Sketch any function with $f(1) = 5$ but $\lim_{x\to 1} f(x) = 2$. Nudges: **F1** jams, factor. **F2** same degree, who leads? **F3** each side has its own rule, compute both. **F4** nonzero over zero-squared, which direction? **F5** growth race. **F6** you need a curve heading to one height and a lonely dot at another; which figure from the pen test is this?

Answers (checked twice, but wrestle first). **P1:** factor to $x+4$, limit $=8$. **P2:** conjugate $\sqrt{x+9}+3$, collapse to $\frac{1}{\sqrt{x+9}+3}$, limit $=\frac{1}{6}$. **P3:** they're right exactly for continuous functions, and wrong wherever the function jams at the target, which is every interesting case in calculus, starting with $\frac{0}{0}$ fractions. **P4:** left piece heads to $1+k$, right piece heads to 4 (and lands there); equal requires $k=3$. **F1:** 6. **F2:** $\frac{5}{2}$. **F3:** left $\to 2$, right $\to 4$, **DNE**. **F4:** $+\infty$ (both sides explode upward; as a number, DNE). **F5:** 0, the log always loses. **F6:** a curve aiming at height 2 with an open circle at $(1,2)$, plus a lone filled dot at $(1,5)$: panel C of the pen test, drawn by you.

Next up: *Derivatives*, where we point the limit machine at the slope of a curve, resolve the speedometer paradox for real, and invent instantaneous change.

➜ Want it on video?

Professor Leonard, *Calculus 1 Lecture 1.1: An Introduction to Limits* and *1.2: Properties of Limits*, at https://www.youtube.com/@ProfessorLeonard. Full, unhurried lectures, a perfect second voice. For L'Hôpital once you have derivatives, his *Calculus 2 Lecture 6.7*.

Derivatives

The second big idea in calculus • Chapter 2

Measuring how fast something is changing right now

❧ Chapter 2. Time to pay off Chapter 1's debt.

Remember where we left the speedometer? It reads 43 at a single instant, but speed is distance over time, and an instant contains zero of both. The number on the dashboard was mathematically illegal. Then Chapter 1 handed you the escape hatch: when a question jams at a point, don't ask *at* the point, creep *toward* it and read off the destination.

This chapter aims that trick at motion, and out comes the **derivative**: the exact, legal, honest answer to "how fast, right now?" You're going to compute one yourself, from a table, before I show you a single rule, same deal as last time. And when the answer appears under your own hands, the official definition will just be a name for what you did.

Fair warning in the other direction: there are shortcut rules in this chapter, and you'll use them forever. But you'll learn them *after* you've seen where they come from, because a rule you can rebuild is yours for life, and a rule you only memorized is gone two weeks after the final. You know my whole philosophy by now: the formulas are rentals, the understanding is the thing you own.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say why derivatives exist and name real things they measure (speed, growth, cost, learning),
- compute an instantaneous rate yourself, by table, before any formula shows up,
- explain what the limit definition is actually doing (it's Chapter 1 aimed at slopes),
- differentiate fast with the power, product, quotient, and chain rules, and know where each comes from,
- never drop the chain rule on a function tucked inside another,
- read a graph's slopes to tell where it climbs, peaks, and falls.

Why this exists, and what it's for

The promise of this book is that nothing shows up without a reason, so here's the reason, up front.

The world is made of things that change: a car's position, a bank balance, the number of users on your app, the temperature of your coffee. For any changing thing, the single most useful question you can ask is *how fast is it changing right now?* Not on average over the last hour. Now. That question is worth money and lives: it's the difference between "the patient's fever averaged 101 today" and "the fever is spiking *this minute*."

And until calculus, that question was unanswerable, because "rate right now" divides zero by zero. The derivative is the machine that answers it anyway. Once you have the machine, it turns out to be everywhere:

- **Motion:** the derivative of position is velocity; the derivative of velocity is acceleration. Your speedometer, finally legal.

- **Geometry:** the derivative is the *slope of a curve at a point*, which is what lets us find peaks and valleys, and that is the entire basis of optimization in the next chapter (biggest profit, least material, fastest route).
- **Everywhere else:** growth rate of a population, marginal cost in economics (what producing one more unit costs), and, for my programmer readers, this one: every neural network you have ever heard of learns by computing derivatives, millions of them, to figure out which direction to nudge its numbers. The "gradient" in gradient descent is a pile of derivatives. This chapter is the engine room of modern machine learning.
- **Unfinished business:** L'Hôpital's rule from Chapter 1 was an IOU, "redeemable after Chapter 2." Today it gets redeemed.

One idea, purchased once, spendable everywhere. Let's go earn it.

Compute one yourself, before any formula

Here's a concrete setup. Drop a rock off a cliff; physics says that after t seconds it has fallen $s(t) = 5t^2$ meters (close enough on Earth, and clean numbers). After 1 second it has fallen 5 meters, after 2 seconds 20 meters, after 3 seconds 45 meters. Now the forbidden question: **how fast is the rock falling at exactly $t = 3$?**

We can't divide by a zero-length instant. But Chapter 1 taught us the move: creep toward it. We *can* compute honest average speeds over little windows of time starting at $t = 3$, using plain old distance over time:

$$\text{average speed from } t = 3 \text{ to } t = 3.1 \;=\; \frac{s(3.1) - s(3)}{3.1 - 3} \;=\; \frac{48.05 - 45}{0.1} \;=\; 30.5 \text{ m/s}.$$

Your turn. Shrink the window and watch what happens. Fill in at least two of these with a calculator before reading on, it's two minutes and it's the whole chapter:

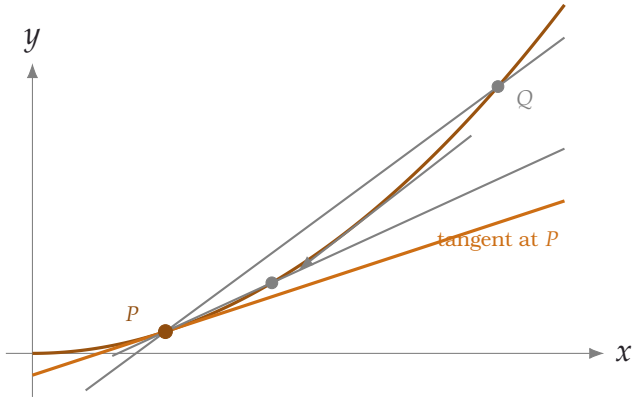
window	average speed $\frac{s(3+h)-s(3)}{h}$
$t = 3$ to 3.1	30.5
$t = 3$ to 3.01	?
$t = 3$ to 3.001	?
$t = 2.99$ to 3	?
$t = 2.999$ to 3	?

Here's what the arithmetic says: 30.05, then 30.005, and from the left side 29.95, then 29.995. Both sides of $t = 3$ are herding the averages toward one number. Say it before I do.

It's **30 m/s**. The averages over shrinking windows have a destination, and that destination is the only sane meaning of "speed at the instant $t = 3$." You didn't divide zero by zero. You walked up to the edge of it, Chapter 1 style, and read off where the averages were headed. That number, the limit of the average rates as the window shrinks to nothing, is called the **derivative** of s at $t = 3$.

The same idea as a picture: slopes

Before the figure, a question to carry into it. An average speed is a *slope*: rise over run, distance over time. On the graph of $s(t)$, the average speed between two moments is the slope of the straight line connecting those two points on the curve (such a line is called a **secant line**). So here's the question: as your time window shrank, your two points slid together. *What did the secant line turn into?* Picture it, then look:



Secants pivoting into the tangent as Q slides home to P.

Walk through it. P is your fixed moment, $t = 3$ in the rock story. Each gray line is a secant through P and a nearby point Q: an average rate, an entry in your table. As Q slides toward P, the secants pivot, and they are visibly settling onto one special line, the bold one, which just grazes the curve at P. That line is the **tangent line**, and its slope is the number your table was converging to. So the derivative wears two outfits that are secretly the same garment: it is *the instantaneous rate of change*, and it is *the slope of the curve at a single point*. Rate when you think in time, slope when you think in pictures. One number, two readings, and you'll use both constantly.

The definition, which you have already done

Now the notation, as names for your own actions. Call the little window width h (your 0.1, then 0.01, then 0.001). The average rate over the window from x to $x+h$ is rise over run:

$$\frac{f(x+h)-f(x)}{h},$$

exactly the fraction you were computing. The derivative is where that heads as the window shrinks:

$$\boxed{f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}}$$

Read $f'(x)$ as "f prime of x": the derivative of f, at the input x. People also write $\frac{d}{dx}f(x)$, Leibniz's notation, which you can read as "the rate of change of f as x changes." Same object, two outfits, and you'll see both everywhere.

Let's use the definition once, for real, on the rock's $s(t)=5t^2$, so you see the $\frac{0}{0}$ actually dissolve. This is Chapter 1 algebra, nothing more:

$$\frac{s(t+h)-s(t)}{h}=\frac{5(t+h)^2-5t^2}{h}=\frac{5t^2+10th+5h^2-5t^2}{h}$$

$$=\frac{10th+5h^2}{h}=\frac{h\,(10t+5h)}{h}=10t+5h.$$

Plugging $h=0$ into the original gave $\frac{0}{0}$, a jam. But factor and cancel, our move from the limits chapter, and the jam clears: as $h\to 0$, the expression $10t+5h$ heads to $10t$. So $s'(t)=10t$, and at $t=3$ that's $s'(3)=30$. **The algebra just reproduced your table.** That agreement is why you can trust the algebra: it's the creeping-numbers story, compressed.

You'll compute a derivative from the raw definition a handful of times in your life, to feel the gears turn. Then never again, because people ground through that limit for every standard function once, found the patterns, and left us the shortcuts. Using them isn't cheating, it's inheriting.

The shortcut rules, with their origin stories

Look at what we just derived: the derivative of $5t^2$ came out $10t$. The exponent 2 came down as a multiplier, and the power dropped by one. Grind the same limit for x^3 and you get $3x^2$. For x^4: $4x^3$. The pattern is unmistakable, and it's the single most-used fact in calculus:

★ The greatest hits

$$\frac{d}{dx}x^n = n\,x^{n-1} \quad \text{(power rule: exponent hops down front, power drops by one)},$$

$$\frac{d}{dx}(c) = 0 \quad \text{(a constant is a flat line, and flat means slope zero)},$$

$$\frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}, \quad \frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x.$$

Two combiners that behave exactly the way you'd hope: constants slide out front, $(c\,f)' = c\,f'$, and sums split, $(f+g)' = f' + g'$. (That e^x line, a function that *is* its own rate of change, is the reason e shows up everywhere growth does; it's essentially the only function, up to a constant multiple, whose speed always equals its size.)

In action, term by term:

$$\frac{d}{dx}\left(x^4 - 3x^2 + 7\right) = 4x^3 - 6x + 0 = 4x^3 - 6x.$$

The $+7$ vanished, and it should: adding 7 lifts the whole graph without tilting it anywhere, and the derivative only reports tilt.

Products: your first guess is wrong, and that's the lesson

Here's a rule I want you to *guess* before I give it to you. Derivatives split over sums. So surely they split over products too: $(fg)' = f'g'$, right? Don't take my word, test it, you have everything you need. Try the simplest product alive: $f(x) = x$ and $g(x) = x$, so $fg = x^2$.

Truth: $(x^2)' = 2x$. The guess: $f' \cdot g' = 1 \cdot 1 = 1$.

The guess says the slope of x^2 is the constant 1 everywhere, which is nonsense (you can *see* x^2 steepening). So the pretty guess is dead, and you killed it yourself in one line, which is the right way to learn it. The real rule:

$$(fg)' = f'g + fg'.$$

Why that shape? A product grows in two ways at once: the first factor can grow while the second holds (that's $f'g$), and the second can grow while the first holds (that's fg'). Both happen together, so you add the contributions. Think of a rectangle whose width and height are both growing: new area comes from a strip on the right *and* a strip on top.

➤ Product rule in action

$$\frac{d}{dx}(x^3 e^x) = \underbrace{3x^2}_{f'} \cdot e^x + x^3 \cdot \underbrace{e^x}_{g'} = x^2 e^x (3 + x).$$

Differentiate the first, keep the second; keep the first, differentiate the second; add. Same chant every time.

For quotients there's a sibling rule:

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \qquad \text{chanted as "low d-high minus high d-low, over low squared."}$$

It isn't a new axiom, and you could rebuild it yourself: write $\frac{f}{g}$ as the product $f \cdot g^{-1}$ and run the product and chain rules you already own; the algebra collapses to exactly this fraction. Rules compounding, not multiplying.

The chain rule: rates multiply

The most important rule in the toolkit, and the one with the cleanest everyday logic, so let's get the logic first.

You're on a bike. Your pedals spin, the chain turns the wheel, the wheel moves the bike. Suppose the gearing means the wheel turns **3 times as fast** as the pedals, and the wheel's size means the bike moves **2 meters per wheel turn**. Question, answer it before reading on: how fast does the bike move *per pedal turn*?

Six meters, obviously: 2×3. When one thing drives a second thing and the second drives a third, **the rates multiply**. That's the entire chain rule. In symbols, if $y = f(g(x))$, a function with another function tucked *inside* it, then

$$\frac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big) \cdot g'(x),$$

read it as: the outer function's rate (evaluated at the inside, left untouched), *times* the inside's own rate. Output responds to the inside, inside responds to x, so output responds to x at the product of the two rates. Pedals, wheel, bike.

➤ Chain rule in action

$$\frac{d}{dx}(3x^2+1)^5 = 5(3x^2+1)^4 \cdot \underbrace{6x}_{\text{the inside's rate}} = 30x(3x^2+1)^4.$$

The outside is "something to the 5th," whose rate is "5 times that something to the 4th," inside left alone. Then you pay the toll: multiply by the inside's derivative, $6x$. Sanity-check the shape of this habit: if you'd answered $5(3x^2+1)^4$ and stopped, you'd be claiming the inside $3x^2+1$ changes at the same speed as plain x, which it clearly doesn't.

☞ Watch out

The wrong ideas almost everyone carries out of this chapter, named and retired:

- *"$(fg)' = f'g'$."* You personally executed this one yourself a few pages back. Rectangle grows two ways, two terms, added.
- *"Forgot the inside."* Any time something sits *inside* something else, $(\dots)^5$, $\sin(\dots)$, $e^{(\cdots)}$, you owe a multiply by the inside's derivative. This is the single most common wrong derivative on Earth.
- *"The derivative is the tangent line."* Close but no: it's the tangent line's *slope*, a number per point, not a line.
- *"$f'(3)$ means f changed by that much."* It's a rate, not an amount: at that instant, f is changing at that many units per unit of x. Speedometer, not odometer.

✎ **Quick practice, before you move on**

Practice 1. $\frac{d}{dx}\left(x^5 - 4x^3 + 2x\right)$.

Practice 2. $\frac{d}{dx}\left(x^2 \sin(3x)\right)$.

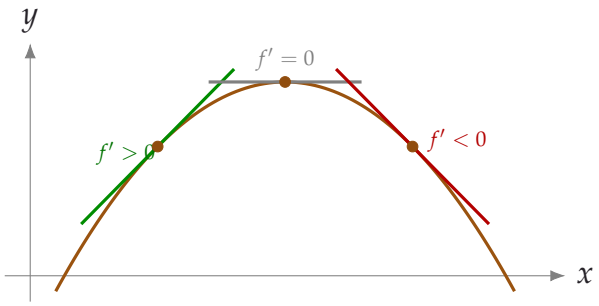
How to think about it: it's two things multiplied, so the product rule runs the show. But the second factor, $\sin(3x)$, has an inside, so differentiating *it* costs a chain-rule toll of 3. Two rules, one problem, one at a time.

Practice 3 (in words, one sentence). Your bike's gearing changes so the wheel turns 4 times per pedal turn, wheel still moves the bike 2 m per turn. What's the new chain-rule product, and which factor changed?

Answers at the end of the chapter. Wrestle first.

Reading a curve through its slopes

The derivative is a number at every point, and the *sign* of that number narrates the graph. Before the picture, predict: on a hill-shaped curve, what must the slope be doing on the way up, at the very top, and on the way down?



One curve, three tangents: climbing, flat at the crest, falling.

Exactly what you predicted: positive slope uphill, *zero* at the crest (the tangent goes flat for one instant), negative downhill. That little observation, "interesting things happen where $f' = 0$," is not a footnote. It is the entire engine of the next chapter: to find the best possible anything, the biggest area, the cheapest can, the maximum profit, you hunt for the flat spots. Hold onto it.

And one loose end, paid in full: L'Hôpital. Chapter 1 told you that a $\frac{0}{0}$ race is settled by which function is falling faster, and promised a gadget that measures "how fast." You now own the gadget. When a limit jams at $\frac{0}{0}$ or $\frac{\infty}{\infty}$, replace top and bottom by their

derivatives and ask again: $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$ (provided that new limit exists). The two mystery rewrites from Chapter 1, "$\ln x$ becomes $\frac{1}{x}$" and "x becomes 1," are now just two lines from your greatest-hits box. Trusted, then verified, as promised.

Carrying this forward

★ Leaving Chapter 2 with this

A derivative is Chapter 1's limit machine pointed at rates: average rates over shrinking windows have a destination, and that destination is the instantaneous rate, which is also the slope of the curve at that point. The definition is $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$, and the shortcut rules are that limit, pre-computed and inherited: power rule, product rule (two growing strips), chain rule (rates multiply, pedals-wheel-bike). Where f' is positive the curve climbs, negative it falls, zero it's momentarily flat, and the flat spots are where treasure gets buried. Next chapter we go dig it up.

✎ The chapter workout (get stuck on purpose)

Cover the answers at the chapter's end, fight all five, then check. Watch for hidden insides.

F1. $\frac{d}{dx}(x^4 - 3x^2 + 7)$ **F2.** $\frac{d}{dx}(x^3 e^x)$

F3. $\frac{d}{dx}(2x+1)^6$ **F4.** $\frac{d}{dx}\sin(x^2)$

F5. The rock from the fill-in table falls $s(t) = 5t^2$ meters in t seconds. How fast is it falling at $t = 4$? And in one sentence: why is your answer a legal number when "distance over time at an instant" is $\frac{0}{0}$?

Nudges: **F1** power rule, term by term. **F2** product rule. **F3** chain, inside is $2x+1$. **F4** chain, inside is x^2. **F5** you already know $s'(t)$.

Answers (wrestle first). **P1:** $5x^4 - 12x^2 + 2$. **P2:** $2x\sin(3x) + 3x^2\cos(3x)$. **P3:** $2 \times 4 = 8$ meters per pedal turn; the inside's rate (wheel turns per pedal turn) changed from 3 to 4. **F1:** $4x^3 - 6x$. **F2:** $x^2e^x(3 + x)$. **F3:** $12(2x + 1)^5$. **F4:** $2x\cos(x^2)$. **F5:** $s'(t) = 10t$, so $s'(4) = 40$ m/s; it's legal because it isn't a division at the instant at all, it's the *limit* of average speeds over shrinking windows, and limits never touch the forbidden point.

Next up: *Applications of Derivatives*, where "flat spots hide treasure" becomes a method: optimization (the best possible anything) and related rates (when two changing things are chained together).

➜ Want it on video?

Professor Leonard, *Calculus 1 Lecture 2.1: Introduction to the Derivative*, then *2.2* (shortcut rules), *2.3* (product and quotient), and *2.6* (the chain rule), at https://www.youtube.com/@ProfessorLeonard.

Applications of the Derivative

Calculus 1 • Chapter 3

Where the slope stops being abstract and becomes a superpower

❧ Chapter 3. Now we get to actually *use* this thing.

Last chapter ended with a treasure map: interesting things happen where the slope hits zero. This chapter digs. You're going to use that one little fact to find the best possible version of things, the biggest pen, the cheapest can, the highest profit, and to connect the speeds of objects that move together.

A heads-up about what makes this chapter hard, because it isn't the calculus. The derivatives in here are easy, Chapter 2 easy. What trips people is the *setup*: turning a paragraph of words into one clean function. That's a skill, it's learnable, and it's mostly a checklist, which I'll hand you. When a problem feels impossible, it's almost never the math. It's that you haven't drawn the picture yet.

So: draw the picture, name the variables, get stuck, stay stuck a minute, and watch the problem crack. Same deal as always. The stuck is the workout.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why these two skills exist and what they're used for in the real world,
- read a graph's whole shape, climbs, peaks, bends, straight from its derivatives,
- find where a function is biggest or smallest, including the sneaky endpoint cases,
- solve optimization word problems with a checklist instead of panic,
- handle related rates, where two changing quantities are chained together,
- predict answers with your gut first, then beat your gut with the math.

Why this exists, and what it's for

Two questions drive this whole chapter, and both are worth real money.

Question one: what's the best possible X? The largest area from a fixed amount of fence. The least aluminum for a can that still holds 12 ounces. The price that maximizes profit. The shortest route, the lightest beam, the fastest schedule. Businesses, engineers, and nature itself are all running optimization problems all day long (soap bubbles minimize surface area; light takes the fastest path). Before calculus, your options were "try a bunch of values and hope." With the derivative, there's a method: the best value of a smooth function lives where the slope is zero or at the edge of what's allowed, and you learned to find flat spots last chapter. That's the whole secret, and this chapter turns it into a routine.

Question two: when two things are linked, how do their speeds relate? A ladder's foot slides away from a wall, so its top must slide down, how fast? A balloon inflates, so its radius and its volume both grow, how are those growth rates tied? Any time two quantities are locked in one equation, their rates are locked together too, and the chain rule from Chapter 2 is exactly the tool that links them. Air traffic control, medical imaging, anything where one measurement has to be inferred from the rate of another, runs on this.

One tool from last chapter. Two superpowers. Setup is the price of admission.

The derivative reads the shape of a graph

First, the reading skills, because both superpowers depend on them. You know from Chapter 2 that the sign of f' narrates the graph: positive means climbing, negative means falling, zero means momentarily flat. Two more pieces complete the kit.

Critical points. A spot where $f'(x) = 0$ (or where f' doesn't exist, like a sharp corner) is called a **critical point**. It's the prime suspect for a peak or a valley, because you can't be at the top of a smooth hill while still tilted.

The second derivative. Differentiate f' again and you get f'', the rate of change *of the slope*, which tells you which way the curve *bends*. Predict before you read on: if the slope itself is increasing as you walk right, what shape is the road? It's curving upward like a cup. So: $f'' > 0$ means **concave up** (cup, holds water), $f'' < 0$ means

concave down (cap, spills water), and a point where the bend flips is an **inflection point**.

★ How to tell a max from a min

Found a critical point? Two ways to classify it, pick whichever is easier that day. **Second-derivative test**: if $f'' < 0$ there, the curve is bending down like a cap, so you're on a local *max* ("local" meaning the top of its own hill; the tallest hill overall is called the *global* max, and step 5 of the coming recipe is how you crown it); if $f'' > 0$, bending up like a cup, a local *min*. **First-derivative test**: walk through the point; if the slope flips from $+$ to $-$, you went up then down, a max; from $-$ to $+$, a min.

Before the worked example, try to beat it. The function is $f(x) = x^3 - 3x$. Without plotting anything, where do *you* think its flat spots are? (You know the move: differentiate, set to zero.)

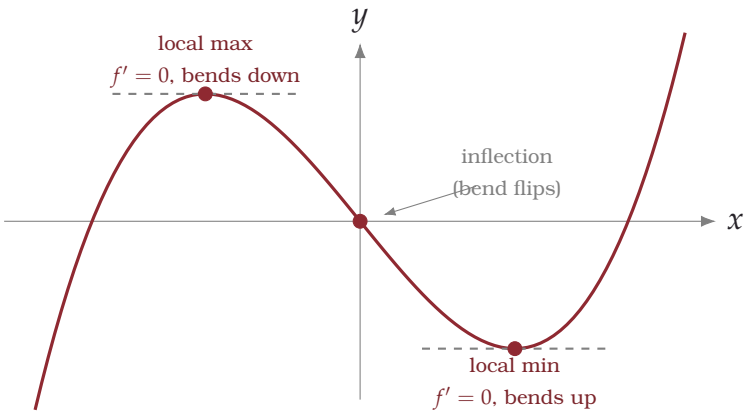
➤ Reading a whole graph from its derivatives: $f(x) = x^3 - 3x$

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 0 \implies x = -1 \text{ and } x = 1 \quad \text{(the critical points)}.$$

Classify with $f''(x) = 6x$: at $x = -1$, $f'' = -6 < 0$, bending down, a local **max**. At $x = 1$, $f'' = 6 > 0$, bending up, a local **min**. And f'' changes sign at $x = 0$: an **inflection point**, where the bend flips.

Assemble the story: climbs to a peak at $x = -1$, falls through the flip at $x = 0$, bottoms out at $x = 1$, climbs forever after. You just sketched a cubic without plotting one point.

The picture below exists only to confirm what the derivatives already told you. Before looking, sketch your own version from the story above, even a two-second scribble. Then compare:



$f(x) = x^3 - 3x$, confirming the derivative's story point for point.

Match up what you see with what you computed: the dashed flat tangents sit exactly at $x = \pm 1$ where $f' = 0$; left of the inflection the curve is a cap, right of it a cup. If your scribble had the same hills in the same order, you've got the skill.

Optimization: finding the best possible anything

The classic: you have 100 ft of fencing and a barn. The barn wall serves as one entire side of a rectangular pen, so your fence only needs to cover the other three sides. What dimensions give the animals the most room?

Gut first, always. A lot of people guess "make it a square, squares are efficient." Hold that thought and let's spy on the problem numerically before any calculus, because you can. Let x be each of the two sides sticking out from the barn; then the side parallel to the barn is whatever fence is left, $y = 100 - 2x$, and the area is $x \cdot y$. Try a few:

x (ft)	20	25	30	33.3
$y = 100 - 2x$	60	50	40	33.3
area $x\,y$	1200	**1250**	1200	1111

Interesting on two counts. First, there's clearly a peak hiding near $x = 25$, the numbers rise then fall. Second, look at the last column: the square-ish pen (33.3×33.3) *loses*. The gut was wrong. Now let's catch the exact peak with calculus instead of guess-and-check.

★ The optimization recipe

1. Name the thing you want to make biggest or smallest (the *objective*) and write it as a formula.
2. Two variables in that formula? Use the *constraint* (the "you only have 100 ft" fact) to eliminate one, so the objective depends on a single variable.
3. Write down the allowed range for that variable (lengths can't go negative).
4. Differentiate, set to zero, solve: those are the critical points.
5. Compare the objective's value at each critical point *and at the endpoints* of the range. Biggest wins a max, smallest wins a min.

Steps 1 to 3 are the hard part and where all the panic lives. The calculus in 4 and 5 is the easy bit.

➤ The fence problem, finished properly

Objective: area $A = xy$. Constraint: $2x + y = 100$, so $y = 100 - 2x$. One variable now:

$$A(x) = x(100 - 2x) = 100x - 2x^2, \qquad 0 \le x \le 50.$$

Differentiate and find the flat spot:

$$A'(x) = 100 - 4x = 0 \implies x = 25, \qquad y = 100 - 50 = 50.$$

Compare all candidates: the endpoints $x = 0$ and $x = 50$ both give area 0 (a pen with no inside), the critical point gives $25 \times 50 = 1250$. And $A'' = -4 < 0$ confirms a genuine peak. Best pen: $\mathbf{25} \times \mathbf{50} = \mathbf{1250}$ square feet, exactly where the table was pointing.

Why not a square? Because the barn donates one side for free. The optimal shape spends the donation: the side running along the barn (50) is double each paid side (25). When all four sides cost fence, the square really does win, and you'll prove that yourself in the practice.

☞ Watch out

The wrong ideas that wreck optimization problems, named and retired:

- *"$f' = 0$, so that's the answer."* Not until you've checked the **endpoints**. The best value can sit on the boundary of what's allowed, where the slope is anything it likes. Flat spots and edges, always both.
- *"Two variables, I'm stuck."* You're not, you're missing the constraint. Every real problem hands you a sentence like "you only have 100 ft," and that sentence is the equation that eliminates a variable.
- *"A critical point must be a max or a min."* It can be neither (an inflection with a flat tangent, like x^3 at 0). Classify, don't assume.

✎ Quick practice, before you move on

Practice 1. Find the critical point of $f(x) = x^2 - 6x + 5$ and say whether it's a max or a min.

Practice 2. A farmer has 40 ft of fence for a rectangular garden, all four sides fenced this time. Prove the gut right for once: show the best shape is the square.
How to think about it: constraint $2x + 2y = 40$, solve for y, plug into $A = xy$, then the recipe. Compare with the barn problem: what changed and why does that rescue the square?

Answers at the end of the chapter. Table your guesses first if you're unsure; it's never cheating to look at numbers.

Related rates: when two things change together

Now the second superpower. Setup for you to predict on: a 10 ft ladder leans against a wall. Someone drags the foot *away* from the wall at a steady 2 ft/s, so the top slides down. Gut question, commit to an answer before the math: does the top also slide at 2 ft/s?

Most people say yes, it feels like the ladder should just "pass the speed along." Let's find out.

The key observation: the two moving quantities are chained together by one unbreakable fact, the ladder's length. If x is the foot's distance from the wall and y is the top's height, then at *every* moment,

$$x^2 + y^2 = 10^2 = 100.$$

Both x and y are changing in time, but this equation holds the whole way through, like a law. And here's the move of the whole section: **differentiate the law itself with respect to time.** Since x and y each depend on t, the chain rule collects a toll on each (that's $\frac{dx}{dt}$ and $\frac{dy}{dt}$, Leibniz notation from Chapter 2: "the rate x changes per second"):

$$2x\,\frac{dx}{dt} + 2y\,\frac{dy}{dt} = 0.$$

Read that equation like a sentence: the two rates are not free, they're locked in a see-saw, weighted by the current positions x and y.

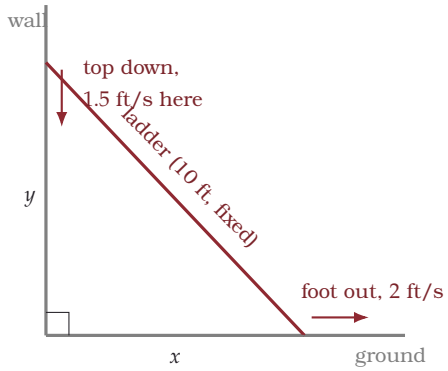
➤ The sliding ladder, snapshot at $x = 6$

At the moment the foot is 6 ft out: $y = \sqrt{100 - 36} = 8$, and we're told $\frac{dx}{dt} = 2$. Solve the see-saw for the rate we want:

$$\frac{dy}{dt} = -\frac{x}{y}\,\frac{dx}{dt} = -\frac{6}{8}\,(2) = -1.5 \text{ ft/s}.$$

The minus sign means downward, which we expected. The *size* is the surprise: 1.5, not 2. The gut was wrong again, and in the best way: the top's speed isn't even constant. Run the same numbers at $x = 8$ (so $y = 6$): $\frac{dy}{dt} = -\frac{8}{6}(2) \approx -2.7$ ft/s, now *faster* than the foot. The further the ladder tips, the faster the top falls, even though the foot never speeds up. (Push it to the limit and the formula predicts absurd speeds as $y \to 0$; that's the math's polite way of saying a real ladder's top leaves the wall before then.)

Before the diagram: you should be able to say, in one sentence, where the "$-\frac{x}{y}$" in that formula came from. (From differentiating the Pythagorean law in time and solving.) Now the picture to pin it all down:



One fixed law, $x^2 + y^2 = 100$, two linked speeds.

The picture is the setup ritual for every related-rates problem you'll ever meet: a drawing, variables on the moving parts, and one equation that stays true the whole time. Everything else is differentiating that equation and plugging in the snapshot.

★ The related-rates recipe

1. Draw it. Label everything that moves with a *variable*, not a number.
2. Write the one equation that ties the variables together at all times (Pythagoras, a volume formula, similar triangles).
3. Differentiate that whole equation with respect to time t. Every variable pays its chain-rule toll.
4. *Only now* plug in the snapshot numbers, and solve for the rate you want.

Plugging numbers in before differentiating is the classic blunder: a number like "6 ft" is only true for one instant, and if you freeze it early, its rate gets recorded as zero and your answer is quietly wrong. Variables until after the derivative. Always.

✎ **Quick practice, before you move on**

Practice 3. Where is $f(x) = x^3 - 3x$ increasing, and where decreasing?

Practice 4. A spherical balloon inflates so its radius grows at 0.5 cm/s. How fast is its *volume* growing at the moment $r = 4$ cm?

How to think about it: the always-true law here is the volume formula $V = \frac{4}{3}\pi r^3$. Differentiate it with respect to time (chain-rule toll on r), then plug the snapshot.

Practice 5 (gut-check, one sentence). In Practice 4, will the volume's growth rate at $r = 8$ be twice the rate at $r = 4$, or more than twice? Decide from the formula's shape before computing.

Answers at the end of the chapter.

Carrying this forward

★ **Leaving Chapter 3 with this**

One tool, three reads. The sign of f' tells the shape, the zeros of f' (plus the endpoints, never forget the endpoints) locate the best and worst, and differentiating a relationship through time links the speeds of anything chained together. Notice what both superpowers had in common: the calculus took one line, the *setup* took the rest, and the setup always started with a drawing and honest variable names. That habit transfers to every quantitative problem you will ever meet, including the ones with no calculus in them at all. Next, the whole machine goes into reverse.

✎ The chapter workout (get stuck on purpose)

F1. Find and classify every critical point of $f(x) = x^3 - 12x$.

F2. Find the maximum value of $f(x) = 6x - x^2$, and say how you know it's a max and not a min.

F3. An open-top box is made by cutting squares of side x from the corners of a 12-inch square of cardboard and folding up the sides. Its volume is $V(x) = x(12 - 2x)^2$. What x gives the biggest box, and what's the allowed range of x you must check?

F4. A stone dropped in a pond makes a circular ripple whose radius grows at 3 ft/s. How fast is the circle's *area* growing when $r = 10$ ft?

Nudges: **F1** $f' = 3x^2 - 12$, then the second-derivative test. **F2** one critical point, check its bend. **F3** expand or product-chain your way to V', solve $V' = 0$, and notice x can't reach 6 (why?). **F4** law: $A = \pi r^2$, differentiate in time, snapshot last.

Answers (wrestle first). **P1:** $f' = 2x - 6 = 0$ at $x = 3$; $f'' = 2 > 0$, a *min* (it's an upward parabola, the bottom of the cup). **P2:** $y = 20 - x$, $A = 20x - x^2$, $A' = 20 - 2x = 0$ at $x = 10$, so 10×10, area 100, and $A'' = -2 < 0$ confirms the peak; the square wins now because no side is free, so there's no donated edge to overspend on. **P3:** $f' = 3(x^2 - 1)$: increasing on $x < -1$ and $x > 1$, decreasing between. **P4:** $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = 4\pi(16)(0.5) = 32\pi \approx 100.5$ cm^3/s. **P5:** more than twice: the rate rides on r^2, so doubling r quadruples it (128π at $r = 8$). **F1:** $f' = 3x^2 - 12 = 0$ at $x = \pm 2$; $f'' = 6x$: max at $x = -2$ (value 16), min at $x = 2$ (value -16). **F2:** $f' = 6 - 2x = 0$ at $x = 3$, $f(3) = 9$; $f'' = -2 < 0$, bending down, a max. **F3:** $V = 144x - 48x^2 + 4x^3$, $V' = 144 - 96x + 12x^2 = 12(x - 2)(x - 6) = 0$ at $x = 2$ or $x = 6$; the range is $0 < x < 6$ (at $x = 6$ the cardboard is all cut away, no base left), so $x = 2$: a $2 \times 8 \times 8$ box, volume 128 in^3. **F4:** $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(10)(3) = 60\pi \approx 188.5$ ft^2/s.

Next up: *Integrals & the Fundamental Theorem*, where instead of breaking change into instantaneous slopes, we add infinitely many slivers back into a whole, and discover the two directions are secretly inverses.

➜ Want it on video?

Professor Leonard, *Calculus 1*, the curve sketching, optimization, and related rates lectures (his 3.x applications sequence), at https://www.youtube.com/@ProfessorLeonard. His optimization walkthroughs are famously patient.

Integrals & the Fundamental Theorem

The reverse gear of calculus • Chapter 4

Adding up infinitely many tiny pieces to measure a whole

❧ Chapter 4. The machine goes into reverse, and the subject snaps together.

Two chapters ago you took a position and extracted a speed. Natural question, and be honest, it's been sitting there: can you go the other way? If you know the *speed* the whole time, can you recover the *distance*? Your odometer thinks so. This chapter builds the machine that does it, the **integral**, and it runs on the same single idea as everything else here: chop the problem into pieces so tiny they're easy, then add them all up with a limit.

And then comes my favorite moment in all of Calc 1. Slopes and areas, the two halves of this subject that look like they live on different planets, turn out to be *the same machine running in opposite directions*. That fact has a grand name, the Fundamental Theorem of Calculus, and this time you won't just be told it. You'll see *why* it has to be true, in one paragraph, with no symbols fancier than the ones you already own.

By the end of this chapter you'll have the complete Calc 1 toolkit, which means you'll have everything Calc 2 is built on. Take it slow. This is load-bearing.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why integrals exist: recovering totals from rates, and measuring curved area, and why those are the same job,
- trap an area between rectangle estimates and watch the vice close on the true value,
- read $\int_a^b f(x)\,dx$ like a sentence, and know why area below the axis counts negative,
- run derivative rules backward to find antiderivatives (and never forget the $+C$),
- explain where the Fundamental Theorem *comes from*, then use it to skip the rectangles forever,
- undo the chain rule with u-substitution, the move half of Calc 2 depends on.

Why this exists, and what it's for

Start with the odometer, because it makes the whole chapter obvious.

Your speedometer is the derivative: it reads your rate, moment by moment. Your odometer answers the reverse question: given all those moments of speed, how much distance *piled up*? If you drive a steady 60 mph for 2 hours, no calculus needed: $60 \times 2 = 120$ miles. Notice what that multiplication is, geometrically: on a graph of speed versus time, a constant 60 for 2 hours is a flat line, and 60×2 is the area of the *rectangle under it*. Height times width. Distance *is* area under the speed curve.

But real speed isn't constant, the line wiggles, and then what? Same move as always: over a tiny sliver of time, speed is *basically* constant, so the sliver of distance is basically a skinny rectangle, rate times tiny width. Add up all the slivers. Total distance equals total area under the wiggly curve, exactly, once the slivers become infinitely thin.

This total-from-a-rate job is everywhere. Distance from speed. Water in the tank from the flow rate at the pipe. Total cost from marginal cost. Charge from current. And, as a pure geometry bonus, the same machine finally answers a question your shape formulas never could: what is the area of a region with a *curved* edge? One machine, two readings, just like the derivative was slope and rate at once. The machine is called the integral. Let's build it with our own hands first.

Trap the area: rectangles, and the closing vice

Target: the area under $y = x^2$ from $x = 0$ to $x = 2$. No formula from school covers it, the top edge bends. So we surround it.

Slice the base $[0, 2]$ into 4 strips of width 0.5 and stand a rectangle on each. If each rectangle's height is the curve's value at the strip's *right* edge, the rectangles poke above the curve, an overestimate. Using each strip's *left* edge, they fall short, an underestimate. Compute both (heights are $f(x) = x^2$ at the sampled points; do a couple of these yourself):

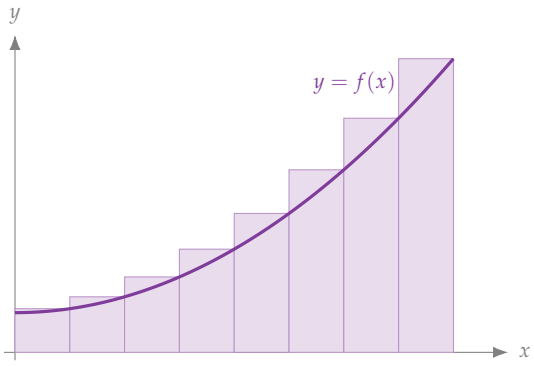
$$\text{right: } (0.25 + 1 + 2.25 + 4) \times 0.5 = 3.75, \qquad \text{left: } (0 + 0.25 + 1 + 2.25) \times 0.5 = 1.75.$$

So the true area is pinned: somewhere between 1.75 and 3.75. Wide pin, sure. Now squeeze: more strips, thinner rectangles. Here's the vice closing (take $n = 8$ on faith or check a couple of heights; $n = 100$ is calculator work):

strips n	left sum	right sum	the truth is…
4	1.75	3.75	in between
8	2.19	3.19	in between
100	2.627	2.707	in between

Both jaws of the vice are closing on one number, something near 2.67. **Remember that number.** In three sections, it's going to reappear from a completely different direction, and when it does, you'll witness the best theorem in the subject catch it exactly. These stacks of rectangles, by the way, have a name, **Riemann sums**, after the mathematician who made the squeezing precise.

Before the picture, a question you can already answer: what should a stack of rectangles under a curve look like, and where exactly does the error hide?



Eight rectangles, already hugging the curve.

Look at where the error lives: only in the little slivers between each rectangle's top and the curve. Thinner rectangles, smaller slivers. In the limit of infinitely many, infinitely thin rectangles, the staircase *becomes* the curve and the error is exactly zero. That limiting total is the area, not approximately, exactly. (Chapter 1 again: we never "finish" the staircase, we read off where the estimates are heading. Limits are still the main character.)

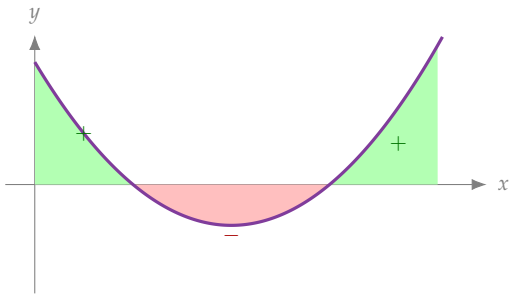
The symbol, read like a sentence

The limiting total gets a name and a notation:

$$\int_a^b f(x)\,dx = \text{the exact signed area under } f \text{ from } a \text{ to } b.$$

Every part of the symbol is a fossil of the rectangles. The $\int$ is a stretched-out letter S, for *sum*, Leibniz drew it that way in the 1600s and it stuck. $f(x)$ is a sliver's *height*. The dx is the sliver's infinitely thin *width* (the same "tiny step in x" from the derivative's $\frac{d}{dx}$). And a, b are where the adding starts and stops. Height times width, summed from a to b: the whole construction, written in four marks.

One honest subtlety before anything else, because it surprises everyone. What if the curve dips *below* the axis? Think about it through the odometer: if the "speed" goes negative, you're driving backward, and those moments *reduce* your net distance from home. The integral keeps the books the same way.



Green adds, red subtracts: the integral is net *area.*

So the integral measures **signed** area: regions above the axis count positive, regions below count negative, and the answer is the net. An integral can come out negative, or exactly zero, with real area visibly on the page. That's not a bug; the pluses and minuses cancelled, the same way forward and backward driving can cancel.

Antiderivatives: the derivative, asked backward

Those rectangle sums give exact answers, but computing them by hand is miserable. The escape route starts with a purely backward question: *what function, when differentiated, produces this one?*

Such a function is called an **antiderivative**. Since $\frac{d}{dx}x^3 = 3x^2$, an antiderivative of $3x^2$ is x^3. But notice: $x^3 + 7$ also works, and $x^3 - 100$, because constants die under differentiation. Antiderivatives come in families, all vertical shifts of each other, so we write the family with a $+C$:

$$\int 3x^2\,dx = x^3 + C.$$

(Yes, the same $\int$ symbol with no endpoints, called the *indefinite* integral, it asks for the antiderivative family rather than a number. Why the same symbol for two jobs? The next section is exactly that story.)

★ The integration greatest hits (derivative rules, read right to left)

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int \frac{1}{x}\,dx = \ln|x| + C,$$

$$\int e^x\,dx = e^x + C, \qquad \int \cos x\,dx = \sin x + C, \qquad \int \sin x\,dx = -\cos x + C.$$

Power rule backward: *raise the exponent by one, divide by the new exponent.* And the $n = -1$ exception exists because raising -1 by one would make you divide by zero; nature routes that one case through the logarithm instead.

Here's the habit that makes integration nearly unfailable, and I want it installed early: **every antiderivative is self-checkable**. Differentiate your answer; you must get the original back. $\frac{d}{dx}\left(\frac{x^4}{4} + C\right) = x^3$, confirmed. Nobody ever has to wonder whether an integral is right. Check it going downhill.

The Fundamental Theorem, and *why* it's true

Time to connect the two halves. Here is the one-paragraph argument, and it's the heart of the chapter.

Define $A(x)$ to be the area accumulated under the curve f, from the start up to position x, the odometer reading, if f is your speed. Question: how fast does that accumulated area grow as x moves right? Nudge x forward by a tiny step h. The new area gained is one skinny sliver: height $f(x)$, width h, so the gain is about $f(x) \cdot h$. Rate of gain = gain over step = $\frac{f(x)\,h}{h} = f(x)$. Shrink h to nothing and "about" becomes "exactly":

$$A'(x) = f(x).$$

Read it slowly, because this is the whole theorem: **the area function's derivative is the original curve**. Accumulating and differentiating undo each other, odometer and speedometer, two directions of one machine. And it hands us a miracle of a shortcut: since A is an antiderivative of f, and all antiderivatives differ only by constants, take *any* antiderivative F you can find; the area from a to b is just the change in F:

★ The Fundamental Theorem of Calculus

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad \text{where } F' = f.$$

To compute an exact area: find an antiderivative, evaluate at both ends, subtract. No rectangles, ever again. (Why it's true, in one line: the area function is itself an antiderivative of f, because each new sliver of area arrives at the rate "height of the curve.")

Now the moment I told you to wait for. The vice from the rectangle section was closing on a number near 2.67 for the area under x^2 from 0 to 2. Watch:

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3} = 2.666\ldots$$

(the bracket $[F(x)]_a^b$ means $F(b) - F(a)$: top value minus bottom value)

Dead center of the vice, $2.627 < \frac{8}{3} < 2.707$, caught exactly, by two subtractions. A hundred rectangles pinned down two digits (average the two jaws and you'd scrape out a third); one antiderivative got you infinitely many. That's the Fundamental Theorem earning its name, and that cross-check, rectangles agreeing with antiderivatives, is why you can trust it with your exams and your bridges.

u-substitution: the chain rule, backward

One more tool and the kit is complete, and Calc 2 will use this one constantly, so plant it deep.

The chain rule said: differentiating $f(g(x))$ costs a multiply by the inside's derivative. So when you're integrating and you *see* a function with an inside, times something that looks suspiciously like that inside's derivative, you're staring at chain-rule exhaust. Run it backward. Try

$$\int 2x\,(x^2+1)^5\,dx.$$

The inside is begging: let $u = x^2 + 1$. Then u's differential is $du = 2x\,dx$, and look, the leftover pieces of the integral are *exactly* $2x\,dx$. Swap everything:

$$\int \underbrace{(x^2+1)^5}_{u^5}\,\underbrace{2x\,dx}_{du} = \int u^5\,du = \frac{u^6}{6} + C = \frac{(x^2+1)^6}{6} + C.$$

Check it downhill, always: differentiating $\frac{(x^2+1)^6}{6}$ gives $\frac{6(x^2+1)^5}{6} \cdot 2x = 2x(x^2+1)^5$. The chain rule's toll pops back out as exactly the $2x$ we absorbed. That's the whole trick: *u-substitution is spotting the toll already sitting in the integrand.* A quicker one: $\int \cos(3x)\,dx$: let $u = 3x$, $du = 3\,dx$, so $dx = \frac{du}{3}$, giving $\frac{1}{3}\sin(3x) + C$. The inside's rate was 3, so the antiderivative owes a $\frac{1}{3}$ to cancel it.

☞ Watch out

The wrong ideas that wreck integrals, named and retired:

- *"Forgot the $+C$."* An indefinite integral is a *family*; omitting $+C$ claims there's only one member. (Definite integrals don't need it, the C cancels in the subtraction.)
- *"Power rule on $\frac{1}{x}$."* Add one to -1 and you divide by zero. That single case is $\ln|x|$, no exceptions to the exception.
- *"My integral came out negative, I must be wrong."* Signed area. Below the axis subtracts. Negative can be the right answer.
- *"The derivative of my answer isn't the integrand, but it's probably fine."* It's never fine. The downhill check is free and binary: match or wrong.
- *"Substitution without the du."* If you set $u = x^2 + 1$ you must also convert dx using $du = 2x\,dx$; swap half the integral and the result is gibberish in two alphabets.

✎ **Quick practice, before you move on**

Practice 1. $\int \left(4x^3 + 1\right) dx.$

Practice 2. $\int \left(x^2 + \frac{1}{x} + e^x\right) dx.$

How to think about it: term by term from the greatest hits; the $\frac{1}{x}$ term is the famous exception; one $+C$ for the whole family at the end.

Practice 3. $\int x\, e^{x^2}\, dx.$

How to think about it: something has an inside (x^2), and its derivative ($2x$) is *almost* sitting out front; you're an adjusting constant away.

Answers at the end of the chapter.

Carrying this forward

★ **Leaving Chapter 4, and Calculus 1, with this**

Four chapters, one idea wearing four costumes. **Limits** ask where things are heading. **Derivatives** are a limit measuring slope and rate. **Integrals** are a limit adding slivers into totals and areas. And the **Fundamental Theorem** is the revelation that the last two are inverses: accumulation's rate is the curve itself, so areas fall to antiderivatives. You also packed u-substitution, the chain rule in reverse. That's the complete foundation. All of Calculus 2, every technique and application ahead, is just getting better at using this kit on harder targets. You built the whole thing from a speedometer that divided zero by zero. Worth being a little proud of that.

✎ **The chapter workout (get stuck on purpose)**

F1. $\int (3x^2 - 2x)\,dx$ **F2.** $\int_1^3 2x\,dx$

F3. $\int \left(\frac{1}{x} + e^x\right) dx$ **F4.** $\int_0^{\pi} \sin x\,dx$

F5. The rock from Chapter 2 falls with speed $v(t) = 10t$ m/s. Use an integral to find how far it falls in the first 3 seconds, and check the answer against Chapter 2's position function $s(t) = 5t^2$.

Nudges: **F1** power rule backward, termwise. **F2** antiderivative of $2x$, subtract ends. **F3** greatest hits. **F4** antiderivative of sin is $-\cos$; watch signs at π. **F5** distance is area under speed: $\int_0^3 10t\,dt$.

Answers (wrestle first). **P1:** $x^4 + x + C$. **P2:** $\frac{x^3}{3} + \ln|x| + e^x + C$. **P3:** $u = x^2$, $du = 2x\,dx$, so $\frac{1}{2}\int e^u du = \frac{1}{2}e^{x^2} + C$ (differentiate it: back to xe^{x^2}, confirmed). **F1:** $x^3 - x^2 + C$. **F2:** $[x^2]_1^3 = 9 - 1 = 8$. **F3:** $\ln|x| + e^x + C$. **F4:** $[-\cos x]_0^{\pi} = -(-1) - (-1) = 2$. **F5:** $\int_0^3 10t\,dt = [5t^2]_0^3 = 45$ m, and $s(3) = 5(9) = 45$: the odometer rebuilt from the speedometer, exactly, which is the Fundamental Theorem doing its day job.

Next up (Calculus 2 begins): *Integration by Parts*, the first of the power techniques, for integrals where the pieces are multiplied together and substitution alone can't crack them.

➜ Want it on video?

Professor Leonard, *Calculus 1*, the lectures on the definite integral, antiderivatives, the Fundamental Theorem, and substitution (his 4.x and 5.x range), at https://www.youtube.com/@ProfessorLeonard.

Integration by Parts

Calculus 2 • Techniques of Integration, Chapter 1

The product rule, run in reverse, to crack integrals you can't do by sight

❧ Welcome to Calc 2. Here's what this whole first unit is really about.

Calc 1 ended with a complete idea: integrals measure totals and areas, and antiderivatives compute them. What Calc 1 politely didn't mention is that most functions you meet in the wild don't have antiderivatives you can see by staring. The greatest-hits table covers maybe a dozen shapes. Everything else needs to be *transformed* into one of those shapes first, and the transformations are this unit.

I want to set your expectations the right way, because this unit has a reputation for being a grab bag of tricks to memorize. It isn't, or at least it doesn't have to be. Every technique in the next five chapters is an old friend from Calc 1 wearing work clothes: today's is literally the product rule read backward. If you know where each tool comes from, you can rebuild it when memory fails, and you know my rule by now: the formulas are rentals, the understanding is yours.

One more thing. You will get stuck in this unit, more than before, because picking the right tool is a judgment call and judgment only grows from scars. Good. Sit in the stuck. That's the rep that counts.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why this technique needs to exist (you'll watch u-substitution fail first),
- rebuild the by-parts formula from the product rule in three lines,
- choose the right "u" every time, and know *why* the LIATE list is ordered the way it is,
- crack $\int xe^x\,dx$, $\int \ln x\,dx$, and their cousins,
- use the tabular shortcut when by parts repeats, and the boomerang when it circles back,
- check any answer yourself by differentiating it.

Why this exists: watch the old tools fail

Here's the integral that forces the issue:

$$\int x\,e^x\,dx.$$

Before any new machinery, try to kill it with what you own. It's not on the greatest-hits table. So try u-substitution, and actually try it, don't just nod: if $u = x$, then $du = dx$, and the e^x left over becomes e^u? No, u *is* x, that renamed nothing. Try $u = e^x$: then $du = e^x dx$, fine, but the integrand has a lone x hanging there that converts into nothing simpler ($x = \ln u$, and you can't integrate $\ln u$ yet, that's today's lesson). Substitution needs the integrand to be chain-rule exhaust, an inside function with its derivative standing by. A plain *product* of two unrelated species, a power times an exponential, just isn't that shape. The wall is real. Feel it.

And products like this are not rare curiosities. "Rate times time-weight" integrals ($t\,e^{-t}$ and friends) run through probability and statistics, physics is full of position-times-force products, and signal processing lives on integrals of x against sines and cosines. A tool for integrating products isn't a luxury; whole fields sit on it.

So where would such a tool come from? Think about it before I show you: what rule do you already own that *produces* products when you run it forward?

Where it comes from: three honest lines

The product rule, of course. For two functions u and v:

$$(uv)' = u'v + uv'.$$

Integrate both sides (the left side just undoes its derivative):

$$uv = \int u'v\,dx + \int uv'\,dx.$$

Now solve for the second integral on the right:

$$\int uv'\,dx = uv - \int u'v\,dx.$$

That's the whole derivation, three lines, nothing hidden. Written in the usual shorthand ($du = u'\,dx$, $dv = v'\,dx$):

★ The by-parts formula

$$\boxed{\int u\,dv = uv - \int v\,du}$$

Split your integrand into two pieces: a u (which you will *differentiate*) and a dv (which you will *integrate*). The formula then trades your integral for a different one, $\int v\,du$. It's not a free win, it's a **trade**. The entire game is making sure the new integral is simpler than the old one.

Small print worth knowing: when you integrate dv to get v, don't add a $+C$ there; any one antiderivative works and the constant would cancel out of the formula anyway. Take the simplest v on offer.

The whole game is choosing u, so let's choose wrong on purpose

The formula gives you a choice: in $\int xe^x dx$, either piece could be u. Let's deliberately pick badly and watch what happens, because the failure teaches the rule better than the rule does.

Take $u = e^x$ and $dv = x\,dx$. Then $du = e^x dx$ and $v = \frac{x^2}{2}$, so the formula says

$$\int xe^x\,dx = \frac{x^2}{2}e^x - \int \frac{x^2}{2}e^x\,dx.$$

Look at the trade we just made: we came in with x times e^x and we're leaving with x^2 times e^x. The power went *up*. The new integral is strictly worse than the one we started with, and another round would make it worse still. Legal move, losing strategy.

Now flip it: $u = x$, $dv = e^x dx$, so $du = dx$ and $v = e^x$:

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

The x got differentiated down to a plain 1 and the leftover integral collapsed to a greatest-hit. Same formula, opposite outcome. Check it the Calc 1 way, downhill: $\frac{d}{dx}\big(e^x(x-1)\big) = e^x(x-1) + e^x = xe^x$. Confirmed, and that check is always available to you.

The moral, stated as a principle: u **should be the piece that improves when differentiated, and** dv **the piece that doesn't get worse when integrated.** Powers of x improve under differentiation (they shrink toward constants). Exponentials and sines don't care what you do to them, differentiate or integrate, they just cycle. Logs improve *dramatically* when differentiated ($\ln x$ becomes the tame $\frac{1}{x}$) and are nasty to integrate. Rank everything by that logic and you get the standard cheat sheet:

★ LIATE: who should be u?

L – Logarithmic ($\ln x$)
I – Inverse trig ($\arctan x$, $\arcsin x$)
A – Algebraic (powers like x, x^2)
T – Trigonometric ($\sin x$, $\cos x$)
E – Exponential (e^x)

Whichever species appears *earliest* on the list becomes u; the rest is dv. This isn't an arbitrary acronym to obey, it's the "improves under differentiation" ranking you just derived by losing on purpose: top of the list desperately wants to be differentiated, bottom of the list doesn't mind being integrated forever. Treat it as a strong default, not a law of physics.

➤ The sneaky one: $\int \ln x \, dx$

One function, no visible product, so where's the dv? Manufacture it: let $dv = dx$, the invisible factor of 1. Then $u = \ln x$ (L, top of the list), $du = \frac{1}{x}dx$, $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Nobody guesses that antiderivative cold; by parts manufactures it. And notice why it worked: the horrid-to-integrate $\ln x$ got *differentiated* into the harmless $\frac{1}{x}$, which then cancelled the x. The list knew.

✎ Quick practice, before you move on

Practice 1. $\int x \cos x \, dx$.

Practice 2. $\int x^2 \ln x \, dx$.
How to think about it: there's an L and an A in the room. L wins the u job even though the x^2 looks like the main character. Trust the ranking, then watch the ln melt into $\frac{1}{x}$.

Practice 3 (one sentence). In Practice 1, what exactly would have gone wrong if you'd chosen $u = \cos x$ instead?

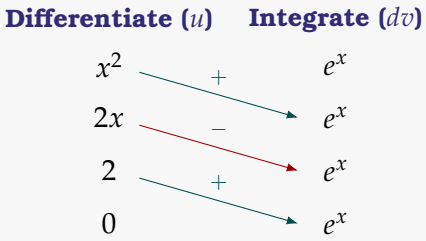
Answers at the end of the chapter. Fight first.

When by parts repeats: the tabular shortcut

Try $\int x^2 e^x \, dx$ with the rule you now own: $u = x^2$ knocks the power down to $2x$, but the leftover integral $\int 2x \, e^x \, dx$ still has a product in it, so you run by parts *again*. It works, it's just bookkeeping, and bookkeeping begs for a better layout. When u is a power of x (so repeated differentiation will grind it down to 0) and dv is a self-recycling integrand like e^x or $\sin x$, the **tabular method** does all the rounds at once.

Two columns: differentiate the u side down to zero, integrate the dv side the same number of times. Before you look at the table, predict how many rounds x^2 needs to die. (Two differentiations to reach the constant 2, a third to hit 0.)

➤ **Tabular method:** $\int x^2 e^x \, dx$



Multiply along each diagonal arrow, with signs alternating $+, -, +, \ldots$ down the page:

$$\int x^2 e^x \, dx = x^2 e^x - 2x\, e^x + 2e^x + C = e^x\left(x^2 - 2x + 2\right) + C.$$

What the table actually is: every row is one full round of by parts, pre-chewed. The alternating signs are the repeated "$-\int v\, du$" flipping each round. Differentiate the answer and you really do get $x^2 e^x$ back; do that check once by hand and you'll trust the table forever.

Two more patterns worth meeting

The inverse-trig move (the I in LIATE). Same manufacture-a-dv trick as $\ln x$:

➤ $\int \arctan x \, dx$

$u = \arctan x$, $dv = dx$, so $du = \frac{1}{1+x^2}dx$, $v = x$:

$$\int \arctan x \, dx = x \arctan x - \int \frac{x}{1+x^2}\, dx = x \arctan x - \frac{1}{2}\ln\left(1 + x^2\right) + C.$$

The leftover integral is a Calc 1 substitution ($u = 1 + x^2$, and its derivative $2x$ is half-present, hence the $\frac{1}{2}$). Two chapters of tools, one answer.

The boomerang (when by parts circles back). Here's a strange one: $\int e^x \sin x \, dx$. Neither piece ever dies, the exponential recycles and the sine just rotates through cosine and back. Predict what happens if you run by parts twice. Something funny:

➤ $\int e^x \sin x \, dx$

Round one ($u = \sin x$, $dv = e^x dx$): $\int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx$. Round two on the new integral ($u = \cos x$): $\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx$. Substitute back, writing I for the original integral:

$$I = e^x \sin x - e^x \cos x - I.$$

The original integral walked back in the door, the boomerang. Don't despair, *solve for it* like the algebra it is: $2I = e^x(\sin x - \cos x)$, so

$$\int e^x \sin x \, dx = \frac{e^x}{2}(\sin x - \cos x) + C.$$

An integral computed without ever actually integrating it to the end. (Note the tabular method is useless here, nothing ever reaches 0; boomerangs are their own species.)

☞ Watch out

The wrong ideas that wreck by-parts problems, named and retired:

- *"Dropped the minus."* The formula is uv **minus** $\int v \, du$, and that sign is the single most common casualty. Write the formula fresh each time; it's ten characters.
- *"LIATE is a law."* It's a default that wins the overwhelming majority of the time. If your trade is making the integral worse (powers climbing, logs multiplying), stop and re-deal, whatever the acronym said.
- *"I need by parts!"* Check for plain u-substitution first. $\int xe^{x^2} dx$ looks like today's problem but it's yesterday's: the x outside is the inside's derivative in disguise. By parts on it is legal and pointlessly painful.
- *"Boomerang panic."* If the original integral reappears, that's not failure, that's the equation handing itself to you. Name it I and solve.

Carrying this forward

★ Leaving this chapter with this

By parts is the product rule run backward: $\int u\,dv = uv - \int v\,du$, a *trade* of one integral for another. Choose u as the piece that improves under differentiation (that's all LIATE is), let the tabular method do the bookkeeping when a power has to grind down to zero, and when the integral boomerangs back, solve for it algebraically. For definite integrals the same formula carries bounds: $\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$. And every answer is self-checkable by differentiating, use that power. Next chapter: a different stubborn family, powers of sines and cosines, tamed by a couple of identities and one clever save-a-factor move.

✎ The chapter workout (get stuck on purpose)

F1. $\int xe^{2x}\,dx$ **F2.** $\int x\sin x\,dx$

F3. $\int x\ln x\,dx$ **F4.** $\int x^3e^x\,dx$

F5 (judgment call, one sentence each). For each of these, say which tool you'd reach for and why, no need to compute: (i) $\int x\,e^{x^2}dx$ (ii) $\int x^2\cos x\,dx$ (iii) $\int e^{2x}\cos x\,dx$.

Nudges: **F1** $u = x$; integrate the dv carefully. **F2** A beats T. **F3** L beats A. **F4** the power is 3: tabular, five rows counting the final 0. **F5** look for a disguised substitution, a dying power, and a boomerang.

Answers (wrestle first). **P1:** $u = x$, $dv = \cos x\,dx$: $x\sin x + \cos x + C$. **P2:** $u = \ln x$, $v = \frac{x^3}{3}$: $\frac{x^3}{3}\ln x - \frac{x^3}{9} + C$. **P3:** you'd be integrating $x\,dx$ into $\frac{x^2}{2}$, raising the power, the same losing trade as the choose-wrong-on-purpose demo. **F1:** $\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C$. **F2:** $-x\cos x + \sin x + C$. **F3:** $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$. **F4:** tabular with x^3: $e^x\left(x^3 - 3x^2 + 6x - 6\right) + C$. **F5:** (i) plain u-sub, the x outside is $\frac{1}{2}$ of the inside's derivative; (ii) by parts, tabular, the x^2 dies after two rounds of parts (three differentiations in the table); (iii) boomerang, neither factor ever dies, run parts twice and solve for I.

Next up: *Trigonometric Integrals*, where powers of sine and cosine fall to two identities and a save-one-factor trick.

➜ Want it on video?

Professor Leonard, *Calculus 2 Lecture 7.1: Integration by Parts*, at `https://www.youtube.com/@ProfessorLeonard`. He runs the tabular method on camera too.

Trigonometric Integrals

Calculus 2 • Techniques of Integration, Chapter 2

Taming powers of sine and cosine with a couple of identities

❧ Chapter 2. Integrals that look much worse than they are.

$\int \sin^3 x \cos^2 x \, dx$ has the silhouette of a nightmare. It's a kitten. Every integral in this chapter cracks with the same two identities you half-remember from trig class, plus the substitution skill you already own, and part of the fun is that you can usually *see* which identity a problem wants before touching your pencil.

The mindset matters here, so let me say it straight: you will not brute-force anything in this chapter. You'll *rearrange* each integral until it becomes a plain u-substitution, and the rearranging follows exactly two patterns. Learn to spot the pattern and the rest is Calc 1. This is also your warmup for the next chapter, where trig stops being the obstacle and becomes the weapon.

Get stuck on the practice before peeking, as always. Spotting patterns is a muscle, and reading about a muscle has never once grown it.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why anyone needs to integrate powers of sines and cosines (your wall outlet is involved),
- crack $\int \sin^m x \cos^n x \, dx$ whenever a power is *odd*, by peeling off one factor,
- explain why that trick dies when all powers are even, because you'll watch it die,
- grind even powers down with the power-reduction identities, and *see* on a graph why those identities are true,
- sanity-check a trig integral with an average-value argument instead of faith.

Why this exists, and what it's for

Sines and cosines aren't just triangle trivia. They are the mathematics of *anything that oscillates*: sound waves, light, radio, your heartbeat on an EKG, the alternating current in your wall. And one of the great facts of applied math (Fourier's insight, which you'll brush against again in the series unit) is that essentially *every* signal can be broken into a pile of sines and cosines. Working with signals, their energy, their averages, how they interfere, means integrating products and powers of sines and cosines, all day, forever. This chapter is that skill.

Here's a concrete one you've touched today. The power an alternating current delivers rides on $\sin^2$, and the "120 volts" printed near your outlet is not the peak voltage (that's about 170). It comes from averaging the $\sin^2$-shaped power, the exact integral you'll do in the power-reduction section. When you finish that example, you'll know where the number on the wall comes from.

And within this book, the chapter has a second job: next chapter we'll deliberately *inject* trig into integrals to destroy square roots, and that only pays off if you can integrate the trig that comes out. Consider this the training arc.

Case 1: an odd power, and you'll discover the trick yourself

Here's the opening boss: $\int \sin^3 x \cos^2 x \, dx$. Before any recipe, let's hunt for a substitution like it's Calc 1, because the discovery is the lesson.

Try $u = \cos x$. Then $du = -\sin x \, dx$, so to swap into u-world, the integrand must donate exactly one $\sin x \, dx$ as the toll. Look at what you have: *three* sines. Pay one as the toll. That leaves $\sin^2 x$ unconverted, and here's the question that unlocks the whole chapter: can you rewrite $\sin^2 x$ purely in terms of cosine? You've known the answer since trig class:

$$\sin^2 x + \cos^2 x = 1 \qquad \Longrightarrow \qquad \sin^2 x = 1 - \cos^2 x.$$

Everything converts. Watch the whole thing go:

➤ Odd power of sine: $\int \sin^3 x \cos^2 x\, dx$

Peel one sine off as the future toll, convert the remaining pair:

$$\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x)\, \sin x,$$

so the integral is

$$\int (1 - \cos^2 x)\, \cos^2 x\, \sin x\, dx.$$

Substitute $u = \cos x$, $du = -\sin x\, dx$ (mind that minus, it's the most-dropped sign in the chapter):

$$= -\int (1 - u^2)\, u^2\, du = -\int (u^2 - u^4)\, du = -\frac{u^3}{3} + \frac{u^5}{5} + C = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C.$$

A scary trig integral turned into integrating a polynomial, which is the entire strategy of this chapter in one example. (Differentiate the answer if you want the downhill check; the chain rule regenerates the $\sin x$.)

★ The odd-power move

If sin or cos carries an *odd* power: peel off one copy of the odd one to serve as du's partner, convert the now-even leftovers to the *other* function via $\sin^2 x + \cos^2 x = 1$, and substitute. If *both* powers are odd, you choose; peel whichever leaves less algebra. The odd power is a gift: it's the spare factor that pays the substitution toll.

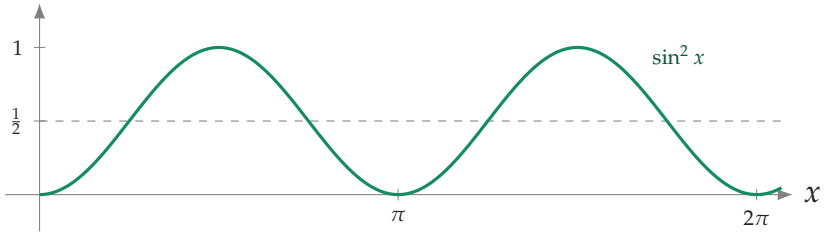
Watch the trick die: why even powers need a new idea

The book's rule is that you should see *why* a method exists, so let's break the odd-power move in front of witnesses. Try it on the innocent-looking $\int \sin^2 x\, dx$. Peel off one sine: $\sin^2 x = \sin x \cdot \sin x$, substitute $u = \cos x$, pay the toll with one $\sin x\, dx$… and now the *leftover* is a single $\sin x = \sqrt{1 - \cos^2 x}$. A square root. We've traded a clean square for a root, strictly worse. The peel trick needs the leftovers to form an *even* power (so they convert to whole powers of cosine); with everything even, there's no spare factor to peel without wrecking the parity. The trick is dead. We need a genuinely different idea.

The different idea: if you can't *convert* the square, *lower* it. There are identities that rewrite a squared trig function with no square at all, at the cheap price of doubling the angle:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These are the **power-reduction** (or half-angle) identities, and rather than ask you to trust them, let me show you one, because it's hiding in plain sight on a graph. Question first: the function $\sin^2 x$ never goes negative, it wobbles between 0 and 1. If you had to guess its *average* height, and the shape of its wobble, what would you guess? Now look:



$\sin^2 x$: a perfect wave of half the amplitude, riding the line $y = \frac{1}{2}$, at double the frequency.

Study what the graph is confessing. $\sin^2 x$ is itself just a cosine wave: centered on $\frac{1}{2}$, swinging $\frac{1}{2}$ up and down, completing its cycle *twice* as fast as $\sin x$. Write that sentence in symbols, remembering the wave starts at its *bottom* (since $\sin^2 0 = 0$, which is what pins down the minus sign), and you get $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$, exactly the identity. It isn't algebraic sorcery; it's a description of the picture. (And the average height is clearly $\frac{1}{2}$, by the symmetry of the wobble around the dashed line. Hold that thought, it's about to compute your wall outlet.)

➤ The even workhorse: $\int \sin^2 x\,dx$

$$\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

(The $\cos 2x$ integrates to $\frac{\sin 2x}{2}$, the inside's rate of 2 charging its toll, Calc 1 substitution.) Sanity-check it with the picture: over a full cycle 0 to 2π, the wobble term $\frac{\sin 2x}{4}$ returns to where it started and contributes nothing, leaving $\frac{x}{2}\Big|_0^{2\pi} = \pi$. Area π over a base of 2π means average height $\frac{1}{2}$. Exactly what the graph promised. And that $\frac{1}{2}$ is the outlet: a current's power rides on $\sin^2$, average power is half the peak power, and the "120 V" rating is the steady voltage that would deliver that same average (peak ≈ 170, and $170/\sqrt{2} \approx 120$). You just derived the number on the wall.

Higher even powers: same hammer, swung more than once.

➤ Both even, higher: $\int \cos^4 x\,dx$

$$\cos^4 x = \left(\cos^2 x\right)^2 = \left(\frac{1+\cos 2x}{2}\right)^2 = \frac{1}{4}\left(1 + 2\cos 2x + \cos^2 2x\right),$$

and that lingering $\cos^2 2x$ is still a square, so reduce it too: $\cos^2 2x = \frac{1+\cos 4x}{2}$. Collect everything:

$$\cos^4 x = \frac{3}{8} + \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x,$$

and now every term is a greatest-hit:

$$\int \cos^4 x\,dx = \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C.$$

The pattern: keep reducing until no squared trig survives, then integrate term by term. Each round doubles the angle and halves the drama.

One mixed case worth meeting: $\int \sin^2 x \cos^2 x\,dx$. You could reduce both factors, but there's a slicker door: $\sin x \cos x = \frac{1}{2}\sin 2x$, so the integrand is $\frac{1}{4}\sin^2 2x$, one power-reduction away from done. Keep an eye out for that doubling shortcut; it shows up on exams because graders like elegance.

☞ Watch out

The wrong ideas that wreck trig integrals, named and retired:

- *"*$\int \sin^2 x\,dx = \frac{\sin^3 x}{3} + C$*."* The reverse power rule only applies to powers of x. Differentiate that guess: the chain rule spits out an extra $\cos x$ that was never in your integrand. Downhill check, always.
- *"Dropped the minus on* $du = -\sin x\,dx$*."* The single most common sign error in the unit. Write the du line every time.
- *"Both powers odd, now what?"* Now you're spoiled for choice: peel either one. Pick the one that leaves the simpler polynomial.
- *"The half-angle identities are random facts to memorize."* You *saw* one on the graph: a squared wave is a half-amplitude, double-speed wave riding its average. Rebuild it from that picture whenever memory flakes.

✎ Quick practice, before you move on

Practice 1. $\int \sin x \cos x\,dx$.
How to think about it: easiest problem in the chapter, and there are *three* correct routes: $u = \sin x$, or $u = \cos x$, or the doubling shortcut. Take any one; if you're feeling strong, take two and confirm they agree (the answers can differ by a constant, which is what $+C$ is for).

Practice 2. $\int \sin^5 x\,dx$.
How to think about it: sine's power is odd, so peel one off and convert the leftover $\sin^4 x = (1-\cos^2 x)^2$. Then it's a polynomial in $u = \cos x$.

Practice 3 (one sentence). Without computing, say what the average value of $\cos^2 x$ over a full cycle must be, and how you know without any integral.

Answers at the end of the chapter.

A note on the cousins: tangent and secant

This chapter's spotlight is sine and cosine, but their cousins tan and sec get tamed in exactly the same spirit, with their own matched pair of tools: the identity $\sec^2 x = 1 + \tan^2 x$ plays the role of Pythagoras, and the derivative pairs $\frac{d}{dx}\tan x = \sec^2 x$, $\frac{d}{dx}\sec x = \sec x \tan x$ tell you which factor to peel off as the toll. Same game, different uniforms: spot the odd or spare factor, peel, convert, substitute. If your course runs through $\int \tan^3 x \sec x \, dx$ and friends, you already have the reflexes.

Carrying this forward

★ Leaving this chapter with this

Two patterns cover the whole family. **Odd power anywhere**: peel one copy for the toll, convert the even leftovers with $\sin^2 + \cos^2 = 1$, substitute, integrate a polynomial. **All powers even**: no spare factor exists (you watched the peel produce a square root and die), so lower the squares instead with power reduction, a squared wave is a half-amplitude double-speed wave riding its average, and repeat until everything is a greatest-hit. Check answers downhill or by average-value sanity (your outlet says $\sin^2$ averages $\frac{1}{2}$). Next chapter flips the relationship: instead of trig being the problem, we'll *import* trig on purpose, to assassinate square roots.

✎ The chapter workout (get stuck on purpose)

Decide the case first (odd peel, or power reduction), then execute.

F1. $\int \cos^3 x\,dx$ **F2.** $\int \sin^3 x\,dx$

F3. $\int \sin^2 x \cos^2 x\,dx$ **F4.** $\int_0^{2\pi} \cos^2 x\,dx$

F5 (spot the trap, one sentence). A classmate writes $\int \sin^3 x\,dx = \frac{\sin^4 x}{4} + C$. What single check exposes the error, and what did they forget?

Nudges: **F1** cosine is odd: peel one, $u = \sin x$. **F2** sine is odd: peel, $u = \cos x$. **F3** the doubling shortcut $\sin x \cos x = \frac{1}{2}\sin 2x$, then one reduction. **F4** you already know the average height of $\cos^2$; base times average beats algebra. **F5** go downhill.

Next up: *Trigonometric Substitution*, where a right triangle turns $\sqrt{a^2 - x^2}$ and its relatives into pure trig, which, thanks to this chapter, you can now integrate.

➜ Want it on video?

Professor Leonard, *Calculus 2 Lecture 7.2: Integrals of Trigonometric Functions*, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** $u = \sin x$ gives $\frac{\sin^2 x}{2} + C$; $u = \cos x$ gives $-\frac{\cos^2 x}{2} + C$; the doubling route gives $-\frac{\cos 2x}{4} + C$. All three differ only by constants, same family. **P2:** $\int (1 - \cos^2 x)^2 \sin x \, dx$ with $u = \cos x$: $-\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C$. **P3:** $\frac{1}{2}$: the graph of $\cos^2$ is the same half-amplitude wave riding $y = \frac{1}{2}$ as $\sin^2$, just shifted, and a symmetric wobble averages to its centerline. **F1:** $\sin x - \frac{\sin^3 x}{3} + C$. **F2:** $-\cos x + \frac{\cos^3 x}{3} + C$. **F3:** $\frac{1}{4}\int \sin^2 2x \, dx = \frac{x}{8} - \frac{\sin 4x}{32} + C$. **F4:** average $\frac{1}{2}$ times base 2π gives π (the wobble term cancels over a full cycle). **F5:** differentiate it: you get $\sin^3 x \cos x$, an uninvited $\cos x$, because the reverse power rule needs the inside's derivative present; sine is odd here, so the peel-and-convert route was the correct one.

Trigonometric Substitution

Calculus 2 • Techniques of Integration, Chapter 3

Hiding a right triangle inside your integral to kill an ugly square root

❧ Chapter 3. The chapter where you finally earn πr^2.

You've been quoting the area of a circle since you were ten. Has anyone ever shown you why it's true? Not waved at it, actually shown you? By the end of this chapter, you'll derive πr^2 yourself, with your own hands, and the tool that does it is the strangest-feeling move in the course: when an integral contains an ugly square root, you deliberately make it *more* trigonometric. It sounds backward. It works like a charm.

Here's the honest pitch. Roots like $\sqrt{a^2 - x^2}$ shut down every technique you own; you'll verify that yourself in a minute. The escape is to stop describing the problem with lengths and start describing it with an *angle*, and the moment you do, the Pythagorean identity reaches in and dissolves the root. The device that keeps all of it organized is nothing fancier than a right triangle drawn in the margin.

This technique has a reputation for feeling like black magic. It isn't, and my whole job in this chapter is to show you the picture that makes it feel inevitable instead. Magic you understand is just technique.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say why these integrals matter (circles, spheres, orbits, lenses, anything round),
- watch your current tools fail on $\sqrt{a^2 - x^2}$, so the new one is earned, not decreed,
- see where the substitution $x = a \sin\theta$ *comes from* (it's a change of viewpoint, not a trick),
- match each of the three root shapes to its substitution and its triangle,
- back-substitute cleanly by reading side ratios off the triangle,
- prove the area of a circle. For real. Yourself.

Why this exists, and what it's for

Where do integrals with $\sqrt{a^2 - x^2}$ even come from? From circles, and the world is full of circles.

Take the circle of radius a centered at the origin: $x^2 + y^2 = a^2$. Solve for the top half: $y = \sqrt{a^2 - x^2}$. That's not an exotic function; that is *the* function whose graph is a circular arc. So the moment you ask any area or length question about anything round, a wheel, a dome, a lens cross-section, a planet's disk, the water in a tilted round pipe, you are integrating expressions with $\sqrt{a^2 - x^2}$ in them. The other two shapes in this chapter, $\sqrt{a^2 + x^2}$ and $\sqrt{x^2 - a^2}$, come from the circle's siblings (hyperbolas and distance formulas), and show up in physics anywhere a distance gets computed from components.

So the practical stakes are plain: no tool for these roots, no calculus on round things. And the chapter's finale will be the roundest result of all: the area formula you memorized in grade school, finally derived instead of decreed.

Watch your tools fail, then find the idea

Target: $\int \sqrt{1-x^2}\,dx$, the area-under-a-circular-arc integral in its simplest outfit. Attack it with everything you own, for real, before reading on.

Greatest hits? Not on the table. u-substitution with $u = 1 - x^2$? Then $du = -2x\,dx$, and you need a spare x in the integrand to pay the toll. There isn't one, the root stands alone. (Important contrast, file it away: $\int x\sqrt{1-x^2}\,dx$ *does* have the spare x, and falls to plain Calc 1 substitution in two lines. Always check for that before deploying this chapter's artillery.) By parts? Try it; the pieces circle without improving. The wall is real.

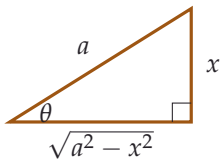
Now the idea, and I want you to see it as a *change of viewpoint*, because that's what it is. The expression $\sqrt{1-x^2}$ is the height of a point on a unit circle sitting above position x. But there are two natural ways to say *where* a point on a circle is: by its horizontal position x, or by its **angle** θ around the circle. And trig class taught you the dictionary between them: the point at angle θ sits at $x = \sin\theta$ with height $\cos\theta$ (measuring the angle from the top). Rewrite the problem in the angle's language, and the height $\sqrt{1-x^2}$ becomes

$$\sqrt{1-\sin^2\theta} = \sqrt{\cos^2\theta} = \cos\theta.$$

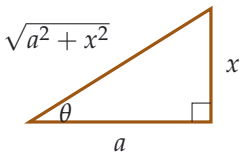
No root. Not simplified, *gone*. The Pythagorean identity $1 - \sin^2\theta = \cos^2\theta$ is a root-annihilation machine, and the "substitution" $x = \sin\theta$ was never a trick: it's just describing the circle the way a circle wants to be described, by angle. That's the whole chapter. Everything else is bookkeeping, and the bookkeeper is a right triangle.

The three shapes, their triangles, and the dictionary

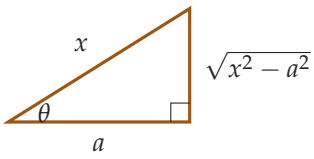
There are exactly three root shapes, each dissolved by one Pythagorean identity. Before the picture, here's what to look for in it: in each triangle, find which side is x, which is a, and which side carries the *root*. The triangle IS the dictionary between x-language and θ-language.



$x = a\sin\theta$



$x = a\tan\theta$



$x = a\sec\theta$

Walk the first triangle with me, the other two run on the same rails. Hypotenuse a, opposite side x: that's a triangle built so that $\sin\theta = \frac{x}{a}$, in other words $x = a\sin\theta$, the substitution. And Pythagoras instantly labels the remaining leg: $\sqrt{a^2 - x^2}$. So the triangle holds, in one picture, the substitution going *in* (the sine ratio) and every conversion you'll need coming back *out* (any trig function of θ is just a ratio of two labeled sides). Same deal for the middle triangle, where $\tan\theta = \frac{x}{a}$ puts the root on the hypotenuse, and the right one, where $\sec\theta = \frac{x}{a}$ puts it on a leg.

★ The three substitutions

$$\sqrt{a^2 - x^2}: \quad x = a\sin\theta \;\Rightarrow\; \sqrt{a^2 - x^2} = a\cos\theta \quad (1 - \sin^2 = \cos^2)$$

$$\sqrt{a^2 + x^2}: \quad x = a\tan\theta \;\Rightarrow\; \sqrt{a^2 + x^2} = a\sec\theta \quad (1 + \tan^2 = \sec^2)$$

$$\sqrt{x^2 - a^2}: \quad x = a\sec\theta \;\Rightarrow\; \sqrt{x^2 - a^2} = a\tan\theta \quad (\sec^2 - 1 = \tan^2)$$

How to remember which goes where without memorizing: match the *signs*. Minus-after-a^2 is the circle, sine's home. Plus means tangent (the only identity with a plus). Minus-before-a^2 is secant. Or just sketch the triangle; it refuses to be drawn wrong.

☞ Watch out

Honest fine print: writing $\sqrt{a^2 - x^2} = a\cos\theta$ rather than $a|\cos\theta|$ quietly assumes θ stays in the standard range of the matching inverse function, where the relevant trig values are positive. Every problem in a first course honors that assumption, so run with it, but know it's there; it earns its keep in the secant case with negative x, where signs genuinely need care.

The three cases, worked

➤ Case 1 (sine): $\int \frac{dx}{\sqrt{9-x^2}}$

Shape: minus after the constant, so sine, with $a = 3$. Substitute $x = 3\sin\theta$, and don't forget the root's little brother: $dx = 3\cos\theta\, d\theta$ (the dx must convert too, it's the most-forgotten step in the chapter). The root dissolves:

$$\sqrt{9-9\sin^2\theta} = 3\cos\theta, \qquad \text{so} \qquad \int \frac{3\cos\theta\, d\theta}{3\cos\theta} = \int 1\, d\theta = \theta + C.$$

Back to x-language: $x = 3\sin\theta$ means $\theta = \arcsin\frac{x}{3}$. Answer: $\arcsin\frac{x}{3} + C$. (Downhill check: differentiate $\arcsin\frac{x}{3}$ and the $\frac{1}{\sqrt{9-x^2}}$ reappears. It does.)

➤ Case 2 (tangent): $\int \frac{dx}{\sqrt{x^2+4}}$

Plus sign: tangent, $a = 2$. Sub $x = 2\tan\theta$, $dx = 2\sec^2\theta\, d\theta$, and $\sqrt{x^2+4} = 2\sec\theta$:

$$\int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta = \ln|\sec\theta + \tan\theta| + C$$

(that last antiderivative is a classic: multiply $\sec\theta$ top and bottom by $\sec\theta + \tan\theta$ and the numerator becomes exactly the derivative of the denominator, a one-line u-sub into a log; derive it once, then pocket it). Now the triangle pays off: from the middle triangle with x opposite and $a = 2$ adjacent, read $\tan\theta = \frac{x}{2}$ and $\sec\theta = \frac{\sqrt{x^2+4}}{2}$. Substitute and fold the constant $-\ln 2$ into C:

$$\int \frac{dx}{\sqrt{x^2+4}} = \ln\left|x + \sqrt{x^2+4}\right| + C.$$

➤ **Case 3 (secant):** $\int \frac{dx}{\sqrt{x^2-9}}$

Minus before the constant: secant, $a = 3$. Sub $x = 3\sec\theta$, $dx = 3\sec\theta\tan\theta\,d\theta$, and $\sqrt{x^2-9} = 3\tan\theta$:

$$\int \frac{3\sec\theta\tan\theta\,d\theta}{3\tan\theta} = \int \sec\theta\,d\theta = \ln|\sec\theta + \tan\theta| + C.$$

Triangle: $\sec\theta = \frac{x}{3}$, $\tan\theta = \frac{\sqrt{x^2-9}}{3}$. Fold the constant:

$$\int \frac{dx}{\sqrt{x^2-9}} = \ln\left|x + \sqrt{x^2-9}\right| + C.$$

Notice cases 2 and 3 landed on near-twin answers; the two shapes are that closely related.

The finale: earning πr^2

Time to collect on the pep talk's promise. The area of a circle of radius r: take the quarter-circle in the first quadrant under $y = \sqrt{r^2 - x^2}$, from $x = 0$ to $x = r$, and multiply by 4. Before the computation, predict what the quarter-circle integral must come out to if grade school was telling the truth: $\frac{\pi r^2}{4}$. Let's see if the machinery delivers.

➤ $4\int_0^r \sqrt{r^2-x^2}\,dx = ?$

Sine substitution, $a = r$: $x = r\sin\theta$, $dx = r\cos\theta\,d\theta$, root becomes $r\cos\theta$. Convert the bounds too (cleanest with definite integrals): $x = 0$ is $\theta = 0$, and $x = r$ is $\theta = \frac{\pi}{2}$.

$$\int_0^r \sqrt{r^2-x^2}\,dx = \int_0^{\pi/2} r\cos\theta \cdot r\cos\theta\,d\theta = r^2\int_0^{\pi/2}\cos^2\theta\,d\theta,$$

and that's last chapter's workhorse, power reduction:

$$r^2\left[\frac{\theta}{2}+\frac{\sin 2\theta}{4}\right]_0^{\pi/2} = r^2\left(\frac{\pi}{4}+0\right) = \frac{\pi r^2}{4}.$$

Multiply by 4: $\mathbf{A = ßr^2}$. There it is. A formula you've repeated your whole life, rebuilt from scratch by you, using the limit (the integral), a viewpoint change (the substitution), and a wave identity (power reduction). Three chapters of tools, one ancient truth. This is what "you can rebuild it from the picture" means in practice.

☞ Watch out

The wrong ideas that wreck trig substitution, named and retired:

- *"Forgot to convert dx."* The substitution isn't just for the root; dx becomes $a\cos\theta\,d\theta$ (or its cousins) every single time. Write the dx line before anything else.
- *"Answered in θ."* $\theta + C$ is half an answer. The question was asked in x; draw the triangle and translate back (or, for definite integrals, convert the *bounds* and never return at all).
- *"Heavy artillery on a u-sub problem."* If a spare x sits next to the root ($\int x\sqrt{1-x^2}\,dx$), it's a two-line Calc 1 substitution. Scan for the spare factor first; this chapter is for roots standing *alone*.
- *"Which sub goes with which shape again?"* Don't memorize, match signs (minus-after: sine; plus: tangent; minus-before: secant), or draw the triangle and let Pythagoras assign the sides.

✎ Quick practice, before you move on

Practice 1. $\displaystyle\int \frac{dx}{\sqrt{1-x^2}}$.

Practice 2. $\displaystyle\int \sqrt{4-x^2}\,dx$.

How to think about it: sine form, $a = 2$. After the dust settles you'll face $\int 4\cos^2\theta\,d\theta$, last chapter's power reduction. Convert back with the triangle: you'll need θ, and $\sin\theta\cos\theta$, both readable from the sides.

Practice 3 (tool choice, one sentence each, no computing). Which tool, and why: (i) $\int x\sqrt{9-x^2}\,dx$ (ii) $\int \frac{dx}{\sqrt{9-x^2}}$ (iii) $\int \frac{x^2}{\sqrt{9-x^2}}\,dx$.

Answers at the end of the chapter.

Carrying this forward

★ Leaving this chapter with this

Roots that stand alone fall to a change of viewpoint: describe the geometry by angle instead of by length, and the Pythagorean identity erases the radical. Three shapes, three subs, signs tell you which: $\sqrt{a^2 - x^2}$ takes $x = a\sin\theta$, $\sqrt{a^2 + x^2}$ takes $x = a\tan\theta$, $\sqrt{x^2 - a^2}$ takes $x = a\sec\theta$. Convert the dx, integrate the trig (last chapter's job), and translate back through the triangle. You used the whole stack to prove πr^2, which should permanently change how you look at "memorized" formulas: most of them are just integrals somebody did once. Next chapter, a new kind of stubborn: fractions of polynomials, and the art of un-adding them.

✎ The chapter workout (get stuck on purpose)

Name the form first, pick the substitution, draw the triangle.

F1. $\displaystyle\int \frac{dx}{\sqrt{25 - x^2}}$ **F2.** $\displaystyle\int \frac{dx}{x^2 + 9}$

F3. $\displaystyle\int \sqrt{1 - x^2}\,dx$ **F4.** $\displaystyle\int \frac{dx}{\sqrt{x^2 - 16}}$

F5 (the payoff check). Compute $\displaystyle\int_{-1}^{1} \sqrt{1 - x^2}\,dx$ using your F3 answer, and say in one sentence why you already knew what the number had to be.

Nudges: **F1** sine, $a = 5$. **F2** no root, but the $+$ shape still wants tangent, $a = 3$. **F3** sine then power reduction, like Practice 2. **F4** secant, $a = 4$, twin of the Case 3 example. **F5** what shape is the region?

➜ Want it on video?

Professor Leonard, *Calculus 2 Lecture 7.3: Integration Using Trigonometric Substitution*, at https://www.youtube.com/@ProfessorLeonard. Watching him draw the triangles in real time is what makes this click for a lot of people.

Answers (wrestle first). **P1:** $a = 1$, $x = \sin\theta$: $\theta + C = \arcsin x + C$. **P2:** $x = 2\sin\theta$ leads to $\int 4\cos^2\theta\, d\theta = 2\theta + 2\sin\theta\cos\theta + C = 2\arcsin\frac{x}{2} + \frac{x\sqrt{4-x^2}}{2} + C$. **P3:** (i) plain u-sub, the spare x pays the toll for $u = 9 - x^2$; (ii) trig sub, the root stands alone; (iii) trig sub, that lone x^2 upstairs is no toll-payer (du needs $x\,dx$, and the pieces don't match), so sine substitution it is. **F1:** $\arcsin\frac{x}{5} + C$. **F2:** $\frac{1}{3}\arctan\frac{x}{3} + C$. **F3:** $\frac{1}{2}\left(\arcsin x + x\sqrt{1-x^2}\right) + C$. **F4:** $\ln\left|x + \sqrt{x^2-16}\right| + C$. **F5:** $\frac{1}{2}\left(\arcsin 1 - \arcsin(-1)\right) = \frac{1}{2}\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = \frac{\pi}{2}$; it had to be, the region is half a unit disk, and you proved the disk's area in this very chapter.

Partial Fractions

Calculus 2 • Techniques of Integration, Chapter 4

Breaking one ugly fraction into simple pieces you can integrate one at a time

❧ Chapter 4. The technique you're going to invent in the first five minutes.

This chapter is different: I'm not going to teach you the trick. You're going to perform the trick, in the wrong direction, with grade-school algebra, and then we'll simply turn around and walk it backward. Within your first few minutes here you'll have built the chapter's core example yourself without noticing.

Fair warning about this chapter's personality: it's more algebra than calculus. The actual integration at the end is the easiest in the whole unit, nearly everything becomes a logarithm. The work, and the thing worth getting stuck on, is the decomposition: seeing which simple pieces a complicated fraction secretly came from, and pinning down the numbers on top.

And when something is "just algebra," people's eyes glaze and they make small mistakes, so this is a small-mistakes chapter. You know what I've learned the hard way about small mistakes: slow is smooth, smooth is fast. Check your constants with a plug-in, every time. It takes ten seconds and it catches almost everything.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say what this technique is for, inside calculus and out (viral growth is in this chapter, for real),
- run fraction-addition backward, which you'll understand because you'll run it forward first,
- check whether a fraction is ready to split, and long-divide when it isn't,
- solve for the unknown constants fast with the root-plugging move, and know why it works,
- handle the two curveballs: top-heavy fractions and unbreakable quadratics (where arctangent is born), plus the template for repeated factors,
- verify your decomposition numerically in ten seconds flat.

Why this exists, and what it's for

The targets of this chapter are **rational functions**: one polynomial divided by another, things like $\frac{x+5}{x^2+x-2}$. They're everywhere once you start noticing. The logistic equation that models how an epidemic spreads, or how a product goes viral, or how a population fills its habitat, hands you the integral $\int \frac{dP}{P(1-P)}$, a rational function, and this chapter's technique is literally how that model gets solved. Control theory, the math that keeps planes level and server loads balanced, runs on decomposing rational functions all day (engineers call it by this chapter's exact name). And one more customer is coming from inside the book: in the series unit, "telescoping sums" collapse because of a partial-fraction split. This isn't a niche trick; it's infrastructure.

The calculus problem is plain to state: $\int \frac{x+5}{x^2+x-2}\,dx$ matches nothing in your kit. No greatest hit, no clean substitution, parts goes nowhere, no square root for a triangle. New shape, new tool. Except, surprise: you've owned this tool since fifth grade.

Run it forward first: you already know this

Do this with a pencil, it's thirty seconds, and it's the entire chapter. Add these two baby fractions the way you'd add $\frac{1}{2} + \frac{1}{3}$, common denominator and all:

$$\frac{2}{x-1} - \frac{1}{x+2} = ?$$

Common denominator is $(x-1)(x+2)$, so:

$$\frac{2(x+2)}{(x-1)(x+2)} - \frac{(x-1)}{(x-1)(x+2)} = \frac{2x+4-x+1}{(x-1)(x+2)} = \frac{x+5}{(x-1)(x+2)}.$$

Look at what you produced: the exact monster from the opening section (its denominator factors as $x^2 + x - 2 = (x-1)(x+2)$). So that "impossible" integrand isn't a fundamental object at all. It's two trivial fractions that somebody added together, and each original piece integrates instantly: $\int \frac{2}{x-1}dx$ is a log, $\int \frac{1}{x+2}dx$ is a log.

So here's the whole strategy of the chapter, in one sentence: **every rational function is a sum of simple fractions in disguise, and we're going to un-add them.** The only genuine question is: given the combined fraction, how do you recover the pieces, specifically, those numbers 2 and -1 on top, when nobody tells you them in advance? That's the skill. Let's build it.

Un-adding: the decomposition

➤ **The core example: $\int \frac{x+5}{(x-1)(x+2)}\,dx$, pretending we don't know the answer**

We know the shape the pieces must have, one simple fraction per factor downstairs, but not the numbers on top. So name them:

$$\frac{x+5}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}.$$

Multiply everything by the full denominator to clear all fractions:

$$x+5 = A(x+2) + B(x-1).$$

This must hold for *every* x. And that's a superpower, because you get to choose x-values that make terms die. Before reading on, look at that equation and pick the smartest possible value of x yourself. What single choice erases B entirely? $x = 1$, because it zeroes the $(x-1)$ riding with B:

$$x = 1: \quad 6 = A(3) \;\Rightarrow\; A = 2. \qquad x = -2: \quad 3 = B(-3) \;\Rightarrow\; B = -1.$$

There are the 2 and -1 from your own forward addition, recovered blind. Ten-second verification, the habit this book keeps preaching: plug a neutral value like $x = 0$ into both sides of the original split. Left side: $\frac{5}{(-1)(2)} = -2.5$. Right side: $\frac{2}{-1} - \frac{1}{2} = -2.5$. Match. Now integrate the easy pieces:

$$\int \frac{x+5}{(x-1)(x+2)}\,dx = 2\ln|x-1| - \ln|x+2| + C.$$

Why did root-plugging work? You answered this when you picked $x = 1$ yourself: each root silences every term except one, so the constants pop out one at a time, no system of equations, no tears. (Keep the caveat in your pocket: this clean kill only finishes the job for *distinct linear* factors. Repeated factors and unbreakable quadratics, coming below, each leave one constant the roots can't reach, and then you plug in any other convenient x, or match coefficients.)

★ The partial-fractions recipe

1. **Degree check.** The top's degree must be *less* than the bottom's. If not, polynomial long division first; decompose only the leftover fraction.
2. **Factor the denominator** completely.
3. **Write the template**: one piece per factor (forms in the next box), with unknown letters on top.
4. **Clear denominators and solve**: plug in roots to kill terms; mop up any survivors with other x-values or coefficient matching.
5. **Verify with a ten-second plug-in**, then **integrate** the simple pieces (logs, and sometimes an arctangent).

★ Which piece for which factor

$$\text{distinct linear } (x-a): \quad \frac{A}{x-a}$$

$$\text{repeated linear } (x-a)^2: \quad \frac{A}{x-a}+\frac{B}{(x-a)^2} \quad \text{(one piece per power)}$$

$$\text{irreducible quadratic } (x^2+bx+c): \quad \frac{Ax+B}{x^2+bx+c} \quad \text{(linear top)}$$

"Irreducible" means it won't factor over the reals; the test is the discriminant, $b^2-4c<0$ means no real roots, so it stays whole (that's the quadratic formula's b^2-4ac with $a=1$, since our quadratic is monic).

✎ **Quick practice, before you move on**

Practice 1. $\int \frac{1}{(x-2)(x+1)}\,dx.$

Practice 2. $\int \frac{x^2}{x^2-1}\,dx.$

How to think about it: run the recipe's step 1 before anything else. Top degree 2, bottom degree 2: not ready. Long-divide (x^2-1 goes into x^2 once, remainder 1), then decompose only the leftover over $(x-1)(x+1)$.

Practice 3 (ten seconds, no integration). A classmate claims $\frac{3x+1}{x(x+1)} = \frac{1}{x} + \frac{2}{x+1}$. Check it with one plug-in and give your verdict.

Answers at the end of the chapter.

The two curveballs

Curveball 1: the top is too big. Decomposition only works when the numerator's degree is strictly smaller. When it isn't, the fraction is hiding a polynomial inside, and long division evicts it:

$$\frac{x^2}{x^2-1} = 1 + \frac{1}{x^2-1} \qquad (\text{check: } 1\cdot(x^2-1)+1 = x^2).$$

The evicted 1 integrates to x on its own; the remainder fraction is now legal and splits normally. Degree check *first*, always; it's the most-skipped step and it poisons everything after it.

Curveball 2: a quadratic that won't factor. This is where the arctangent in your greatest-hits table finally finds its birthplace.

➤ **The arctangent case:** $\int \frac{dx}{(x-1)(x^2+1)}$

x^2+1 is irreducible (discriminant $0-4<0$), so it stays whole and gets a linear top:

$$\frac{1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}.$$

Clear denominators: $1 = A(x^2+1) + (Bx+C)(x-1)$. The one real root $x=1$ gives $1 = 2A$, so $A = \frac{1}{2}$. No more real roots to plug, so match coefficients: expanding, $1 = (A+B)x^2 + (C-B)x + (A-C)$, forcing $B = -\frac{1}{2}$ and $C = -\frac{1}{2}$. (Ten-second check at $x=0$: left side $\frac{1}{(-1)(1)} = -1$; right side $\frac{1/2}{-1} + \frac{-1/2}{1} = -1$. Match.) Integrate piece by piece, splitting the quadratic piece's top:

$$\frac{1}{2}\ln|x-1| - \frac{1}{2}\int \frac{x}{x^2+1}\,dx - \frac{1}{2}\int \frac{1}{x^2+1}\,dx,$$

where the middle is a u-sub ($u = x^2+1$, half its du present) and the last is the greatest-hit $\arctan x$:

$$\int \frac{dx}{(x-1)(x^2+1)} = \frac{1}{2}\ln|x-1| - \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\arctan x + C.$$

So *that's* where arctangents come from in integration: they're the toll an unbreakable quadratic charges. When your answer to a rational integral sprouts an arctan, it's because some x^2+c downstairs refused to factor.

☞ Watch out

The wrong ideas that wreck partial fractions, named and retired:

- *"Skipped the degree check."* Decomposing a top-heavy fraction produces confident nonsense. Degrees first, division if needed, every time.
- *"Put a plain A over the quadratic."* An irreducible quadratic carries a *linear* top, $Ax + B$. Forget the Bx and the coefficient-match will refuse to balance, that's your smoke alarm.
- *"A constant came out negative, I must have erred."* Negative constants are routine (our very first B was -1). The plug-in check is the judge, not your gut.
- *"Wrote $\ln(x-1)$ without absolute values."* The antiderivative of $\frac{1}{x}$ is $\ln|x|$; those bars are load-bearing for negative x.
- *"Repeated factor, one piece."* $(x-a)^2$ owes *two* pieces, one for each power. Shortchange it and, again, the check catches you, if you run the check. Run the check.

Carrying this forward

★ Leaving this chapter with this

Every rational function is a sum of simple fractions that somebody added together, and integration-wise the pieces are trivial (logs, plus an arctangent whenever an unbreakable quadratic is in the room). The method: degree-check (divide if top-heavy), factor, template one piece per factor, kill constants with root-plugging, mop up with coefficient matching, verify with a ten-second plug-in, then integrate. Pure reverse-engineering of grade-school addition. Next chapter the integrals themselves misbehave, running to infinity or blowing up mid-interval, and Chapter 1 of Calc 1 comes back to referee.

✎ The chapter workout (get stuck on purpose)

Degree-check first, factor, template, solve, verify, integrate.

F1. $\displaystyle\int \frac{dx}{(x-3)(x+1)}$ **F2.** $\displaystyle\int \frac{x-1}{(x+1)(x+2)}\,dx$

F3. $\displaystyle\int \frac{3x+1}{x^2+x}\,dx$ **F4.** $\displaystyle\int \frac{x^3+x+1}{x^2+1}\,dx$

F5 (the chapter in one sentence). Your friend asks why anyone would ever "make one fraction into two." Answer them in a sentence that mentions what each little piece turns into.

Nudges: **F1** two distinct roots, double root-plug. **F2** same, roots -1 and -2. **F3** factor the bottom as $x(x+1)$ first. **F4** top-heavy! Long-divide: write x^3+x+1 as $x\cdot(\text{the denominator})+\text{leftover}$, then notice what's left over needs no decomposition at all, just a greatest hit.

Next up: *Improper Integrals*, where the region under a curve stretches to infinity and the area, sometimes, politely stays finite.

➜ Want it on video?

Professor Leonard, *Calculus 2 Lecture 7.4: Integration by Partial Fractions*, at https://www.youtube.com/@ProfessorLeonard. He grinds through the repeated-factor case on camera if you want extra reps past this chapter.

Answers (wrestle first). **P1:** $A = \frac{1}{3}$, $B = -\frac{1}{3}$: $\frac{1}{3}\ln|x-2| - \frac{1}{3}\ln|x+1| + C$. **P2:** $x + \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$. **P3:** plug $x = 1$: left side $\frac{4}{2} = 2$, right side $1+1 = 2$, looks fine, so try one more, $x = 2$: left $\frac{7}{6}$, right $\frac{1}{2} + \frac{2}{3} = \frac{7}{6}$. It checks twice, verdict: the classmate is right. (Two different plug-ins passing is strong evidence; one can be a coincidence.) **F1:** $\frac{1}{4}\ln|x-3| - \frac{1}{4}\ln|x+1| + C$. **F2:** $A = -2$, $B = 3$: $-2\ln|x+1| + 3\ln|x+2| + C$. **F3:** $A = 1$, $B = 2$: $\ln|x| + 2\ln|x+1| + C$. **F4:** division gives $x + \frac{1}{x^2+1}$, so $\frac{x^2}{2} + \arctan x + C$, no decomposition needed, the degree check did all the work. **F5:** because the big fraction matches nothing in our kit, while every little piece integrates on sight, each linear piece becomes a log, and an unbreakable quadratic piece becomes an arctangent.

Improper Integrals

Calculus 2 • Techniques of Integration, Chapter 5

When the region runs forever but the area can still be finite

❧ Chapter 5. Place your bet before you read anything else.

Here's the question, and I want an actual answer from you, out loud or in the margin, before the math weighs in. Take the curve $y = \frac{1}{x^2}$, and the region trapped under it from $x = 1$ rightward, *forever*. The region never ends. If you had to paint it, is the amount of paint finite or infinite?

Almost everyone's gut says infinite, endless shape, endless paint. The gut is wrong, and not by a little: the area is exactly 1. One square unit of paint covers an infinitely long region. And before you decide infinity is simply broken, know this: change the curve to $\frac{1}{x}$, which looks nearly identical and also dives to zero, and the area really *is* infinite. Two curves, both fading to nothing, one finite, one not, and the difference between them is a razor's edge you'll locate precisely in this chapter.

The tool for all of it is the oldest one in the book, Chapter 1's limit. We never "plug in infinity." We stop at a finite fence post, measure honestly, then slide the post toward forever and watch what the measurements are heading toward. By now that move should feel like home.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why integrals over infinite regions matter (rockets and probability both depend on them),
- spot the two kinds of "improper": an infinite bound, or a blow-up inside the interval,
- rewrite an improper integral as a limit of honest ones, and read off converge vs diverge,
- watch the $\frac{1}{x}$ vs $\frac{1}{x^2}$ race in a table and locate the razor's edge at $p = 1$,
- retire the most dangerous wrong idea in the unit: "the function dies, so the area is finite,"
- dodge the interior blow-up trap that produces impossible negative areas.

Why this exists, and what it's for

"Integrals to infinity" sound like a math-department indulgence. Two real customers say otherwise.

Rockets. To launch something off Earth for good, you must pay the energy to lift it against gravity from the surface *all the way to infinity*. That's the improper integral $\int_R^\infty \frac{GMm}{r^2}\,dr$, gravity's pull added up over an endless journey. Here's the miracle your gut wouldn't have granted: because gravity fades like $\frac{1}{r^2}$, that infinite journey costs *finite* energy (the integral comes out to $\frac{GMm}{R}$). Finite cost means a finite launch speed covers it: about 11.2 km/s from Earth. **Escape velocity exists because an improper integral converges**, which is the word for "settles on a finite value"; the precise definition is one section away. If gravity faded like $\frac{1}{r}$ instead, the integral would diverge, no speed would ever be enough, and nothing would ever leave any planet.

Probability. The bell curve, the one your exam scores and heights and measurement errors live on, stretches over the whole number line. The statement "all probabilities total 100%" is the statement $\int_{-\infty}^{\infty}(\text{bell curve})\,dx = 1$: an improper integral converging. Every continuous probability model in statistics stands on this chapter.

So: infinite regions with finite totals aren't a curiosity. They're why rockets fly and statistics works. Let's learn to tell finite from infinite.

Type 1: an infinite bound, and the race you should watch

The definition first, built from the fence-post idea in the pep talk:

$$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^{t} f(x)\,dx.$$

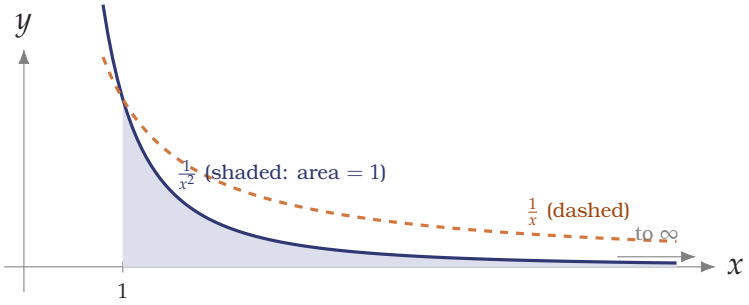
Integrate honestly up to a movable post t, then ask Chapter 1's question: as t runs off, where is the running total *heading*? A finite destination means the integral **converges** to it. No finite destination means it **diverges**, and that covers both "blows up to ∞" and "never settles" (the running total of $\int_1^{\infty} \cos x\,dx$ seesaws forever, settling on nothing; that's divergence too).

Now the bet from the pep talk, settled with your own eyes. Run the fence post out for both curves and tabulate the area collected so far (check a value or two: the antiderivatives are $-\frac{1}{x}$ and $\ln x$):

post at $t =$	10	1,000	10^6	heading toward…
area of $\frac{1}{x^2}$ so far $(1 - \frac{1}{t})$	0.9	0.999	0.999999	1
area of $\frac{1}{x}$ so far $(\ln t)$	2.3	6.9	13.8	∞, at a crawl

Study that second row a moment, because it's subtle and it will matter again in the series unit. $\frac{1}{x}$'s area doesn't blast off to infinity; it *trudges* there, gaining only about 7 units per factor of 1,000. But it never stops gaining. Slow and unbounded is still unbounded. Meanwhile $\frac{1}{x^2}$'s area is being herded to exactly 1, just like the pep talk promised.

Before the picture, a question to carry into it: both curves fade to zero, so what, visually, could possibly be different about them? Look at the *tail*:



Two fading tails. The lower one is thin enough to have finite area; the dashed one, barely fatter, is not.

Walk the tails with your eye. Past every point, $\frac{1}{x}$ sits *above* $\frac{1}{x^2}$, and by a widening ratio: at $x = 100$, it's a hundred times taller. The shaded curve's tail thins out fast enough that each new mile of region adds almost nothing, the leftovers past t total exactly $\frac{1}{t}$, which dies. The dashed tail stays just fat enough that each new *factor* of distance (10 to 100, 100 to 1000) deposits the same fixed chunk of area, about 2.3, forever. Same destination for the curves, opposite destinations for their areas. "Fades to zero" tells you nothing by itself; *how fast* it fades is the whole story.

➤ The two verdicts, in full

Converges:

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{t\to\infty}\left[-\frac{1}{x}\right]_1^t = \lim_{t\to\infty}\left(1-\frac{1}{t}\right) = 1.$$

Diverges:

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{t\to\infty}\left[\ln x\right]_1^t = \lim_{t\to\infty}\ln t = \infty.$$

One power's difference, opposite fates. The exact watershed between them is coming in the razor's-edge section.

Type 2: the function blows up

The other way an integral goes improper: the *bound* is fine, but the function detonates somewhere on the interval, a vertical asymptote. Same medicine: sneak up on the explosion with a limit instead of standing inside it.

➤ Blow-up at the edge: $\int_0^1 \frac{1}{\sqrt{x}}\,dx$

The integrand explodes as $x \to 0^+$, so approach from the safe side:

$$\int_0^1 x^{-1/2}\,dx = \lim_{t\to 0^+}\left[2\sqrt{x}\right]_t^1 = \lim_{t\to 0^+}\left(2-2\sqrt{t}\right) = 2.$$

Converges to 2: the spike at zero shoots infinitely high, but it's so *thin* that it holds finite area. (An infinitely tall shape with finite area, the sideways twin of the pep talk's infinitely long one. Infinity keeps losing to thinness.)

☞ **Watch out**

The interior blow-up trap, and the impossible negative area. If the explosion sits *inside* $[a,b]$ rather than at an edge, you must split the integral at the bad point and give each half its own limit; if either half diverges, the whole thing diverges. Skip the check and arithmetic will happily hand you garbage: plugging endpoints blindly into $\int_{-1}^{1} \frac{1}{x^2}\,dx$ (explosion dead center at 0) gives $\left[-\frac{1}{x}\right]_{-1}^{1} = -2$. A *negative* area, for a function that's positive everywhere. That absurd minus sign is the method screaming that you integrated across a detonation as if it weren't there. Done properly, split at 0, each half diverges. The same split-and-both-must-converge rule governs double-infinite integrals $\int_{-\infty}^{\infty}$.

✎ **Quick practice, before you move on**

Practice 1. $\displaystyle\int_1^{\infty} \frac{1}{x^3}\,dx.$

Practice 2. $\displaystyle\int_0^{\infty} e^{-x}\,dx.$

How to think about it: fence post at t, antiderivative $-e^{-x}$, then ask what e^{-t} does as t runs off. (This exact integral, with a constant tuned in, is how "average waiting time" gets computed for servers and queues. Your app's latency math is in here.)

Practice 3 (predict first, one sentence). Before computing anything: $\int_0^1 \frac{1}{x}\,dx$, finite or infinite? Make the call from what the $\frac{1}{x}$ tail did at infinity, then note this is the *other* end of the curve, the spike at zero.

Answers at the end of the chapter.

The razor's edge: the p-test

Run the fence-post computation on the whole family $\frac{1}{x^p}$ and the watershed appears:

★ The p-integral cutoffs (two versions, and they're mirror images)

$$\int_1^\infty \frac{1}{x^p}\,dx : \begin{cases} \text{converges} & p > 1 \\ \text{diverges} & p \le 1 \end{cases} \qquad \int_0^1 \frac{1}{x^p}\,dx : \begin{cases} \text{converges} & p < 1 \\ \text{diverges} & p \ge 1 \end{cases}$$

Why the flip? Out at infinity (left case) the enemy is a *fat tail*, so you need a big power to slim it down. Near a blow-up at zero (right case) the enemy is a *steep spike*, so you need a small power to keep it gentle. And $p = 1$, the curve $\frac{1}{x}$, is the knife's edge that loses both ways: too fat at infinity, too steep at zero. File the left cutoff somewhere permanent, in the series unit it returns, almost word for word, as the "p-series," and this integral is exactly where its authority comes from.

☞ Watch out

The wrong ideas that wreck improper integrals, named and retired:

- *"The function goes to zero, so the area is finite."* The most dangerous sentence in this unit. $\frac{1}{x}$ goes to zero and encloses infinite area; you watched it trudge there in the table. Dying is necessary, dying *fast enough* is the question. (Remember this one; it has a famous twin waiting in the series chapters.)
- *"Just plug in ∞."* Infinity isn't a number (Chapter 1's words). The limit-with-a-post isn't decoration, it's the definition, write it.
- *"The interval looks symmetric, so it cancels."* Only after *both* halves converge on their own. A divergent half can't be "cancelled" by the other; that's trading one infinity against another, which the running totals never actually do.
- *"I integrated straight through the asymptote and got an answer."* An answer, yes; a meaningful one, no. Negative area under a positive curve is the classic tell. Scan the interval for blow-ups before you integrate, every time.

Carrying this forward

★ Leaving this chapter, and the techniques unit

Improper integrals are ordinary integrals with Chapter 1 bolted on: measure to a movable post, then read off where the totals are heading. Finite destination: converges. None: diverges. The $\frac{1}{x^p}$ family splits at the razor's edge $p = 1$, big powers win at infinity, small powers win at a spike, and "the function fades to zero" guarantees exactly nothing, speed of fading is everything. That closes the techniques unit: you can now integrate by parts, through trig, past roots, across broken fractions, and off the edge of the number line. Next unit we stop sharpening tools and start building things with them: areas, volumes, lengths, and honest physical work.

✎ The chapter workout (get stuck on purpose)

Converge or diverge? If it converges, give the value. Rewrite as a limit first, every time.

F1. $\displaystyle\int_2^\infty \frac{1}{x^2}\,dx$ **F2.** $\displaystyle\int_1^\infty \frac{1}{\sqrt{x}}\,dx$

F3. $\displaystyle\int_0^1 \frac{1}{x^2}\,dx$ **F4.** $\displaystyle\int_0^\infty 2e^{-2x}\,dx$

F5 (one sentence). Your friend says: "$\frac{1}{x^{0.99}}$ and $\frac{1}{x^{1.01}}$ are basically the same curve, so their integrals from 1 to ∞ basically do the same thing." Deliver the bad news, with the name of the test that settles it.

Nudges: **F1** p-test, then compute the value honestly. **F2** $p = \frac{1}{2}$, which side of the edge? **F3** this is the *spike* version of the p-test, mind the flip. **F4** fence post, antiderivative $-e^{-2x}$. **F5** razor's edge.

Next up (new unit): *Area Between Curves*, the first real construction project: trapping the region between two functions and measuring it with strips.

➜ Want it on video?

Professor Leonard, *Calculus 2 Lecture 7.6: Improper Integrals*, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** $\left[-\frac{1}{2x^2}\right]_1^t \to \frac{1}{2}$, converges. **P2:** $\left[-e^{-x}\right]_0^t = 1 - e^{-t} \to 1$, converges. **P3:** infinite: at infinity $\frac{1}{x}$'s tail was exactly fat enough to diverge, and at zero its spike is exactly steep enough to diverge too, $p = 1$ loses on both ends (compute if you like: $\lim_{t\to 0^+}[\ln x]_t^1 = +\infty$). **F1:** converges, $\left[-\frac{1}{x}\right]_2^\infty = \frac{1}{2}$. **F2:** $p = \frac{1}{2} \leq 1$, diverges. **F3:** spike version needs $p < 1$, here $p = 2$: diverges. **F4:** $\left[-e^{-2x}\right]_0^\infty = 1$, converges (that's a complete probability distribution, the exponential one, totaling 100%). **F5:** completely different fates: the p-test puts 0.99 on the divergent side of the edge and 1.01 on the convergent side; near $p = 1$, "basically the same curve" means nothing, the edge is exact.

Area Between Curves

Calculus 2 • Applications of Integration, Chapter 1

Measuring the region trapped between two graphs

❧ New unit. The tools go to work now.

You spent five chapters sharpening integration until it could cut anything. This unit is the payoff arc: integrals pointed at actual stuff, areas, volumes, lengths, physical work. And I'll let you in on the structure before we start, because seeing it coming makes everything easier: **every single chapter in this unit is the same move**. Chop the thing into slices so thin they're simple, write down one slice's contribution, add them all up with an integral. Area by strips. Volume by disks. Length by tiny steps. Work by tiny lifts. One idea, four costumes. By the second chapter you'll see the costume coming off before I do.

Today's version is the gentlest: the area trapped between two curves. You already own 95% of it from Calc 1, and the last 5% you're going to derive yourself in about four lines. Most of the chapter is really about the two judgment calls, where does the region start and stop, and who's on top, because that's where every point actually gets lost.

There's also a booby trap in this chapter that produces a confident, clean, completely wrong answer of zero. You'll walk into it on purpose, because springing a trap in practice is how you never spring it on an exam.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say what areas-between-curves measure out in the world (profit and inequality, among others),
- derive the top-minus-bottom formula yourself from Calc 1,
- find a region's start and stop by locating the crossings, and settle who's on top with a test point,
- survive the swap trap, where blind integration returns zero for a region you can see,
- flip to horizontal strips when the region is sliced more naturally sideways.

Why this exists, and what it's for

"The area between two curves" sounds like geometry homework until you notice that a lot of the world's important quantities *are* one.

Profit. Plot a company's *marginal* revenue and *marginal* cost curves against output, what one more unit brings in, and what one more unit costs to make. The vertical gap between them is the profit on that next unit, and the *area* between the curves across a production range is the total profit that range adds. "Grow the area between those two curves" is the sentence "make money" wearing graph clothes.

Inequality. The Gini coefficient, the number economists quote to compare inequality across countries, is built from the area between two curves (twice that area, to be exact): the line of perfect equality and the actual income curve (the Lorenz curve). The bigger the trapped region, the more unequal the society. When you read "inequality rose," someone integrated a gap.

Everything comparative. Two treatments' effect curves, two servers' load curves, projected versus actual anything: whenever you care about the *accumulated difference* between two trajectories, you are computing this chapter.

Derive it yourself: what's the height of a strip?

Here's the whole derivation, and you're doing it, not me. Calc 1 taught you that $\int_a^b f(x)\,dx$ is the area between the curve f and the floor $y = 0$, computed by slicing into skinny vertical strips of height $f(x)$ and width dx. One question: *what was special about the floor being $y = 0$?* Commit to an answer before reading on.

Nothing. The axis is just some curve that happens to be flat. So suppose the floor is itself a curve $g(x)$, with a ceiling $f(x)$ above it. Slice vertically, same as always. How tall is the strip at position x? It runs from height $g(x)$ up to height $f(x)$, so its height is the gap, $f(x) - g(x)$. Multiply by the width dx, add up all the strips. You've just written the formula:

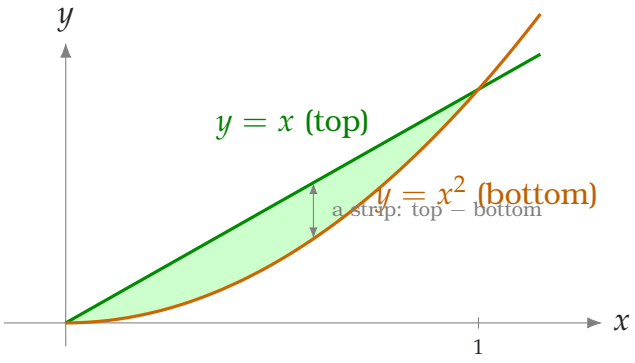
★ Area between two curves

If $f(x)$ rides above $g(x)$ across $[a, b]$,

$$A = \int_a^b \big[\, f(x) - g(x) \,\big]\, dx \qquad \text{(top minus bottom, strip by strip).}$$

Calc 1's area-under-a-curve is just the special case where the bottom curve is the flat one, $g = 0$. Nothing new was invented; the floor got promoted.

Before the picture, predict what you'll see: a strip, trapped between two curves, whose height is the gap. Check yourself:



The line and the parabola, with one strip shown earning its height.

Walk the region with your eye before any computation, because estimation is a superpower: the whole unit square from 0 to 1 has area 1, and this sliver between the line and the parabola looks like, what, somewhere around a sixth of it? Hold that guess.

➤ The worked example: between $y = x$ and $y = x^2$

Where does the region start and stop? Where the curves cross: $x = x^2$ gives $x(x-1) = 0$, so $x = 0$ and $x = 1$. **Who's on top in between?** Test one point, $x = \frac{1}{2}$: line says $\frac{1}{2}$, parabola says $\frac{1}{4}$. The line wins. Integrate the gap:

$$A = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

And check it against your eyeball estimate: a sixth of the unit square. The guess and the calculus shook hands, which is how you know nobody fumbled a sign.

★ The setup checklist

1. **Find the crossings** (set the curves equal); those are usually your limits.
2. **Settle who's on top** with one test point inside the interval.
3. **Integrate top minus bottom**, and if the curves swap roles partway through, split at the swap, which brings us to the trap.

The trap: a visible region, an answer of zero

Do this one blind, the way a hurried student would, so you feel the spring snap. Curves $y = x$ and $y = x^3$ between -1 and 1. They visibly enclose two little leaf-shaped regions, real area you could shade with a pencil. Integrate "top minus bottom" in one go, taking the line as top:

$$\int_{-1}^1 (x - x^3)\,dx = \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_{-1}^1 = \left(\frac{1}{2} - \frac{1}{4}\right) - \left(\frac{1}{2} - \frac{1}{4}\right) = 0.$$

Zero. For a region you can *see*. Before reading on, figure out what happened, you have all the pieces.

Here's the autopsy: the curves *trade places* at $x = 0$. On $[0, 1]$ the line is on top, but on $[-1, 0]$ the *cubic* is on top (test $x = -\frac{1}{2}$: cubic gives -0.125, line gives -0.5, and -0.125 is higher). So on the left half, our integrand "$x - x^3$" was *bottom minus top*, a

negative gap, and it exactly cancelled the honest positive gap from the right half. The integral did precisely what we asked; we asked the wrong question.

➤ The trap, disarmed

Split at the swap and keep top-minus-bottom honest on each piece:

$$A = \int_{-1}^{0} (x^3 - x)\,dx + \int_{0}^{1} (x - x^3)\,dx = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Real region, real answer. The ritual that saves you: find *all* the crossings first, and if any crossing sits strictly inside your interval, that's a swap point, split there.

✎ Quick practice, before you move on

Practice 1. Area between $y = x^2$ and $y = x^3$ on $[0, 1]$. (Eyeball first: which is bigger on that interval?)

Practice 2. Area between $y = x^2$ and $y = 2x$.
How to think about it: no interval given, so the region defines itself: crossings first, then a test point, then the integral.

Practice 3 (one sentence). A classmate integrates two crossing curves over one unbroken interval and proudly reports an area of exactly zero. Without seeing their work, what happened, and what's the fix?

Answers at the end of the chapter.

Sideways regions: horizontal strips

Some regions fight vertical strips. Try to describe the region between $x = y^2$ and $x = y + 2$ with vertical strips and you'll find the "top" and "bottom" curves keep changing identity, because these curves are naturally written sideways, x in terms of y. So rotate your whole mental picture a quarter turn: slice *horizontally*, and a strip's length becomes **right curve minus left curve**, integrated in y:

$$A = \int_{c}^{d} \left[\, x_{\text{right}}(y) - x_{\text{left}}(y) \,\right] dy.$$

Same derivation as before, the gap is just horizontal now.

➤ Horizontal strips: between $x = y^2$ and $x = y + 2$

Crossings: $y^2 = y + 2$ gives $(y - 2)(y + 1) = 0$, so $y = -1$ and $y = 2$ (the y-limits). Who's on the right? Test $y = 0$: line says $x = 2$, parabola says $x = 0$; the line is the right-hand boundary. Then

$$A = \int_{-1}^{2} \left[(y + 2) - y^2\right] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2} = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{9}{2}.$$

Far minus near, sliced sideways. Choosing the slicing direction is a real decision, and choosing well can turn three split integrals into one clean one.

☞ Watch out

The wrong ideas that wreck area problems, named and retired:

- *"My answer is zero but the region is clearly there."* Curves swapped mid-interval and the signed gaps cancelled. Find every crossing; split at interior ones. You sprang this trap on purpose today, never again by accident.
- *"My area came out negative."* You had top and bottom backwards the whole way. Swap them (or take the absolute value and privately fix your test point).
- *"Part of the region is below the x-axis, so I need extra sign care."* You don't. Top minus bottom measures the *gap*, and gaps don't care where the axis sits. The axis only matters when it's one of your two curves.
- *"Always slice vertical."* If the boundaries are given as x in terms of y, or vertical strips keep changing their ceiling, rotate. The strips work for you, not the other way around.

Carrying this forward

★ Leaving this chapter with this

One move: slice, write one strip's contribution, integrate. For areas, a strip's contribution is its gap, top minus bottom (or right minus left, sideways), across the interval the crossings define, splitting wherever the curves trade places. You derived the formula from Calc 1 yourself, you sprang the zero-trap in a controlled setting, and you eyeballed an answer before computing it, keep all three habits. Next chapter, the same strips *spin*: rotate a region around an axis and the strips sweep out disks, and suddenly we're measuring volume.

✎ The chapter workout (get stuck on purpose)

Crossings, test point, top minus bottom. Split if they swap.

F1. between $y = x$ and $y = x^3$ on $[0, 1]$ **F2.** between $y = x^2$ and $y = 8 - x^2$

F3. between $y = \sqrt{x}$ and $y = x$ on $[0, 1]$

F4. between $x = y^2$ and $x = 2y$ (slice sideways)

F5 (estimate, then verify). For F3: the region lives inside the unit square. Eyeball what fraction of the square it covers, write your guess down, then compare with your computed answer.

Nudges: **F1** no interior crossing on $[0, 1]$, one piece. **F2** set them equal for the limits; the downward parabola is the ceiling. **F3** $\sqrt{x}$ rides above x on $[0, 1]$. **F4** crossings where $y^2 = 2y$; test which boundary is right-hand. **F5** sixths keep showing up in this chapter for a reason.

Next up: *Volumes of Revolution*, where a flat region spins around an axis into a solid, and strips become disks, washers, and shells.

➜ Want it on video?

Professor Leonard, *Calculus 2*, the Area Between Curves lecture (start of his applications sequence), at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** $x^2 \geq x^3$ there, $\int_0^1 (x^2 - x^3)\,dx = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. **P2:** crossings $x = 0, 2$; $2x$ on top; $\int_0^2 (2x - x^2)\,dx = 4 - \frac{8}{3} = \frac{4}{3}$. **P3:** their curves swapped somewhere inside the interval and the positive and negative gaps cancelled; the fix is to split the integral at the interior crossing and integrate honest top-minus-bottom on each piece. **F1:** $\int_0^1 (x - x^3)\,dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. **F2:** crossings $x = \pm 2$; $\int_{-2}^2 (8 - 2x^2)\,dx = \frac{64}{3}$. **F3:** $\int_0^1 (\sqrt{x} - x)\,dx = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$. **F4:** $y = 0, 2$; right curve is $x = 2y$; $\int_0^2 (2y - y^2)\,dy = \frac{4}{3}$. **F5:** about a sixth of the square, and the computation agrees, $\frac{1}{6}$; same sliver-shape, same answer as the worked example, with the two curves' roles mirrored.

Volumes of Revolution

Calculus 2 • Applications of Integration, Chapter 2

Spin a flat region into a solid, then measure what you built

❧ Chapter 2. Flat goes 3D, and you already know the move.

Last chapter I told you this whole unit is one move in different costumes: slice thin, write one slice's contribution, add with an integral. You used it on areas. Now take a flat region, grab an axis, and *spin* the region around it, like clay on a potter's wheel. It sweeps out a solid: a bowl, a cone, a vase, a donut. New question, same move. What does one thin slice of that solid look like, and what's its volume?

That's honestly the entire chapter. There are three named "methods" coming, disks, washers, shells, and I want you to refuse, from the start, to treat them as three formulas to memorize. They are three answers to the single question "what shape is my slice?": a coin, a coin with a hole, or a tube. Know what your slice looks like and you can rebuild any of the formulas in the margin of an exam, which is the only kind of knowing that survives exam adrenaline.

And a bonus for the builders among you: the slicing idea in this chapter is literally how 3D printers think. Keep that image around; it makes everything concrete.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say where volumes-by-slices show up outside the classroom (printers, scanners, vases, engines),
- build the disk formula yourself from one coin, then use it,
- handle washers (coins with holes) without falling for the square-then-subtract trap,
- slice sideways into shells, and know *when* tubes beat coins,
- keep your bearings when the axis moves off the coordinate axes,
- sanity-check answers against geometry formulas you already trust.

Why this exists, and what it's for

Rotational solids are everywhere humans make things, because spinning is how we make them: pottery on a wheel, parts on a lathe, bottles, bowls, nozzles, engine cylinders, rocket nose cones. "How much does this vase hold?" and "how much steel is in this part?" are this chapter's questions verbatim.

Two modern versions you'll like. A **3D printer** builds a solid exactly the way this chapter computes one: as a stack of thin horizontal slices, one layer at a time; the slicer software's whole job is to work out each layer's cross-section, which is precisely the "what does one slice look like?" question you're about to practice. And a **CT scanner** runs the idea in reverse: it builds a deck of thin cross-section images of your body (reconstructed from X-ray sweeps), and software integrates the stack back into a 3D volume. Slice, know one slice, sum: medicine, manufacturing, and this chapter all run on the same loop.

Disks: the slice is a coin

Spin a region that touches the axis, then cut the resulting solid *perpendicular* to the axis, like slicing a baguette. Each slice is a thin circular **coin**: a disk. Before I write any formula, write it yourself, you have everything: the coin at position x has some radius $R(x)$ (how far the solid reaches from the axis, which is just the curve's height) and tiny thickness dx. What's the volume of one coin, and what do you do with all of them?

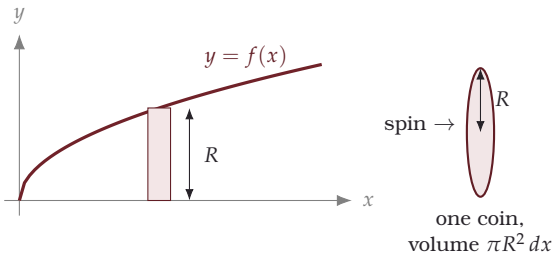
Circle area times thickness, $\pi R^2\,dx$, then sum:

★ The disk method

$$V = \int_a^b \pi\big[R(x)\big]^2\,dx, \qquad R(x) = \text{distance from the axis out to the curve.}$$

Not a formula to memorize; a sentence to re-derive: *coin volume, summed.*

Before the picture: it should show a strip of the flat region, and the coin that strip sweeps when spun. Find both:



The strip and the coin it becomes.

Trace the correspondence: the skinny rectangle standing on the x-axis is one strip of the flat region, exactly the strips from last chapter. Spin it around the axis and its top edge sweeps a circle of radius R (the strip's height), so the strip fattens into the coin on the right. Strip height becomes coin radius; strip width dx becomes coin thickness. Every formula in this chapter is born in that little transformation, so make sure you can see the strip *becoming* the coin before moving on.

➤ Disk: $y = \sqrt{x}$ on $[0,4]$, spun about the x-axis

Coin radius is the curve's height: $R(x) = \sqrt{x}$. Sum the coins:

$$V = \int_0^4 \pi\left(\sqrt{x}\right)^2 dx = \pi \int_0^4 x\, dx = \pi \left[\frac{x^2}{2}\right]_0^4 = 8\pi.$$

The squaring un-did the square root, a kindness this particular curve is famous for.

Washers: the slice is a coin with a hole

Now spin a region that *doesn't* touch the axis, say, the ring of space between two curves. Each slice is a coin with a bite drilled out of its center: a **washer**, as in the little hardware-store ring. Its face area is big circle minus small circle, and here is the single most important warning in this chapter, dressed as arithmetic you can do right now. With outer radius 2 and inner radius 1, compute both of these: $\pi(2^2 - 1^2)$ and $\pi(2-1)^2$. You get 3π and π. *Not the same number.* The first is right (a real washer's face), the second is the most common volume mistake on Earth. Square each radius **separately**, then subtract.

★ The washer method

$$V = \int_a^b \pi\Big(\big[R_{\text{out}}(x)\big]^2 - \big[R_{\text{in}}(x)\big]^2\Big)\, dx,$$

R_{out} from the axis to the far curve, R_{in} to the near curve. Squares first, then subtract. You proved to yourself ten seconds ago why the other order lies.

➤ Washer: between $y = x$ and $y = x^2$, spun about the x-axis

Last chapter's region, promoted to 3D. Crossings at $x = 0, 1$; at $x = \frac{1}{2}$ the line (0.5) rides farther from the axis than the parabola (0.25), so $R_{\text{out}} = x$, $R_{\text{in}} = x^2$:

$$V = \pi \int_0^1 \left(x^2 - x^4\right) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \pi\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{2\pi}{15}.$$

Shells: the slice is a tube

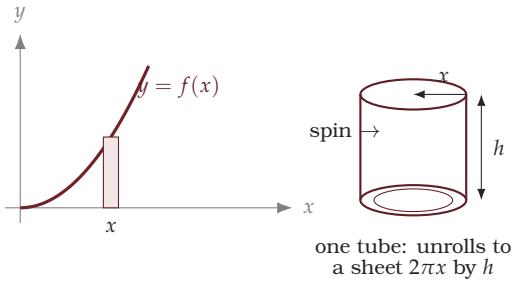
There's a second way to take apart a spun solid: not as a stack of coins, but as *nested tubes*, like tree rings, or those Russian dolls. Take a vertical strip of the flat region at distance x from the axis, with height $h(x)$, and spin it *parallel* to the axis: it sweeps a thin cylindrical **shell**. One slice's volume, think it through before the reveal: unroll the tube into a flat sheet. The sheet's length is the tube's circumference, $2\pi x$; its height is $h(x)$; its thickness is dx. Volume $2\pi x\, h(x)\, dx$.

★ The shell method

$$V = \int_a^b 2\pi\,(\text{radius})(\text{height})\,dx = \int_a^b 2\pi\, x\, h(x)\,dx.$$

One unrolled tube, summed. Shells earn their keep when you spin about the y-axis but your curve is written as $y = f(x)$: coins would force you to invert the function, tubes don't ask.

Before this picture: it should show the strip *parallel* to the spin axis this time, and the tube it sweeps. Compare it against the disk picture; the strip's orientation is the entire difference between the methods:



The strip and the tube it becomes.

Walk it: this strip stands at distance x from the y-axis, and we're spinning around the y-axis, so the strip travels in a circle of circumference $2\pi x$, dragging its full height h along. The disk picture's strip was *perpendicular* to its axis and fattened into a coin; this one is *parallel* and stretches into a tube. Same solid can often be taken apart either way; you'll pick whichever slice is easier to describe.

➤ Shell: $y = x^2$ on $[0, 2]$, spun about the y-axis

Shell at radius x, height $h(x) = x^2$:

$$V = \int_0^2 2\pi x\,(x^2)\,dx = 2\pi \int_0^2 x^3\,dx = 2\pi \left[\frac{x^4}{4}\right]_0^2 = 8\pi.$$

For contrast, try it with coins: you'd slice horizontally into washers, rewrite the curve as $x = \sqrt{y}$, track an outer wall at $x = 2$ and an inner radius $\sqrt{y}$, and arrive, eventually, at the same 8π. Shells exist so you don't have to.

✎ Quick practice, before you move on

Practice 1. Spin $y = x$ on $[0, 3]$ about the x-axis. Before integrating: this builds a cone with base radius 3 and height 3, so *predict* the answer from the grade-school cone formula $\frac{1}{3}\pi r^2 h$. Then earn it with disks and compare.

Practice 2. Spin the region under $y = x^2$ on $[0, 1]$ about the *y-axis*.
How to think about it: y-axis spin, curve written as y in terms of x: that combination is the shell method's home turf. Radius x, height x^2.

Practice 3 (ten seconds, no integral). A washer slice has outer radius 5 and inner radius 3. Your friend computes its face area as $\pi(5-3)^2 = 4\pi$. What's the correct area, and what exactly did they do wrong?

Answers at the end of the chapter.

When the axis moves, and when the variable flips

A shifted axis. Spin about $y = -1$ instead of the x-axis and nothing conceptual changes, but the radius must be re-measured, because radius always means *distance from the axis of rotation*, wherever that axis went.

➤ **Washer about $y = -1$: region between $y = x$ and $y = 0$ on $[0, 1]$**

The axis sits one unit below the floor of the region. Outer radius: from $y = -1$ up to the top curve $y = x$, a distance $x + 1$. Inner radius: from $y = -1$ up to the bottom edge $y = 0$, a distance 1. (Draw the one-unit gap; it's the whole problem.)

$$V = \pi \int_0^1 \left[(x+1)^2 - 1^2\right] dx = \pi \int_0^1 \left(x^2 + 2x\right) dx = \pi \left[\frac{x^3}{3} + x^2\right]_0^1 = \frac{4\pi}{3}.$$

The variable follows the slicing. Coins are perpendicular to the axis, so spinning about the y-axis with coins means *horizontal* coins, integrated in dy, with curves rewritten as x in terms of y:

➤ **A dy disk: region bounded by $y = x^2$, $y = 4$, and the y-axis ($x \geq 0$), about the y-axis**

Horizontal coins from $y = 0$ to $y = 4$, each with radius equal to its x-extent: $y = x^2$ means $x = \sqrt{y}$:

$$V = \pi \int_0^4 \left(\sqrt{y}\right)^2 dy = \pi \int_0^4 y \, dy = \pi \left[\frac{y^2}{2}\right]_0^4 = 8\pi.$$

★ **Coins or tubes? The one-look decision**

Draw one representative strip of the flat region. Strip *perpendicular* to the spin axis: it becomes a coin, disk/washer method, integrate in the axis's own variable. Strip *parallel* to the spin axis: it becomes a tube, shell method. And when one choice would force you to solve the curve for its inverse, that's the universe telling you to pick the other slice.

☞ Watch out

The wrong ideas that wreck volume problems, named and retired:

- *"$\pi(R_{out} - R_{in})^2$."* You disproved this with radii 2 and 1 in ten seconds: 3π versus π. Squares first, subtract second, forever.
- *"The radius is the function."* The radius is the *distance to the spin axis*. They coincide only while the axis is the x-axis; move the axis and the radius becomes a difference, like $x + 1$ above.
- *"Spun about the y-axis, so I integrate dy."* Only if you're slicing coins. Shells around the y-axis integrate happily in dx. The *slice* dictates the variable, not the axis alone.
- *"I'll just memorize which method goes with which axis."* There's no such table; there's only the strip. Draw it, see whether it's a coin or a tube, done.

Carrying this forward

★ Leaving this chapter with this

One question runs the whole chapter: what does one slice look like? A coin ($\pi R^2\,dx$), a coin with a hole ($\pi(R_{\text{out}}^2 - R_{\text{in}}^2)\,dx$, squares before subtraction), or an unrolled tube ($2\pi\,(\text{radius})(\text{height})\,dx$). Radius always means distance to the spin axis, the slice's orientation picks the method and the variable, and a geometry formula you already trust (the cone) is sitting right there to sanity-check your setups. Next chapter we put down the solids and ask something sneakier about a curve: not what's under it or around it, but how *long* it is.

✎ The chapter workout (get stuck on purpose)

Decide coin, washer, or tube first; then set up; then integrate.

F1. $y = x^2$ on $[0,2]$, about the x-axis **F2.** $y = x$ on $[0,2]$, about the y-axis

F3. between $y = \sqrt{x}$ and $y = x$ on $[0,1]$, about the x-axis

F4 (setup only, no integration). The region under $y = \sqrt{x}$ on $[0,4]$ is spun about the line $y = -2$. Write the washer integral, radii only, no evaluating.

F5 (one sentence). Why does the shell formula contain a $2\pi x$ while the disk formula contains a πR^2? Answer with the two slice shapes.

Nudges: **F1** coin, $R = x^2$. **F2** tube: radius x, height x (or horizontal coins in dy if you insist; same answer, more steps). **F3** washer; who's farther from the axis on $[0,1]$? **F4** both radii are distances measured from $y = -2$. **F5** unroll one, don't unroll the other.

Next up: *Arc Length & Surface Area*, where the question becomes how long a curve is, and the answer is the distance formula wearing an integral.

➜ Want it on video?

Professor Leonard, *Calculus 1 Lectures 5.2–5.3* (Volumes by Disks/Washers and by Cylindrical Shells), at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** prediction $\frac{1}{3}\pi(3^2)(3) = 9\pi$; disks: $\pi\int_0^3 x^2\,dx = 9\pi$. Geometry and calculus agree, which is the point of predicting. **P2:** $\int_0^1 2\pi x\,(x^2)\,dx = 2\pi\cdot\frac{1}{4} = \frac{\pi}{2}$. **P3:** $\pi(5^2-3^2) = 16\pi$; they subtracted the radii before squaring, which shrinks every washer and always undercounts (your ten-second demo from the washer section, with bigger numbers). **F1:** $\pi\int_0^2 x^4\,dx = \frac{32\pi}{5}$. **F2:** $\int_0^2 2\pi x\cdot x\,dx = \frac{16\pi}{3}$. **F3:** $\pi\int_0^1 \left(x-x^2\right)dx = \frac{\pi}{6}$. **F4:** $V = \pi\int_0^4 \left[(\sqrt{x}+2)^2 - 2^2\right]dx$ (outer: from -2 up to the curve; inner: from -2 up to the floor at 0). **F5:** a disk is a solid coin, so its face is a full circle's area πR^2; a shell is a hollow tube that unrolls into a flat sheet whose length is its circumference $2\pi x$, no circle face anywhere.

Arc Length & Surface Area

Calculus 2 • Applications of Integration, Chapter 3

Measuring how long a curve really is, and the surface it sweeps

❧ Chapter 3. How long is a curvy line?

A straight line, you can measure with the distance formula, rise and run and Pythagoras, sixth-grade equipment. But how long is a *curve*, something that bends the whole way? You can't lay a ruler along it, and the distance formula needs straightness.

Except, think back two units: what did we do when a region's ceiling was curved and area formulas needed straight edges? Chopped it into pieces so small the curve couldn't bend inside them. Same cure here. Zoom into any smooth curve hard enough and each tiny piece *is* straight, or close enough that the error dies in the limit. Measure every tiny straight piece with the sixth-grade formula, add them with an integral. You already know this move so well that I'd bet you could build this chapter's formula yourself, and in a minute, that's exactly what you'll do.

Then, for a two-for-one: once you can measure a curve's length, spinning that curve around an axis gives you the *area of the surface* it sweeps, a lampshade, a dome, a satellite dish. One idea, two payoffs, and the second one explains how much paint the dome needs.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say where length-of-a-curve problems come from (bridges and cables, mostly),
- build the arc-length formula yourself out of the distance formula,
- compute curve lengths, and sanity-check them against the straight-line chord,
- find the area of a surface of revolution (and know which radius it spins on),
- know the honest truth about which of these integrals are even doable by hand.

Why this exists, and what it's for

The purest customer: **cable**. A suspension bridge's main cables hang in curves, and somebody has to order the steel *by length*. Underestimate and the bridge doesn't close; overestimate by much and you've bought tons of unused cable. "How long is this curve?" is a purchasing question with a dollar sign on it. The same calculation prices the wire on a power line between towers, the length of a curved road through hills (your odometer reads arc length, not map distance), and the track of a roller coaster.

The surface-area sibling has its own customers: how much paint covers a dome, how much aluminum skins a rocket's nose cone, how much reflective coating fills a satellite dish. Dishes and headlight reflectors are parabolas spun around an axis, this chapter's exact object.

And inside mathematics, today's small star, the little diagonal ds, becomes a workhorse: it reappears in physics (line integrals), in the next chapter's setup, and beyond. Worth meeting it properly.

Build the formula yourself

Here's the construction, and your job is to supply each step before reading past it.

Zoom in on a smooth curve until the piece in your window looks straight. That tiny near-straight piece has a tiny run dx and a tiny rise dy. Question one: what's the length of a straight piece with run dx and rise dy? Sixth grade, hypotenuse:

$$ds = \sqrt{dx^2 + dy^2}.$$

Question two: we want to integrate "$\ldots dx$," so factor the dx out of that root yourself. Pull dx^2 out of both terms:

$$ds = \sqrt{dx^2\left(1 + \frac{dy^2}{dx^2}\right)} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx.$$

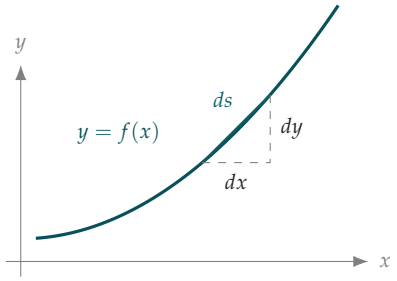
And $\frac{dy}{dx}$ is an old friend wearing its Leibniz coat: the derivative. Add up all the tiny hypotenuses:

★ The arc-length formula

$$L = \int_a^b \sqrt{1 + \left[f'(x)\right]^2}\,dx.$$

It's the distance formula, $\sqrt{(\text{run})^2 + (\text{rise})^2}$, applied to infinitely many tiny straight pieces and summed. You just derived it; it will never need memorizing again.

Before the picture: it should show exactly the triangle you just built, sitting on a zoomed-in piece of curve. Check your mental image against it:



One zoomed-in piece of curve: run dx, rise dy, hypotenuse ds.

Note what the diagram is and isn't claiming: the bold little diagonal ds isn't exactly on the curve, it's the straight shortcut across one tiny step. The whole theory is that as the steps shrink, the shortcuts and the curve become indistinguishable, the same way the rectangle staircase became the area. Limits, still the main character.

➤ Length of $y = \frac{2}{3}x^{3/2}$ on $[0,3]$

First $f'(x) = x^{1/2}$, so $[f'(x)]^2 = x$, and

$$L = \int_0^3 \sqrt{1+x}\,dx.$$

To integrate $\sqrt{1+x}$: let $u = 1+x$ (so $du = dx$), giving $\int \sqrt{u}\,du = \frac{2}{3}u^{3/2}$. Evaluate:

$$L = \left[\frac{2}{3}(1+x)^{3/2}\right]_0^3 = \frac{2}{3}(4^{3/2}-1) = \frac{2}{3}(8-1) = \frac{14}{3} \approx 4.67.$$

Now the sanity check this chapter wants you to make a habit: the straight chord from the curve's start $(0,0)$ to its end $(3, 2\sqrt{3}) \approx (3, 3.46)$ has length $\sqrt{9+12} = \sqrt{21} \approx 4.58$. Our curve came out *slightly longer than its chord*, which is exactly right: a straight line is the shortest path, so an arc can never come in under its chord, it ties only when the "curve" is itself straight, and beats it whenever there's any bend, by only a little when the bend is gentle. If your arc length ever comes out *shorter* than the chord, something broke.

☞ Watch out

An honest confession most textbooks whisper: the examples in this section are *hand-picked* so that $1 + (f')^2$ collapses into something integrable. Out in the wild, even $y = x^2$ produces $\sqrt{1+4x^2}$, doable but ugly, and most curves produce roots with no elementary antiderivative at all (no formula built from the standard functions), which engineers hand to a computer. So the skill being taught is the *setup*, which always works, plus a sharp eye for the rigged simplification, usually a perfect square hiding inside $1 + (f')^2$. If your root refuses to simplify, re-check f' first; the problem was probably built to collapse.

Surface area: spin the curve itself

Volumes chapter: we spun a *region* and got a solid. Now spin just the *curve*, a wire, not a pancake, and it sweeps a thin surface, like a lampshade. What's one slice of that surface? Think it through with the unrolling trick from shells: a tiny piece ds of curve at height $f(x)$ sweeps a thin band around the axis. The band's length is the circle it travels, circumference $2\pi f(x)$; its width is the piece itself, ds. Band area: circumference times width.

★ Surface area (revolving about the x-axis)

$$S = \int_a^b 2\pi\, f(x)\, \sqrt{1 + \left[f'(x)\right]^2}\, dx.$$

Read it as a sentence: $2\pi f(x)$ is the lap each point runs (radius = height, assuming the curve stays above the axis as all of ours do; a dipping curve would need $|f(x)|$), and $\sqrt{1+(f')^2}\,dx$ is our ds, the slanted width of the band. Laps times width, summed.

➤ Revolve $y = x$ on $[0,1]$ about the x-axis: a cone

Here $f' = 1$, so $ds = \sqrt{2}\,dx$, and the radius is $f(x) = x$:

$$S = \int_0^1 2\pi x\sqrt{2}\,dx = 2\pi\sqrt{2}\left[\frac{x^2}{2}\right]_0^1 = \pi\sqrt{2}.$$

Cross-check against geometry, the habit that keeps us honest: a cone's lateral surface is $\pi r\ell$, where ℓ is the *slant height*, the straight distance from tip to rim. Our cone has tip $(0,0)$ and rim point $(1,1)$, so $r = 1$ and $\ell = \sqrt{1^2+1^2} = \sqrt{2}$, giving $\pi r\ell = \pi\sqrt{2}$. The integral and the geometry shook hands again.

☞ Watch out

This formula's $2\pi f(x)$ assumes you're spinning about the x-*axis*, where each point's lap radius is its height. Spin about a different line, the y-axis, or $y = 2$, and the radius changes to *distance from that axis*, exactly like the volumes chapter: the factor becomes $2\pi \cdot$ (distance to the spin axis). Same band, different lap.

✎ Quick practice, before you move on

Practice 1. Arc length of $y = x$ from 0 to 4. Then check yourself against the distance formula, since this "curve" is a straight line and you can score your own answer.

Practice 2. Arc length of $y = \frac{1}{3}(x^2+2)^{3/2}$ on $[0,2]$.
How to think about it: compute f', square it, add 1, and *hunt for the perfect square*: here $f'(x) = x\sqrt{x^2+2}$, so $(f')^2 = x^4 + 2x^2$, and $1 + (f')^2 = x^4 + 2x^2 + 1 = (x^2+1)^2$. The root melts. (This is the rigged collapse from the warning box, caught in the act.)

Practice 3 (one sentence). Your computed arc length for some curve comes out *less* than the straight-line distance between its endpoints. Why is that an automatic red flag?

Answers at the end of the chapter.

Carrying this forward

★ Leaving this chapter with this

Arc length is the distance formula aimed at tiny pieces and summed: $L = \int \sqrt{1 + (f')^2}\, dx$, a formula you built, not borrowed. Every answer must beat its chord (straight lines are shortest), the textbook examples are rigged to collapse via a hidden perfect square, and the wild ones go to a computer with the same setup. Spin the curve and each piece ds runs a lap: $S = \int 2\pi f(x)\, ds$, radius measured to the spin axis, cross-checkable on a cone. Next chapter is the unit's finale and my favorite: integrals that compute physical *work*, including the exact rope-and-pumping problems that show up on every exam, and once showed up on mine.

✎ The chapter workout (get stuck on purpose)

For each: find f', build $1 + (f')^2$, hunt the collapse.

F1. arc length of $y = \frac{2}{3}x^{3/2}$ on $[0, 8]$

F2. surface: $y = x$ on $[0, 3]$ about the x-axis

F3. arc length of $y = 3x$ from 0 to 1

F4 (setup only). Write, but don't evaluate, the arc-length integral for $y = x^2$ on $[0, 1]$, then say in one clause why this one isn't rigged.

F5 (chord check). For F1, compute the chord from $(0, 0)$ to $(8, f(8))$ and confirm your arc length beats it.

Nudges: **F1** $(f')^2 = x$, same u-sub as the worked example. **F2** the cone again, scaled; check with $\pi r \ell$. **F3** a straight line, so you know the answer before integrating. **F4** what does $1 + 4x^2$ refuse to be? **F5** $f(8) = \frac{2}{3} \cdot 8^{3/2}$.

Next up: *Work*, where a changing force gets integrated into a total: springs, hanging ropes, and pumping tanks, the problems your physics class and my Calc 2 exam were built from.

➜ Want it on video?

Professor Leonard, *Calculus 1 Lecture 5.4* (Arc Length) and the Surface Area lecture that follows it, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** $f' = 1$, $L = \int_0^4 \sqrt{2}\,dx = 4\sqrt{2}$; distance formula from $(0,0)$ to $(4,4)$: $\sqrt{32} = 4\sqrt{2}$. Agreement, as it must be for a straight line. **P2:** $\int_0^2 (x^2+1)\,dx = \frac{8}{3} + 2 = \frac{14}{3}$. **P3:** a straight line is the shortest possible path between two points, so no curve through them can be shorter than the chord; an arc length below the chord means an arithmetic or derivative error, guaranteed. **F1:** $\int_0^8 \sqrt{1+x}\,dx = \frac{2}{3}\left[(1+x)^{3/2}\right]_0^8 = \frac{2}{3}(27-1) = \frac{52}{3} \approx 17.33$. **F2:** $\int_0^3 2\pi x\sqrt{2}\,dx = 9\pi\sqrt{2}$ (geometry check: $r = 3$, $\ell = \sqrt{18} = 3\sqrt{2}$, $\pi r\ell = 9\pi\sqrt{2}$). **F3:** $(f')^2 = 9$, $L = \int_0^1 \sqrt{10}\,dx = \sqrt{10}$, matching the chord to $(1,3)$ exactly. **F4:** $L = \int_0^1 \sqrt{1+4x^2}\,dx$; $1+4x^2$ is not a perfect square, so nothing collapses (it yields to a trig substitution or a computer, not to sight). **F5:** $f(8) = \frac{2}{3} \cdot 22.63 \approx 15.08$; chord $= \sqrt{64 + 227.6} \approx 17.08 < 17.33$. The arc beats its chord, as the law requires.

Work

Calculus 2 • Applications of Integration, Chapter 4

Adding up a force that changes as it pushes

❧ Chapter 4. The unit's finale, and a personal one.

In physics, work is force times distance. Push with 10 newtons for 3 meters, you've done 30 joules of work (the joule being the metric unit of energy). One multiplication, done. So why does this topic get a whole chapter and a permanent slot on every Calc 2 exam?

Because the multiplication only works while the force holds still, and real forces don't. A spring fights harder the farther you stretch it. A hanging rope gets lighter to lift as less of it dangles. Pumping a tank, every layer of water rides a different distance to the top. Force times distance with a moving force is exactly the kind of "impossible multiplication" this book keeps meeting, and by now you know the cure without me saying it: slice the job into steps tiny enough that the force can't change inside one, multiply on each step, integrate.

I'll be straight with you about why I care about this chapter: these are the problems from *my* exam, the ropes, the tanks, the cables with buckets. I sat with them, got them wrong, got stuck, sat some more, and they cracked. Every setup in this chapter is one I've personally fought with on a whiteboard. Learn the slicing ritual and these become the most predictable points on the test.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say what work measures and why it needs an integral the moment force varies,
- handle springs with Hooke's law, including stretches that don't start at rest,
- lift ropes and cables, and beat the famous half-the-naive-guess surprise,
- add a hanging load to the cable (the two-jobs-in-one setup exams love),
- pump tanks, including the real case where the slab's size changes with height,
- keep pounds and kilograms from double-crossing you.

Why this exists, and what it's for

Work is the physicist's word for *energy spent moving something against a resistance*, and it's the currency every mechanical question gets priced in. How much energy to compress this shock absorber? To wind this crane cable? To empty this reservoir over the levee, to fill this water tower, to stretch this bow? Each is "force times distance" where one of the two refuses to sit still, so each is an integral.

You've met the deluxe version already: the escape-velocity integral from the improper chapter was a *work* integral, gravity's force summed over an infinite climb. Today's problems are the terrestrial editions, springs, ropes, tanks, and they're beloved by exam writers because they test the one skill this whole unit has been drilling: can you describe *one thin slice of the job* and integrate it? That ritual, slice, price one slice, sum, is the entire chapter.

★ Work, when force varies

$$W = \int_a^b F(x)\,dx.$$

Over a tiny step dx, the force $F(x)$ has no room to change, so that step costs $F(x)\,dx$, an honest multiplication. The integral adds up the steps. (Constant force is the special case where the integral collapses back to force times distance.)

Springs: Hooke's law

Stretch a spring and it pulls back with a force proportional to the stretch: $F = kx$, where x measures stretch *from the natural length* and k is the spring's stiffness. (Robert Hooke, 1676, and it's a good model until you stretch far enough to ruin the spring.) The force grows as you pull, so the work is an integral.

➤ Stretching a spring

A spring takes 10 N to hold at a stretch of 0.2 m. How much work to stretch it from rest to 0.3 m?

Two separate acts, don't blur them. **Act one, find the spring's personality:** $F = kx$ with the given snapshot, $10 = k(0.2)$, so $k = 50$ N/m. **Act two, integrate the force over the actual journey:**

$$W = \int_0^{0.3} 50x\,dx = 50\left[\frac{x^2}{2}\right]_0^{0.3} = 25(0.09) = 2.25 \text{ J}.$$

Quick gut check: the force at the far end is $50(0.3) = 15$ N; if that peak force had acted the whole way the cost would be $15 \times 0.3 = 4.5$ J, and since the force ramped from zero, the true answer should be about half that. 2.25. Exactly half, in fact, because the ramp is linear. Your sanity checks are getting sharper.

✎ **Quick practice, before you move on**

Practice 1. A spring has stiffness $k = 200$ N/m. Work to stretch it 0.1 m from rest?

Practice 2. A spring has $k = 80$ N/m. Work to stretch it *from* 0.1 m *to* 0.3 m? *How to think about it:* same $F = kx$, but the journey doesn't start at rest, so neither does the integral: the limits are the journey's. And predict before computing: should this cost more or less than the same 0.2 m of stretch starting from rest? The spring's already tense when you begin…

Answers at the end of the chapter.

Ropes and cables: the lift that lightens

A 50 ft rope hangs from a rooftop. You haul it all up. Gut question first, commit before computing: the rope weighs 0.5 lb/ft, so 25 lb total, and the top of the building is 50 ft up. Is the work just $25 \times 50 = 1250$ ft·lb?

Feel for the flaw before the math finds it: does every part of the rope travel 50 feet? The bit at the very top travels about zero. Only the far tip travels the full 50. The naive multiplication charges full fare to every inch of rope, so it must be an *overestimate*, and the fix is the ritual: slice the rope.

➤ **Pulling up a hanging rope**

Slice at depth x below the top: that piece has weight $0.5 \cdot dx$ (weight per foot, times a dx of feet) and must rise exactly x feet, its own depth. One slice costs $(0.5\,dx) \cdot x$. Sum over the rope:

$$W = \int_0^{50} 0.5\,x\,dx = 0.5\left[\frac{x^2}{2}\right]_0^{50} = 0.25(2500) = 625 \text{ ft·lb}.$$

Half the naive 1250. And you can now say precisely why: the slices' travel distances run evenly from 0 to 50, so the *average* slice travels 25 ft, and total weight 25 lb times average trip 25 ft is 625. The integral computed that average for you, and it always will.

☞ Watch out

Units, before they bite. Why did the rope's "0.5 lb/ft" go straight into the integral, while the water tanks below will multiply by 9800? Because **pounds are already a force** (a weight), gravity's included. **Kilograms are a mass**, and a metric problem must convert mass to force through g: for water, density 1000 kg/m^3 times $g = 9.8$ gives the weight density $\rho g = 9800$ N/m^3. Rule of thumb: data in lb, integrate as is, answer in ft·lb; data in kg or metric water, owe a factor of g, answer in joules (one joule = one newton-meter).

➤ A cable lifting a load: two jobs in one

A 30 ft cable weighing 3 lb/ft hangs down a shaft with a 200 lb bucket on the end. Total work to wind it all up?

Split the job in two and price each honestly. **The cable** is a hanging-rope problem: slice at depth x weighs $3\,dx$, rises x:

$$W_{\text{cable}} = \int_0^{30} 3x\,dx = 3\left[\frac{x^2}{2}\right]_0^{30} = 1350 \text{ ft·lb}.$$

The bucket is a *constant* 200 lb for the full 30 ft, no integral needed, constant force is the easy case:

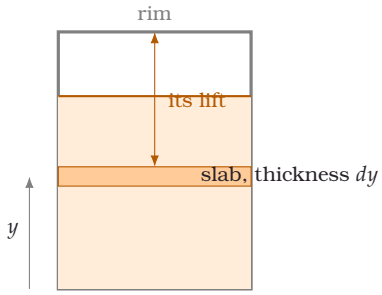
$$W_{\text{bucket}} = 200 \times 30 = 6000 \text{ ft·lb}.$$

Total $1350 + 6000 =$ **7350** ft·lb. The pattern to keep: varying parts get sliced and integrated, constant parts get multiplied, and the jobs add. Nearly every exam cable problem is this template with different numbers.

Pumping a tank: every layer rides its own elevator

The boss fight of the chapter, and the most visual. To pump a tank's water out over the top, you lift every thin horizontal layer from where it sits up to the rim, and layers near the bottom ride a longer elevator than layers near the top. Slice horizontally, price one slab, integrate.

One slab at height y, thickness dy, needs three facts multiplied: its *volume* (cross-section area times dy), its *weight* (volume times 9800, for water, metric), and its *lift* (from height y up to wherever the water exits). Before the picture: predict which of those three ingredients changes from slab to slab in a straight-walled tank. Commit to a pick; the box example below will grade it.



One slab of water and the elevator ride it's about to take.

Read the picture like a work order: the slab lives at height y, the rim is at the top, so this slab's lift is (rim height $-\,y$), and a deeper slab's order says a longer ride. That $(\text{top} - y)$ factor inside the integral is the entire reason pumping problems need calculus at all.

➤ Straight walls: pumping a full box tank

A box tank, base 2 m by 3 m, depth 4 m, full of water. Work to pump everything to the rim?

Slab at height y (from the bottom): area $2 \times 3 = 6$ always, weight $9800 \cdot 6\,dy = 58800\,dy$ N, lift $(4 - y)$:

$$W = \int_0^4 58800\,(4-y)\,dy = 58800\left[4y - \frac{y^2}{2}\right]_0^4 = 58800(16-8) = 470{,}400 \text{ J}.$$

That box was the gentle case: straight walls, so every slab had the same area, and only the lift varied. The tanks that actually get asked about, cones, troughs, bowls, have walls that slope, so *the slab's size changes with height too*, and finding "area as a function of y" becomes step one.

➤ Sloped walls: the triangular trough

A trough is 3 m long; its cross-section is a triangle, 2 m wide at the top, coming to a point at the bottom, 2 m deep. Full of water, pumped to the top edge.

Set $y = 0$ at the bottom point. The water's width grows linearly from 0 (at the point) to 2 (at the top, $y = 2$), so width $= y$. A slab is a thin plank: length 3, width y, thickness dy:

$$\text{weight} = 9800 \cdot 3y\,dy = 29400\,y\,dy, \qquad \text{lift} = (2 - y).$$

Now *two* things vary, the slab's heft and its ride, and the integral carries both:

$$W = \int_0^2 29400\,y\,(2-y)\,dy = 29400\left[y^2 - \frac{y^3}{3}\right]_0^2 = 29400\left(4 - \frac{8}{3}\right) = 39{,}200 \text{ J}.$$

The one new skill: before integrating, find the cross-section's size as a function of y (here by similar triangles; for a cone you'd do the same with the radius). After that it's the same slice-price-sum as every page of this unit.

☞ Watch out

The wrong ideas that wreck work problems, named and retired:

- *"Total weight times total height."* The naive 1250 from the rope section. Only the deepest slice pays full fare; the integral charges each slice its own trip. If your answer matches weight-times-height exactly on a rope or tank problem, you skipped the slicing.
- *"The lift is just y."* The lift is the distance to the *exit*. Water leaving over a rim at height H rides $(H - y)$; and if the problem adds a spout 1 m *above* the rim, every slab rides $(H + 1 - y)$. Read where the liquid actually leaves.
- *"Multiply by g…or was it don't?"* Pounds: force already, no g. Kilograms or metric water: mass, owes g (water: 9800 N/m^3). Mixing these is the quietest way to be off by a factor of about ten.
- *"The spring constant is the force."* k is force *per meter of stretch*; you must calibrate it from the given snapshot first ($F = kx$), then integrate. Two acts, in order.

Carrying this forward

★ Leaving this chapter, and the applications unit

Work is force times distance, rescued by slicing whenever either one won't hold still: springs ramp their force ($\int kx\,dx$), ropes lighten as they shorten (each slice pays its own depth), tanks lift every layer a different ride (and sloped walls make the layer's size vary too). Constant pieces multiply, varying pieces integrate, jobs add. And step back for the unit's full picture: areas, volumes, lengths, surfaces, work, five chapters, one ritual, slice, price one slice, sum. That ritual is what you actually own now, and no exam variation can take it from you. Next, the book changes worlds entirely: infinite lists, infinite sums, and the strange question of when forever adds up to something finite.

✎ The chapter workout (get stuck on purpose)

Slice, price one slice, sum.

F1. A spring needs 30 N to hold a 0.5 m stretch. Work to stretch it 0.4 m?

F2. A 40 ft rope at 2 lb/ft, hauled all the way up.

F3. A box tank, 2×2 m base, 3 m deep, full of water, pumped to the rim.

F4 (predict, then verify). For F2, state the naive weight-times-height number first, then say what fraction of it the true answer must be and why, then integrate and confirm.

F5 (setup only). The trough from the worked example gets a spout 1 m above its top edge. Write the new integral; don't evaluate.

Nudges: **F1** calibrate k from the 0.5 m snapshot, then integrate to the *target* 0.4. **F2** slice at depth x. **F3** slab area 4, weight $39200\,dy$, lift $(3 - y)$. **F4** where do the trip lengths run, and what's their average? **F5** only the lift changes.

Next up (new unit): *Sequences*, infinite lists of numbers and where they're heading, the first step toward sums that never end.

➜ Want it on video?

Professor Leonard, *Calculus 2*, the Work lecture in his applications sequence (springs, ropes and chains, pumping tanks), at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** $\int_0^{0.1} 200x\,dx = 100(0.01) = 1$ J. **P2:** more than the from-rest version, the spring fights hardest late: $\int_{0.1}^{0.3} 80x\,dx = 40(0.09 - 0.01) = 3.2$ J, versus 1.6 J for 0.2 m from rest. **F1:** $k = 30/0.5 = 60$ N/m; $\int_0^{0.4} 60x\,dx = 30(0.16) = 4.8$ J. **F2:** $\int_0^{40} 2x\,dx = 1600$ ft·lb. **F3:** $\int_0^3 39200(3-y)\,dy = 39200(9-4.5) = 176{,}400$ J. **F4:** naive: 80 lb $\times$ 40 ft $= 3200$; trips run evenly from 0 to 40, average 20, so the truth must be half: 1600, and the integral confirms it. **F5:** same slab, longer ride: $W = \int_0^2 29400\,y\,(3-y)\,dy$ (exit now at height $2+1=3$).

Sequences

Calculus 2 • Sequences & Series, Chapter 1

Infinite lists of numbers, and where they're heading

❧ New unit, new world. Same main character.

Everything so far has been about smooth, continuous things: curves, areas, solids. This unit switches to something that looks completely different, *infinite lists of numbers*, and I want to tell you up front why that's not the detour it seems.

Here's the question that runs the whole unit: write down a never-ending list of numbers, where is it *going*? Some lists march toward one value and settle in. Others blow up, or bounce forever and never decide. The settlers *converge*; the rest *diverge*. If that question sounds familiar, it should: it's "what is this heading toward?", the very first question this book asked, the limit, back for its biggest arc. Derivatives were limits of slopes, integrals were limits of sums, and now sequences are limits of *lists*. The main character never left.

Which means you arrive armed. L'Hôpital, dominant terms, the squeeze, the whole Calc 1 limit kit transfers almost unchanged. You're not starting a new subject. You're pointing a familiar telescope at a new patch of sky, and the stars there are infinite sums and Taylor series.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say where sequences show up outside math class (your code is full of them),
- read $\{a_n\}$ notation and the words converge and diverge,
- find a sequence's destination yourself, by table, before any algebra,
- reuse the Calc 1 kit: top-power, L'Hôpital through the continuous version, the squeeze,
- know the behavior of r^n cold, the single fact the whole unit leans on,
- retire the "it never actually gets there" confusion for good.

Why this exists, and what it's for

A **sequence** is an infinite ordered list of numbers, and the reason mathematicians care is that *every step-by-step process produces one*. If you're a programmer, you've been generating sequences your whole career: each pass of a loop emits a value; binary search halves an interval over and over (widths $1, \frac{1}{2}, \frac{1}{4}, \ldots$, a sequence); an iterative algorithm improves an estimate each round. "Does my algorithm converge?" is a sequence question, asked daily, with money on it.

Your calculator runs on sequences too: when it computes $\sqrt{2}$, it generates a list of better and better guesses ($1.5,\ 1.4167,\ 1.41422, \ldots$) and stops when the list settles. Compound interest produces a sequence of balances; populations produce sequences of yearly counts; and famously, the sequence $\left(1+\frac{1}{n}\right)^n$ walks straight to the number e.

And within this book, sequences are the load-bearing floor: next chapter we start *adding up* infinitely many numbers, and the only way to make sense of an endless sum is to watch its running totals, which form a sequence. No sequences, no series, no Taylor. This chapter is the gateway.

Find a destination yourself, before any machinery

The notation, quickly, built from what you know: we write a sequence as $\{a_n\}$, where a_n is the formula for the nth entry, and n runs through $1, 2, 3, \ldots$ It's just a function whose inputs are the counting numbers $1, 2, 3, \ldots$, $f(x)$ with the dial clicking instead of sliding. So $a_n = \frac{1}{n}$ is the list $1, \frac{1}{2}, \frac{1}{3}, \ldots$

Now, the book's favorite ritual. Here's a sequence:

$$a_n = \frac{3n+1}{2n-5}.$$

Where is this list heading as n grows? Don't analyze, *compute*. Grab a calculator and fill in at least two:

n	10	100	1000	heading toward…
a_n	?	?	?	?

The machine says: 2.067, then 1.544, then 1.504. The list is being herded toward 1.5, that is, $\frac{3}{2}$. We write

$$\lim_{n\to\infty} a_n = \frac{3}{2},$$

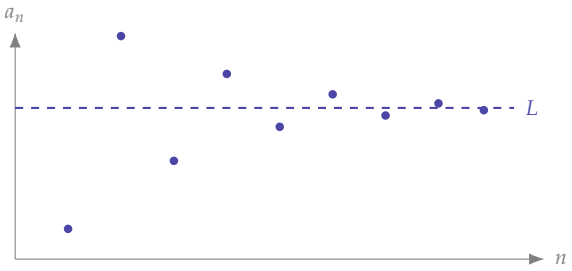
and say the sequence **converges** to $\frac{3}{2}$. A sequence with no single finite destination **diverges**, and that covers two very different fates: blowing up ($a_n = n^2$) and *never deciding* ($a_n = (-1)^n$, the list $-1, 1, -1, 1, \ldots$, which bounces forever between two camps; bounded (it never leaves the range $[-1, 1]$), polite, and still divergent, because "converge" means settling on *one* number).

Why $\frac{3}{2}$? You can see it in the formula with Calc 1 eyes: divide top and bottom by n,

$$a_n = \frac{3 + \frac{1}{n}}{2 - \frac{5}{n}} \longrightarrow \frac{3+0}{2-0} = \frac{3}{2},$$

the little $\frac{1}{n}$ crumbs dying exactly like they did in the horizontal-asymptote days. Table and algebra agree, which is what you always want to see.

Before you look: sketch, in your head, what the dots of your table should do if you plotted them with n along the bottom and a_n up the side. Wobble early, then settle at a height? Call that height L, the limit. Now check your sketch against mine:



A convergent sequence: dots homing in on the dashed line L.

Walk the dots left to right: early entries scatter wildly, and that's fine, convergence never cares how a list *starts*, only how it *ends up*. The herding is the whole definition: eventually the dots crowd as close to the line as you please and stay there. Notice also what the picture doesn't require: no dot ever has to *touch* L. The wall from the very first limits chapter is back, you can know exactly where you're headed without arriving, and "it never actually reaches L" is not an objection to convergence, it's the normal case.

The old kit transfers

➤ L'Hôpital, through the continuous understudy: $a_n = \dfrac{\ln n}{n}$

This is an $\frac{\infty}{\infty}$ race: top and bottom both head to infinity, the exact form L'Hôpital was built for. One formality first: L'Hôpital needs derivatives, and a dotted list isn't differentiable, there's nothing between the dots. The legal move: switch to the *continuous* function $f(x) = \frac{\ln x}{x}$, which passes through every dot and does have a derivative:

$$\lim_{x\to\infty} \frac{\ln x}{x} \overset{\text{L'H}}{=} \lim_{x\to\infty} \frac{1/x}{1} = 0.$$

If the smooth curve settles at 0, the dots riding it settle at 0 too. So $a_n \to 0$: the sequence converges to 0. (The handoff only runs that direction, curve verdicts bind the dots, not vice versa, but that direction is the one you need.)

★ The growth hierarchy (who wins a race to ∞)

When a limit is a race between growing things, the faster one wins, often with no computation:

$$\ln n \ll n^p \ (p>0) \ll a^n \ (a>1) \ll n!$$

Logs lose to powers, powers to exponentials, exponentials to factorials. (Here $n!$, read "n factorial," means $1\cdot 2\cdot 3\cdots n$, and it grows faster than anything else on the list.) So $\frac{\ln n}{n}\to 0$, $\frac{n}{e^n}\to 0$, $\frac{2^n}{n!}\to 0$, read off the hierarchy. (This is the same growth race you watched flip at $x\approx 996$ back in the limits chapter, now in its natural habitat.) When the *faster* grower is on top, the ratio blows up instead.

The fact to own cold: r^n

The entire next chapter balances on one question: what does the list $r, r^2, r^3, \dots$ do? Bet before you look, for $r=\frac{1}{2}$, $r=1$, $r=2$, and $r=-\frac{1}{2}$:

★ What r^n does as $n\to\infty$

$$r^n \longrightarrow \begin{cases} 0 & \text{if } |r|<1 \quad (\frac{1}{2},\frac{1}{4},\frac{1}{8},\dots;\ (-\frac{1}{2})^n\to 0 \text{ too}) \\ 1 & \text{if } r=1 \quad (1,1,1,\dots) \\ \infty & \text{if } r>1 \quad (\text{doubling never stops}) \\ \text{no limit} & \text{if } r\le -1 \quad (\text{flips sign forever without shrinking: diverges}) \end{cases}$$

One line to carry: $r^n\to 0$ *exactly when* $|r|<1$, and that includes negative r like $-\frac{1}{2}$: the sign flips but the size shrinks, so it still dies. Repeated multiplication by something smaller than 1 in size grinds anything to dust; by something bigger, it explodes.

The squeeze: handling a passenger that won't settle

What about $a_n = \frac{\cos n}{n}$? The $\cos n$ upstairs wanders between -1 and 1 forever, no limit of its own, so none of the direct tools grab. But ask the practical question: how big can the whole fraction possibly be? The wanderer is *trapped*, $-1 \le \cos n \le 1$, so

$$-\frac{1}{n} \le \frac{\cos n}{n} \le \frac{1}{n},$$

and both fences collapse to 0. A sequence pinned between two others that share a destination has nowhere else to go: that's the **Squeeze Theorem**, imported intact from Calc 1. Verdict: $\frac{\cos n}{n} \to 0$. The pattern to spot: *bounded wanderer divided by something growing* gets squeezed to zero, every time.

One more import, for sequences you can't compute head-on: the **Monotonic Sequence Theorem** says a list that always moves one direction (never down, with a ceiling it can't pass; or the mirror image, never up, with a floor) *must* converge, it keeps stepping forward in a room with a wall, so it has to pile up somewhere. That's the theorem that proves $\left(1+\frac{1}{n}\right)^n$ converges; its destination is $e \approx 2.718$, and that's one of the standard definitions of e.

☞ Watch out

The wrong ideas that wreck sequence problems, named and retired:

- *"It never actually reaches L, so it doesn't converge."* Convergence asks where the list is *heading*, not whether it arrives. $\frac{1}{n}$ never touches 0 and converges to 0 all day. This is the wall from the very first chapter; trust it.
- *"Diverges means blows up."* Blowing up is one way; *never deciding* is the other. $(-1)^n$ stays politely between -1 and 1 and still diverges, no single destination.
- *"L'Hôpital on n directly."* Dots have no derivative. Step to the continuous understudy $f(x)$, run L'Hôpital there, hand the verdict back.
- *"The first terms look weird, so it diverges."* The first million terms are irrelevant; only the eventual behavior votes. Convergence is a question about the tail, meaning everything after some point, however far out.

✎ **Quick practice, before you move on**

Practice 1. $\lim_{n\to\infty} \dfrac{5n-2}{n+3}$. (Table it if you like; algebra it too; they'd better agree.)

Practice 2. $\lim_{n\to\infty} \dfrac{n}{e^n}$.

How to think about it: it's a growth race, so call the winner from the hierarchy first. Then find a way to *prove* it with a Calc 1 tool (the nudge: a continuous understudy).

Practice 3 (one sentence). Your friend says $a_n = (-1)^n \frac{1}{n}$ diverges "because it keeps flipping sign." Settle it, with a reason. (Careful: this one is different from $(-1)^n$ alone.)

Answers at the end of the chapter.

Carrying this forward

★ **Leaving this chapter with this**

A sequence is an infinite list, and the only question is whether it settles on one finite number (converges) or doesn't (diverges, by explosion *or* by indecision). You find destinations with the Calc 1 kit, top-power division, L'Hôpital via the continuous understudy, the squeeze for trapped wanderers, the growth hierarchy for races, and you keep $r^n \to 0 \iff |r| < 1$ loaded at all times, because next chapter fires it constantly. Next: we stop watching lists and start *adding* them, infinitely many numbers, one finite answer, sometimes.

✎ The chapter workout (get stuck on purpose)

Converge or diverge? Name the destination where there is one.

$$\textbf{F1. } a_n = \frac{2n^2+1}{n^2+n} \qquad \textbf{F2. } a_n = \frac{\sin n}{\sqrt{n}}$$

$$\textbf{F3. } a_n = \left(\tfrac{1}{2}\right)^n \qquad \textbf{F4. } a_n = (-1)^n \frac{n}{n+1}$$

F5 (one sentence). Your calculator's $\sqrt{2}$ routine produces guesses 1.5, 1.4167, 1.41422, ... What, in this chapter's vocabulary, is the routine checking before it stops?

Nudges: **F1** divide by the top power. **F2** bounded wanderer over a grower. **F3** which case of r^n? **F4** the size heads to 1, but watch what the sign does to the destinations. **F5** one word from this chapter.

Next up: *Series & the Divergence Test*, what it means to add infinitely many numbers, the one family you can sum exactly, and the first test every series must face.

➜ Want it on video?

Professor Leonard, *Calculus 2*, his sequences lecture (the opener of the whole series unit), at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** divide by n: $\frac{5-2/n}{1+3/n} \to \mathbf{5}$. **P2:** exponential beats power, so 0; proof: $\lim_{x\to\infty} \frac{x}{e^x} \overset{\text{L'H}}{=} \lim \frac{1}{e^x} = \mathbf{0}$. **P3:** it *converges*, to 0: the flipping sign rides on a size $\frac{1}{n}$ that dies, so both the odd and even entries head to the same place, 0 (squeeze it between $\pm\frac{1}{n}$ if you want it airtight). Flipping only causes divergence when the size *doesn't* shrink. **F1:** divide by n^2: $\to \mathbf{2}$. **F2:** squeezed between $\pm\frac{1}{\sqrt{n}}$: $\to \mathbf{0}$. **F3:** $|r| < 1$: $\to \mathbf{0}$. **F4:** sizes approach 1 but signs alternate, so even entries head to $+1$, odd entries to -1: two camps, no single destination, **diverges**. **F5:** convergence. The routine uses a proxy: it stops when successive guesses differ by less than its tolerance (a tiny cutoff like 0.000001). That proxy is only trustworthy because Newton's method is known to converge here; small steps alone don't prove a list settles, as next chapter will show you in a famous way.

Series & the Divergence Test

Calculus 2 • Sequences & Series, Chapter 2

Adding infinitely many numbers, and when that total is finite

❧ Chapter 2. The idea that shouldn't work, but does.

Here's something that should feel impossible. Add up infinitely many positive numbers, and sometimes you get a *finite* answer:

$$\tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{16} + \cdots = 1.$$

The total is exactly 1, and you can feel why: cover half the floor, then half of what's left, then half of *that*, forever, and you tile exactly one floor. Zeno used this very sum to argue motion was impossible (to cross a room you must first cross half, then half the rest, infinitely many tasks!), and the resolution of his paradox is this chapter: infinitely many steps can have a finite total.

Also, a confession on behalf of your education: you've been trusting infinite series since third grade. The decimal $0.333\ldots$ is the infinite sum $\frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \cdots$, and every time you wrote "$= \frac{1}{3}$," you summed a series to its exact value without being told. Today you learn the machinery you've been using all along.

The difference between students who get this unit and students who don't is almost always one question: do you know *what object you're looking at*? A series converges exactly when its *running totals* settle, and running totals form a sequence, last chapter's object. Keep that one thread in hand and nothing in the next four chapters can shake you off.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say why infinite sums matter (Zeno, decimals, and steady-state drug doses),
- define a series honestly, through its running totals, and watch one settle in a table,
- apply the Divergence Test as your first reflex, and know exactly what it can't tell you,
- recognize a geometric series and sum it exactly, including proving $0.999\ldots = 1$,
- collapse a telescoping series with the partial-fractions split,
- respect the harmonic series, the trap that breaks the naive rule, with numbers to prove it.

Why this exists, and what it's for

Infinite sums sound like philosophy until you notice how often reality runs a process "forever" (or long enough to count): a bouncing ball's total travel is an infinite sum of ever-shorter bounces; a repeating decimal is an infinite sum of shrinking digits; the total distance Zeno walks is an infinite sum of half-steps. Each has a finite, physical answer, so the math had better be able to produce one.

A sneaky-practical example: **medication**. Take a dose every morning while your body clears a fixed fraction of what's present each day, and the amount in your blood is a running total of fading doses, a geometric series. The "steady state" your doctor talks about is that series' sum. Same mathematics prices a perpetuity in finance (a payment that goes on forever, yet has a finite worth today) and tells an engineer the total effect of an infinitely-echoing signal.

And inside this book, the series machine is the road to the summit: power series and Taylor, where functions themselves become infinite sums.

What a series is: watch the running total

Take a sequence and start *adding* it: $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$ (the Σ is the Greek capital S, for Sum). You can't literally add forever, so, last chapter's trick, watch where the process is *heading*: form the **partial sums**, the running totals,

$$S_1 = a_1, \quad S_2 = a_1 + a_2, \quad S_3 = a_1 + a_2 + a_3, \ \ldots$$

Do it by hand right now for the pep-talk series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$, at least three entries:

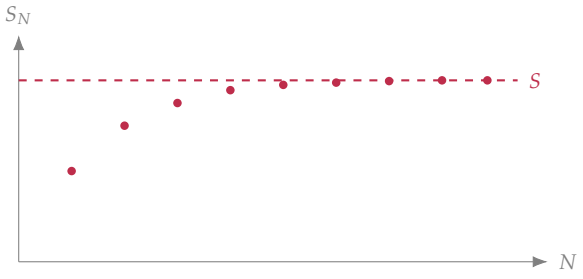
N	1	2	3	4	5	heading toward…
S_N	0.5	0.75	?	?	?	?

You'll get 0.875, 0.9375, 0.96875: a sequence of totals being herded to 1. That herding is the entire definition:

★ Convergence of a series

$\sum a_n$ **converges** to S when its partial sums do: $\lim_{N\to\infty} S_N = S$. No settling, no sum: the series **diverges**. *A series is a sequence in disguise, the sequence of its running totals.* Every series question is secretly last chapter's question asked about S_N.

Before you look: plot your five S_N values in your head, N along the bottom, total up the side. What shape *must* the dots make if the total is going to settle? Now check:



Running totals S_N climbing and settling at the sum S.

Two reads of this picture worth saying aloud. The dots always *climb* when the terms are positive (each new term adds something), so the only question is whether they level off under a ceiling or climb forever. And the gap between consecutive dots is exactly the term a_N being added, which sets up the first test: if the terms you're adding don't shrink to zero, the dots take forever-sized steps and can't possibly settle.

The first reflex: the Divergence Test

That last observation, made formal. Before any clever machinery, check the terms themselves:

★ The Divergence Test (nth-term test)

If $\lim_{n\to\infty} a_n \neq 0$ (or doesn't exist), then $\sum a_n$ **diverges**. You cannot settle a running total while adding non-vanishing chunks.

➤ One-line kill: $\sum \dfrac{n}{2n+1}$

The terms head to $\frac{1}{2}$ (divide by n), not to 0. So the running total climbs by roughly $\frac{1}{2}$ forever. **Diverges**, one line, done. This reflex should fire before any other tool, every time you meet a series.

Now the question that separates the passing grades from the top ones, and I want your vote before the next section: *is the converse true?* If the terms **do** shrink to zero, does the series have to converge? It feels like yes, shrinking steps should mean settling. Lock in your answer; the harmonic section will collect.

Geometric series: the one you can sum exactly

A **geometric** series multiplies by a fixed ratio r each step: $a + ar + ar^2 + \cdots$ (with $a \neq 0$, otherwise there's nothing to add). Guess first: what should $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$ total? Write a number down before reading on. For these series you don't just learn their fate, you get the exact sum, and you can derive it in three lines. Multiply the partial sum by r and subtract, watch everything cancel:

$$S_N = a + ar + \cdots + ar^N, \qquad rS_N = ar + ar^2 + \cdots + ar^{N+1},$$

$$S_N - rS_N = a - ar^{N+1} \quad \Longrightarrow \quad S_N = \frac{a(1 - r^{N+1})}{1 - r}.$$

Now the fact you were told to keep loaded: $r^{N+1} \to 0$ exactly when $|r| < 1$, that is, when r is strictly between -1 and 1 (multiply a number smaller than 1 in size by itself over and over and it shrinks to nothing). Let N run:

★ Geometric series

$$\sum_{n=0}^{\infty} a\,r^n = \frac{a}{1-r} \quad \text{when } |r|<1, \qquad \text{diverges when } |r|\geq 1 \ (a\neq 0).$$

In words: $\dfrac{\text{first term}}{1-\text{ratio}}$. The $|r|<1$ condition is just $r^n \to 0$ from last chapter; the terms must die for the total to settle.

➤ Two classics in one box

The pep-talk sum, certified: first term $\frac{1}{2}$, ratio $\frac{1}{2}$: $\dfrac{1/2}{1-1/2} = 1$. The floor-tiling intuition, now a theorem.

The internet's favorite argument, settled: $0.999\ldots = \frac{9}{10} + \frac{9}{100} + \cdots$ is geometric with first term $\frac{9}{10}$, ratio $\frac{1}{10}$:

$$0.999\ldots = \frac{9/10}{1-1/10} = \frac{9/10}{9/10} = 1.$$

Exactly 1, in the same way your S_N table's destination was. Anyone who fights you on this is arguing with the definition of a decimal. (And your guess for $1+\frac{1}{3}+\frac{1}{9}+\cdots$? First term 1, ratio $\frac{1}{3}$: $\frac{1}{1-1/3} = \frac{3}{2}$.)

☞ Watch out

The formula wants the *first term that actually appears*, not blindly "a." $\sum_{n=0}^{\infty} 3(\frac{1}{2})^n$ starts at 3, sum $\frac{3}{1/2} = 6$. But $\sum_{n=1}^{\infty} 3(\frac{1}{2})^n$ starts at $\frac{3}{2}$, sum $\frac{3/2}{1/2} = 3$. Always plug the starting index in to see who actually leads the parade.

Telescoping: the partial-fractions payoff

Remember the promise from the partial-fractions chapter, that its un-adding trick would collapse infinite sums? Collection time. Before the example, compute S_1, S_2, S_3 for $\frac{1}{n(n+1)}$ by hand: $\frac{1}{2}$, then $\frac{1}{2}+\frac{1}{6}$, then add $\frac{1}{12}$. See a pattern in the totals?

➤ **Telescoping:** $\sum_{n=1}^{\infty} \dfrac{1}{n(n+1)}$

Split the term with partial fractions: write $\frac{1}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1}$, clear denominators to get $1 = A(n+1) + Bn$, and plug in the roots: $n = 0$ gives $A = 1$, $n = -1$ gives $B = -1$. So

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}.$$

Now write out a partial sum and watch the interior annihilate itself:

$$S_N = \left(1 - \tfrac{1}{2}\right) + \left(\tfrac{1}{2} - \tfrac{1}{3}\right) + \left(\tfrac{1}{3} - \tfrac{1}{4}\right) + \cdots + \left(\tfrac{1}{N} - \tfrac{1}{N+1}\right) = 1 - \frac{1}{N+1}.$$

Every middle piece cancels its neighbor, like a collapsing telescope, only the first and last survive. And $1 - \frac{1}{N+1} \to \mathbf{1}$: the series converges to 1, *exactly*, computed by hand. Telescoping and geometric are the only two families whose exact sums you'll find in this unit; until Taylor series arrive, everything else settles for a verdict.

The trap collects: the harmonic series

Time to grade the vote from the Divergence Test section. The test case is the most innocent-looking series alive, the **harmonic series**:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \cdots$$

The terms certainly die: $\frac{1}{n} \to 0$. If your vote was "terms die, so it converges," here's the table that collects the debt, running totals, computed:

N	10	1,000	10^6	heading toward…
S_N	2.93	7.49	14.39	∞, at a crawl

No ceiling. The totals *trudge* upward forever, gaining about 7 per factor of a thousand, slow beyond belief and utterly unstoppable. If this trudge feels familiar, it should: it's the same gait as $\int_1^{\infty} \frac{dx}{x}$ from the improper chapter, the curve whose area out to t grew like $\ln t$, slowly and without limit. Same shape, same fate, and in the next chapter that resemblance gets promoted to an actual theorem.

So the verdict stands, and burn it in: **terms shrinking to zero is necessary (without it, no hope) but nowhere near sufficient (with it, no guarantee).** The Divergence Test can only convict (terms don't die: diverges). It can never acquit. When terms do die, you know *nothing yet*, and the entire next chapter exists to finish the job.

☞ Watch out

The wrong ideas that wreck series problems, named and retired:

- *"The terms go to 0, so it converges."* The harmonic series is the eternal counterexample, dying terms, infinite total. This is the single most-penalized sentence in Calc 2. Never write it.
- *"Sequence, series, same thing."* A sequence is a *list*; a series is a list being *added*. The sequence $\frac{1}{n}$ converges (to 0); the series $\sum \frac{1}{n}$ diverges. Both statements are true at once, about different objects.
- *"The sum is where the* terms *settle."* The sum is where the *running totals* settle. Terms heading to 0 just means the staircase's steps shrink, not where the staircase tops out.
- *"Geometric formula, plug in a from the formula."* Plug in the starting index first; the leader of the parade is the a. You watched 6 become 3 over exactly this.

✎ Quick practice, before you move on

Practice 1. $\displaystyle\sum_{n=0}^{\infty} 5\left(\frac{1}{10}\right)^n$. (And once you have it: which repeating decimal did you just sum? Look at the digits.)

Practice 2. $\displaystyle\sum_{n=1}^{\infty} \frac{3n^2}{n^2+1}$.

How to think about it: first reflex, always: where are the *terms* heading?

Practice 3. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$.

How to think about it: partial-fraction the term, write out S_N, watch the telescope collapse, keep the two survivors.

Answers at the end of the chapter.

Carrying this forward

★ Leaving this chapter with this

A series converges when its *running totals* settle; the terms and the totals are different objects and only the totals vote. First reflex forever: the Divergence Test, terms that don't die convict instantly, but terms that do die prove nothing, the harmonic series trudges to infinity on dying terms. Two families pay out exact sums: geometric ($\frac{\text{first term}}{1-r}$, $|r| < 1$, derived by the multiply-and-subtract cancellation) and telescoping (partial fractions, interior annihilates). Everything else needs a court system for verdicts without sums, and that court convenes next chapter, with the improper-integral razor's edge as its first judge.

✎ The chapter workout (get stuck on purpose)

Verdict on each; exact sum where one exists.

F1. $\displaystyle\sum_{n=0}^{\infty} \left(\tfrac{2}{3}\right)^n$ **F2.** $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n+100}$

F3. $\displaystyle\sum_{n=1}^{\infty} 3\left(\tfrac{1}{2}\right)^n$ **F4.** $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$

F5 (the dosing problem). You take 10 mg of a medication each morning, and your body clears half of whatever is present each day. In the long run, the amount in your blood right after a dose heads toward $10 + 10(\frac{1}{2}) + 10(\frac{1}{2})^2 + \cdots$ What's the steady-state level?

Nudges: **F1** geometric, who leads? **F2** first reflex: the terms. **F3** starting index! **F4** already split for you; write S_N. **F5** geometric, and sanity-check: should the answer be more than one dose? Than two?

Next up: *Integral Test, p-Series & Comparison*, how to judge a series you can't sum, by comparing it to an integral you already understand or a neighbor you've already judged.

➜ Want it on video?

Professor Leonard, *Calculus 2*, his lectures on infinite series and geometric series, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** first term 5, ratio $\frac{1}{10}$: $\frac{5}{9/10} = \frac{50}{9} = 5.555\ldots$, which is the repeating decimal $5.\overline{5}$, you just summed $5 + 0.5 + 0.05 + \cdots$ **P2:** terms $\to 3 \neq 0$: **diverges** by the Divergence Test, one line. **P3:** $\frac{1}{n+1} - \frac{1}{n+2}$; $S_N = \frac{1}{2} - \frac{1}{N+2} \to \frac{1}{2}$: **converges to** $\frac{1}{2}$. **F1:** first term 1, ratio $\frac{2}{3}$: $\frac{1}{1/3} = 3$. **F2:** terms $\to 1 \neq 0$: diverges. **F3:** starts at $n = 1$, first term $\frac{3}{2}$: $\frac{3/2}{1/2} = 3$. **F4:** $S_N = 1 - \frac{1}{N+1} \to 1$. **F5:** geometric, first term 10, ratio $\frac{1}{2}$: $\frac{10}{1/2} = 20$ mg, more than one dose (yesterday's leftovers stack) but it levels off at exactly two doses' worth, which is why your doctor can promise a stable level instead of an ever-growing one. Convergence, prescribed daily.

Integral Test, p-Series & Comparison

Calculus 2 • Sequences & Series, Chapter 3

Judging a series you can't sum, by leaning on ones you understand

❧ Chapter 3. You don't have to sum it to understand it.

Last chapter ended with two aristocrat families, geometric and telescoping, that hand you exact sums, and a cliffhanger: for nearly every other series on Earth, *nobody* can write the exact sum. Not your professor, not anyone. So we lower the question to the one that actually matters: finite or not? Does the running total settle somewhere, or trudge off to infinity?

The move that makes this answerable is one you use in real life daily: **judge the unknown by comparing it to a known**. You can't add up a series with a stubborn $+1$ in its denominator. But it sits just barely under one you *can* judge, and if the bigger one stays finite, the smaller one is trapped under its ceiling with no choice in the matter. The whole chapter is that instinct, made rigorous.

What makes it work is having trustworthy things to compare *against*, and you already own the crown jewel: the improper-integral result that $\int_1^\infty \frac{dx}{x^p}$ converges exactly when $p > 1$, the razor's edge, with $p = 1$ sitting right on the blade. Remember the harmonic totals trudging upward at exactly the pace of $\int \frac{dx}{x}$? That resemblance was a loaded gun on the wall. This chapter fires it.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why verdict-without-a-sum is the right goal, and where it's used,
- see *why* a series and its matching integral share a fate (one picture does it),
- wield the p-series rule, inherited straight from the improper chapter's razor edge,
- trap a series under a convergent ceiling, or over a divergent floor (Direct Comparison),
- match a messy series to its look-alike with the Limit Comparison Test,
- avoid the two comparison directions that prove exactly nothing.

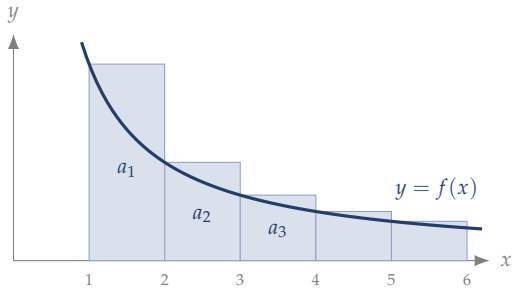
Why this exists, and what it's for

"Does it converge?" with no hope of an exact sum sounds like settling for less. It's actually the professional question. When a physicist expands a force into an infinite series of corrections, the first ask is whether the corrections *pile up finitely*, because if not, adding more terms stops improving the answer and starts wrecking it. When your numerics library sums a series to approximate a function (next unit's whole story), it needs a guarantee the series settles before it trusts any partial sum. And a concrete cautionary tale: stack identical blocks at the edge of a table, each one overhanging the one below by as much as it can without tipping. The overhangs are $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \ldots$, half the harmonic series, so with enough blocks the top one can stick out past the table by *any* distance you like. Nobody believes that until they've seen the series diverge. Knowing *finite or not* is the load-bearing fact; the decimal value, when you need it, comes from a computer adding terms, *after* convergence is certified.

So this chapter builds the court system: reference families whose fates are beyond doubt, and two ways to hold a stranger up against them.

The Integral Test: the trudge becomes a theorem

Last chapter you watched two trudges side by side: the harmonic partial sums after 10, 1000, and a million terms (2.93, 7.49, 14.39) and, a unit earlier, the area under $\frac{1}{x}$ out to those same three distances (2.3, 6.9, 13.8). Same gait, no coincidence, and here's the picture that makes it a theorem. Question to carry into it: the series' terms are drawn as rectangles of width 1, the first one standing on $[1, 2]$ with height a_1; what does the decreasing curve do to them, and what would happen if you slid every rectangle one unit to the left?



The series (rectangles) and the integral (area under the curve), interlocked.

Read it closely, this is the proof. Each rectangle has width 1 and height $a_n = f(n)$, so its area *is* the term a_n, and the stack of rectangles *is* the series. Now look at how the decreasing curve threads them. As drawn, each rectangle's top-left corner touches the curve and the rest of it pokes *over*, so the rectangles together hold *more* area than the curve: series $\geq$ integral. Now slide every rectangle one unit to the *left*. The rectangle of height a_2 now stands on $[1, 2]$, where the curve is at least a_2 high the whole way, so it sits *under* the curve; same for a_3, a_4, and on. Only a_1, which slid off the picture, is unaccounted for. So: $a_2 + a_3 + \cdots \leq$ integral. Put the two together and the series is pinned: integral $\leq$ series $\leq a_1 +$ integral. If the integral's area is finite, the series is trapped below a finite ceiling; if the integral diverges, the series is propping it up from above and diverges too. They live or die together.

★ The Integral Test

If $a_n = f(n)$ for a positive, continuous, *decreasing* f (from some point on), then

$$\sum_{n=N}^{\infty} a_n \quad \text{and} \quad \int_N^{\infty} f(x)\,dx \quad \text{share a fate: both converge or both diverge.}$$

Fate, not value: the integral's number is *not* the series' sum (the rectangles and the area differ by a bounded gap). The integral is the judge, never the accountant.

➤ Integral Test: $\sum_{n=2}^{\infty} \dfrac{1}{n \ln n}$

(Why start at $n = 2$? At $n = 1$, $\ln 1 = 0$ sits in the denominator, undefined. Dropping finitely many terms never changes a convergence verdict, only the total.) The function $f(x) = \frac{1}{x \ln x}$ is positive, continuous, decreasing for $x \geq 2$, so the test applies. Improper integrals are limits, write it that way. The substitution $u = \ln x$, $du = \frac{dx}{x}$ turns the integrand into $\frac{du}{u}$, whose antiderivative is $\ln|u| = \ln(\ln x)$:

$$\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{t \to \infty} \Big[\ln(\ln x) \Big]_2^t = \lim_{t \to \infty} \big(\ln(\ln t) - \ln(\ln 2) \big) = \infty.$$

The integral diverges, so the series **diverges**. Savor the slowness: $\ln(\ln t)$ reaches a measly 3 only around $t \approx 5 \times 10^8$. This series diverges so slowly that no computer could ever catch it in the act, and the Divergence Test is blind to it ($\frac{1}{n \ln n} \to 0$). Only the theorem sees it.

The p-series: the razor edge, inherited

Point the Integral Test at the whole family $\sum \frac{1}{n^p}$ (for $p > 0$, where the terms shrink; for $p \leq 0$ the terms don't even die, and the Divergence Test from last chapter, terms not heading to 0 means divergence, convicts on the spot) and it defers, instantly, to a judge you already know: $\int_1^{\infty} \frac{dx}{x^p}$, the improper chapter's razor's edge. Before reading the box, write the verdicts yourself; you memorized the integral's cutoff two units ago, and the series simply inherits it.

★ The p-series rule

$$\sum_{n=1}^{\infty} \frac{1}{n^p}: \quad \textbf{converges if } p > 1, \qquad \textbf{diverges if } p \leq 1.$$

Straight from the p-integral: when the terms shrink slowly ($p \leq 1$, a "fat tail") the totals trudge to infinity; when they shrink fast ($p > 1$, a "thin tail") the totals settle. The boundary case $p = 1$ is the harmonic series, divergent, as its table of partial sums already testified. This family is your reference shelf; from here on you'll judge strangers by holding them against $\frac{1}{n^p}$.

Instant reads, say each before confirming: $\sum \frac{1}{n^2}$? Converges, $p = 2$. $\sum \frac{1}{\sqrt{n}}$? That's $p = \frac{1}{2} \leq 1$: diverges, even though its terms die, dying slowly is the whole harmonic disease, and $\frac{1}{\sqrt{n}}$ dies even slower than $\frac{1}{n}$.

Direct Comparison: ceilings and floors

Now the courtroom's first procedure. Suppose every term of your series sits *below* the matching term of a series you know converges. Then your running totals sit below a finite ceiling, and (being positive-term totals, always climbing) a climbing staircase under a ceiling must settle, that's the Monotonic Sequence Theorem from the sequences chapter (always-up plus a ceiling forces convergence), earning its keep. Before the box, decide for yourself: every term of $\sum \frac{1}{n^2+1}$ is smaller than the matching term of $\sum \frac{1}{n^2}$. Does that force a verdict? Which one? Mirror image: sit *above* a divergent series and you're dragged up to infinity with it.

★ Direct Comparison Test (for positive terms)

Suppose $0 \le a_n \le b_n$ from some point on.

- $\sum b_n$ converges $\Rightarrow \sum a_n$ converges (trapped under a finite ceiling).
- $\sum a_n$ diverges $\Rightarrow \sum b_n$ diverges (propped above an infinite floor).

And name the two *useless* directions out loud: being *smaller than a divergent* series proves nothing (plenty of room under infinity), and being *bigger than a convergent* one proves nothing (plenty of room above any ceiling). Half of all comparison mistakes are these two.

➤ Direct Comparison: $\sum \frac{1}{n^2+1}$

The $+1$ downstairs makes each term *smaller* than $\frac{1}{n^2}$:

$$\frac{1}{n^2+1} < \frac{1}{n^2},$$

and $\sum \frac{1}{n^2}$ converges ($p = 2$, reference shelf). Smaller than a convergent ceiling: **converges**. Three lines, and notice the courtroom logic: we never learned the sum, only the verdict.

Limit Comparison: match the stranger to its look-alike

Direct comparison needs a clean inequality, and algebra doesn't always cooperate (for $n \ge 2$, $\frac{1}{n^2-1}$ is slightly *bigger* than $\frac{1}{n^2}$, whoops, wrong direction). The friendlier tool asks only: *which reference does this series resemble at large n?* Strip your term to its dominant powers, the limits-chapter move where the biggest power swamps the rest, and verify the resemblance with one limit.

★ Limit Comparison Test (for positive terms)

If $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ with $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ share a fate. Finite positive L means the two series are, eventually, steady multiples of each other, and scaling by a constant never changes finite-versus-infinite. Pick b_n by keeping only each part's biggest power. (One-sided refinements exist for $L = 0$ and $L = \infty$, but the finite-positive case is the workhorse.)

➤ Limit Comparison: $\sum \frac{2n+1}{n^3+n}$

Strip to dominant powers yourself first: top behaves like $2n$, bottom like n^3, so the term behaves like $\frac{2n}{n^3} = \frac{2}{n^2}$, smelling strongly of the convergent $p = 2$ shelf. Certify the smell with $b_n = \frac{1}{n^2}$:

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{(2n+1)\,n^2}{n^3+n} = \lim_{n\to\infty} \frac{2n^3+n^2}{n^3+n} = 2.$$

Finite and positive, so the stranger shares the fate of $\sum \frac{1}{n^2}$: **converges**. (The limit being 2 rather than 1 is irrelevant, any finite positive number seals the match.)

☞ Watch out

The wrong ideas that wreck comparison problems, named and retired:

- *"It's smaller than a divergent series, so..."* So nothing. Under infinity there's room for anything. Only smaller-than-convergent and bigger-than-divergent carry verdicts.
- *"The integral equals the sum."* The integral shares the *fate*, never the *value*; the rectangles and the curve differ by a real gap. Judge, not accountant.
- *"My limit came out 2, not 1, so limit comparison failed."* Any finite positive L is a success; 1 is not special.
- *"Terms go to 0 and it's a fraction of polynomials, so it converges."* The verdict rides on the *degree gap*: $\frac{n}{n^2+1}$ behaves like $\frac{1}{n}$ and diverges. Strip to dominant powers before trusting a smell.

✎ Quick practice, before you move on

Practice 1. Verdicts, straight off the shelf: (i) $\sum \frac{1}{n^3}$ (ii) $\sum \frac{1}{n^{0.5}}$.

Practice 2. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2 - n + 1}$.

How to think about it: dominant powers say it resembles $\frac{1}{n^2}$; the $-n$ makes a direct inequality awkward (wrong direction), which is exactly the situation Limit Comparison was built for.

Practice 3 (one sentence). Your friend proves $\frac{1}{n!} \leq \frac{1}{n}$ for $n \geq 2$ (here $n! = 1 \cdot 2 \cdots n$) and announces $\sum \frac{1}{n!}$ diverges "by comparison with the harmonic series." Name the exact error.

Answers at the end of the chapter.

Carrying this forward

★ Leaving this chapter with this

Most series never surrender a sum, so the professional question is the verdict, and verdicts come from comparison. The Integral Test interlocks a series with its curve (rectangles and area share a fate, never a value), and hands you the p-series shelf: converges exactly when $p > 1$, the razor's edge inherited whole from improper integrals. Hold strangers against the shelf: trap them directly (smaller than convergent, or bigger than divergent, the only two directions that prove anything), or match them to their dominant-power look-alike with Limit Comparison. Next chapter the terms start flipping signs, cancellation enters the story, and two power tools arrive, ratio and root, that will carry you all the way through Taylor series.

✎ The chapter workout (get stuck on purpose)

Verdict and the test that delivered it.

F1. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^4}$ **F2.** $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+1}$

F3. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3+5}$ **F4.** $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ (which tool, and why not the others?)

F5 (one sentence). Why does the Integral Test demand a *decreasing* function? What does the interlocking-rectangles picture lose if the curve can jump back up?

Nudges: **F1** shelf. **F2** dominant powers first, then the right comparison, and mind the verdict's direction. **F3** a clean ceiling is sitting right there. **F4** Divergence Test is blind here, and no p-series matches; what's left? **F5** think about what made the rectangles thread the curve.

Next up: *Alternating Series, Ratio & Root Tests*, sign-flipping sums that converge on cancellation, the two tests that crack factorials and powers, and a strategy for choosing among all of them.

➜ Want it on video?

Professor Leonard, *Calculus 2*, his lectures on the Integral Test, p-series, and the Comparison Tests, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** (i) $p = 3 > 1$, converges; (ii) $p = 0.5 \leq 1$, diverges. **P2:** $\lim \frac{1/(n^2-n+1)}{1/n^2} = \lim \frac{n^2}{n^2-n+1} = 1$, finite positive, so it matches convergent $\sum \frac{1}{n^2}$: **converges**. **P3:** they compared in a useless direction: being *smaller* than a divergent series proves nothing (and in fact $\sum \frac{1}{n!}$ converges, it's hiding under the convergent ceiling $\sum \frac{1}{2^{n-1}}$, or wait one chapter and the Ratio Test ends it in one line). **F1:** shelf, $p = 4$: converges. **F2:** behaves like $\frac{1}{n}$; limit comparison with the harmonic gives $L = 1$: **diverges**. **F3:** $\frac{1}{n^3+5} < \frac{1}{n^3}$, convergent ceiling ($p = 3$): converges. **F4:** Integral Test, because the terms die (Divergence Test blind) and $\frac{1}{n \ln n}$ matches no p-series (it sits between $p = 1$ and every $p > 1$), but its continuous twin integrates cleanly to $\ln(\ln x)$: diverges. **F5:** decreasing is what lets the same rectangles sit under the curve on one side and over it on the other; a curve that could jump back up could dodge the rectangles entirely, and the squeeze, and the shared fate, would collapse.

Alternating Series, Ratio & Root Tests

Calculus 2 • Sequences & Series, Chapter 4

Sign-flipping sums, the two power tools, and how to choose a test

❧ Chapter 4. The toolkit gets finished, and then organized.

Two stories close out the convergence saga, and both are satisfying in different ways.

First, a plot twist: when a series' terms *flip sign*, plus, minus, plus, minus, the series gets help from an unexpected ally, **cancellation**. Each negative term undoes part of the positive one before it, so the running total stops climbing and starts *closing in from both sides*. That's why $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges even though the same sizes with all plus signs, the harmonic series, famously don't. The flipping alone rescues it, and in a few pages you'll see exactly why.

Second, the heavy machinery arrives: the *Ratio* and *Root* tests, which ask one brutal, simple question, "is this series eventually behaving like a geometric one?", and in exchange they chew through the factorials and nth powers that stop every other test cold. They're also, quietly, the engine of the entire next unit, so the reps you put in here pay double. And because you'll then own *seven* tests, this chapter ends with the thing students actually beg for: a decision list for which test to grab, in what order. That judgment is the real finish line of the unit, and it's the skill I most wish someone had handed me before my own exam.

✓ **In this chapter**

Here's what you'll be able to do by the end, in plain words:

- say why alternating series matter (your pendulum, your bank, and your error bars),
- see *why* flipping signs rescues convergence, with a picture of totals closing in,
- apply the Alternating Series Test, and tell absolute from conditional convergence,
- run the Ratio Test on factorials and exponentials, and know *why* it works,
- run the Root Test when the whole term is an nth power,
- choose the right test on sight, with a strategy list you'll actually use.

Why this exists, and what it's for

Sign-flipping sums aren't exotic; they're what happens whenever contributions *alternate direction*. A pendulum overshoots left, then right, each swing shorter than the last: each swing's displacement is a term, the signs alternate, and the running total is where the bob actually is. A bank account that takes a deposit one month and a slightly smaller withdrawal the next, forever, with the amounts shrinking, is an alternating series, and its long-run balance is the sum. In numerical computing, many useful approximations alternate, and alternation hands you a free gift most series don't offer: the error after you stop adding is smaller than the first term you dropped, which is how a program can know when it's safe to stop. (That gift gets cashed properly in the Taylor chapter.)

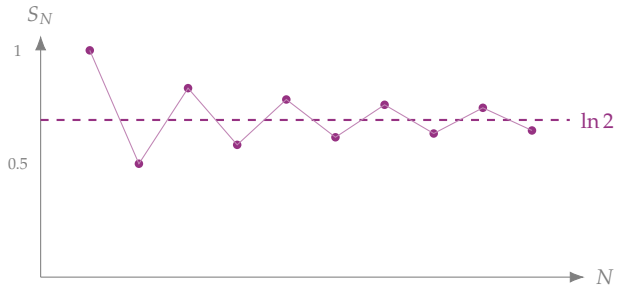
The ratio and root tests have a different clientele: series with $n!$ (factorials) or c^n (a fixed number c raised to the nth power) in them, which are exactly the series the next unit is made of. When a program sums a Taylor series for sin, somebody first certified convergence with the Ratio Test. This chapter is that certification office.

Flipping signs: watch the totals close in

Take the **alternating harmonic series**, $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$, which is $\sum \frac{(-1)^{n+1}}{n}$; the factor $(-1)^{n+1}$ is the sign switch, $+1$ when n is odd and -1 when n is even. Do the ritual: running totals S_N (the sum of the first N terms), by hand, at least four of them, *before* you look at mine. Then check:

N	1	2	3	4	5	6
S_N	1	0.5	0.833	0.583	0.783	0.617

Look at the *shape* of that: up, down, up, down, but each swing **shorter** than the last. The totals aren't climbing a staircase anymore, they're a pendulum losing energy, bouncing between an upper camp and a lower camp that squeeze toward each other. Whatever's trapped between closing walls converges (squeeze logic, old friend). Here's that table as a picture. Before you look: where should the dots land, above 0.5, below 1, and closing in on what?



Swings shrink; the totals trap $\ln 2$ *between them.*

Walk it: the first total is 1, the second drops to 0.5, the third climbs back but not as high, the fourth drops but not as low. Every odd total is an upper fence, every even total is a lower fence, and the fences march toward each other. The dashed line is the number they're trapping, $\ln 2$, and you can see the sum will never escape the gap between the latest two totals. That last observation is the free gift from the first section: stop anywhere, and the true sum is within one more term of where you stopped. The walls close because each new term is *smaller* than the one before and the terms die out, and that's the entire theorem:

★ Alternating Series Test

An alternating series $\sum(-1)^n b_n$ (sizes $b_n > 0$) **converges** if the sizes are *decreasing* and $b_n \to 0$. Shrinking swings, dying out: the pendulum settles. (And the settling point is always between consecutive totals, which is why the leftover error is at most the next term, the free gift.)

So the alternating harmonic series converges (sizes $\frac{1}{n}$ decrease to 0); its limit happens to be $\ln 2 \approx 0.693$, and squint at the table, the totals are visibly circling it. Meanwhile the plain harmonic series, same sizes, no flips, trudges to infinity. Cancellation did real, measurable work.

Absolute vs. conditional: how sturdy is the convergence?

That contrast begs a classification. Some series converge so robustly that even stripping away their minus signs leaves them convergent; others survive *only because* of the cancellation, strip the signs and they blow up.

★ Absolute vs. conditional

- $\sum |a_n|$ converges: the series is **absolutely convergent**, the sturdy kind (absolute convergence automatically implies plain convergence).
- $\sum a_n$ converges but $\sum |a_n|$ diverges: **conditionally convergent**, alive purely on cancellation.

Test in words: *would it still converge if every term pulled the same direction?*

The poster children: $\sum \frac{(-1)^{n+1}}{n}$ is *conditional* (strip signs, get the harmonic, divergence), while $\sum \frac{(-1)^n}{n^2}$ is *absolute* (strip signs, get $p = 2$, still fine). The distinction isn't bookkeeping. Conditionally convergent series are genuinely fragile: rearrange their terms and you can change the sum, a famous piece of mathematical lawlessness (Riemann proved you can rearrange the alternating harmonic to total *any number you like*). Absolute convergence is the certificate that your series behaves like finite addition. Conditional is "converges, handle with care."

The Ratio Test: is it secretly geometric?

Here's the thinking that produces the first power tool, and it starts from the series you understand best: geometric. What made $\sum ar^n$ converge for $|r| < 1$? Each term was a fixed fraction r of the one before. Now take *any* series and ask: what fraction of each term is the next one? Try it before the box: for $\sum \frac{2^n}{n!}$, compute $\frac{a_2}{a_1}$, $\frac{a_3}{a_2}$, $\frac{a_4}{a_3}$ (the terms are $2, 2, \frac{4}{3}, \frac{2}{3}, \ldots$). What's the trend in those ratios? If the ratio *settles toward* some $L < 1$, then from some point on the series is shrinking at least as fast as a geometric series with some ratio r between L and 1, and gets trapped under a convergent geometric ceiling (comparison, last chapter). If $L > 1$ (or the ratio runs off to infinity), the terms eventually *grow*, and the Divergence Test finishes it.

★ Ratio Test

Let $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$. Then:

$$L < 1 \Rightarrow \textbf{converges absolutely}, \qquad L > 1 \text{ or } L = \infty \Rightarrow \textbf{diverges},$$

$$L = 1 \Rightarrow \textbf{no verdict}.$$

$L = 1$ really is silence, not a hint: the harmonic ($L = 1$, diverges) and $p = 2$ ($L = 1$, converges) both land there. When ratio says 1, go back to last chapter's tools. Reach for this test on sight of $n!$ or c^n, whose ratios cancel beautifully.

➤ Ratio Test: $\sum \frac{2^n}{n!}$

Build the ratio and watch the stacked factorials collapse ($\frac{n!}{(n+1)!} = \frac{1}{n+1}$, everything cancels but the newest factor):

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1} \longrightarrow 0.$$

$L = 0 < 1$: **converges absolutely**. And read what the ratio is *saying*: each term is $\frac{2}{n+1}$ of the previous, a shrinkage that gets more savage every step. Factorials outgrow exponentials, the growth hierarchy from sequences, now weaponized. (Spoiler for two chapters from now: started at $n = 0$, this series sums to e^2.)

The Root Test: for wall-to-wall nth powers

Same geometric soul, different grip. When the *entire term* is something to the nth power, taking the nth root peels the exponent off in one motion. Try it first: the fifth term of $\sum \left(\frac{n}{2n+1}\right)^n$ is $\left(\frac{5}{11}\right)^5$. What's its fifth root? That's the whole trick:

★ Root Test

Let $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$. Same verdicts as Ratio: $L < 1$ converges absolutely, $L > 1$ (or $L = \infty$) diverges, $L = 1$ silence. Grab it when the whole term wears an exponent n.

➤ Root Test: $\sum \left(\frac{n}{2n+1}\right)^n$

The whole term is an nth power, so take the root:

$$\sqrt[n]{\left(\frac{n}{2n+1}\right)^n} = \frac{n}{2n+1} \longrightarrow \frac{1}{2}.$$

$L = \frac{1}{2} < 1$: **converges absolutely**. (Try the Ratio Test on this one some rainy afternoon and you'll understand why the Root Test exists.)

The strategy: seven tests, one glance

You now own seven tests: Divergence, Integral, Direct Comparison, Limit Comparison, Alternating Series, Ratio, Root (plus two reference shelves, geometric and p-series, that you compare against rather than test). The real exam skill isn't running a test, it's *choosing* one fast. Run down this list in order; it resolves almost everything you'll meet.

★ How to pick a test (top to bottom)

1. **Terms don't die?** Divergence Test, guilty, done. (Always check this first; it's free.)
2. **Geometric or telescoping?** Don't test, *sum it exactly.*
3. **Ratio of polynomials (or powers/roots of n)?** Strip to dominant powers, then compare against the p-series shelf (the facts $\sum \frac{1}{n^p}$ converges exactly when $p > 1$) via Comparison or Limit Comparison.
4. **Contains $n!$ or c^n?** Ratio Test. **Whole term raised to the nth?** Root Test.
5. **Signs alternating?** Alternating Series Test on the sizes; then ask absolute-or-conditional by stripping the signs and re-running this list.
6. **Looks like a function you could integrate?** Integral Test, the specialist for oddballs like $\frac{1}{n \ln n}$.

☞ Watch out

The wrong ideas that wreck test-choosing, named and retired:

- *"Ratio gave $L = 1$, so it converges."* $L = 1$ is the test shrugging. No verdict, switch tools. (Every p-series lands ratio at exactly 1, which is why comparison exists.)
- *"It alternates, so it converges."* The Alternating Series Test still demands sizes decreasing *to zero*. $\sum(-1)^n \frac{n}{n+1}$ alternates and diverges; its sizes head to 1, so the Divergence Test, step one of the list, convicts it.
- *"Absolutely convergent, conditionally convergent, whatever."* The difference is load-bearing: conditional series can change their sum under rearrangement; absolute ones behave. Exams and real analysis both care.
- *"I'll just try tests at random."* The list exists because order matters: the free checks (terms dying? exactly summable?) come first, the heavy machinery second, the specialist last.

✎ **Quick practice, before you move on**

Practice 1. $\sum \frac{(-1)^n}{n^3}$: converge? Absolutely or conditionally?

Practice 2. $\sum \frac{3^n}{n!}$.

How to think about it: an exponential and a factorial in one term, that's Ratio Test turf by the list. Build $\left|\frac{a_{n+1}}{a_n}\right|$, cancel, limit.

Practice 3 (one sentence). For the alternating harmonic series, you stop adding after 100 terms. The free gift says your error is below what number?

Answers at the end of the chapter.

Carrying this forward

★ **Leaving this chapter with this**

Flipping signs buys cancellation: if the sizes shrink to zero, the totals swing in ever-shorter arcs and settle (Alternating Series Test), with leftover error below the first dropped term. Convergence comes in two grades, absolute (survives de-signing; sturdy) and conditional (alive on cancellation alone; rearrangeable, fragile). The Ratio and Root tests ask "eventually geometric?": $L < 1$ converges absolutely, $L > 1$ diverges, $L = 1$ is silence, and factorials and nth powers are their natural prey. Then the strategy list turns seven tests into one glance. That completes the convergence court system. Next unit, the payoff for building it: series with a *variable* inside, infinite polynomials, and the question of which x they're allowed to swallow, which the Ratio Test answers almost single-handedly.

✎ The chapter workout (get stuck on purpose)

Verdict, test used, and absolute-vs-conditional where signs flip.

$$\textbf{F1.}\ \sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt{n}} \qquad \textbf{F2.}\ \sum_{n=1}^{\infty}\frac{n!}{2^n}$$

$$\textbf{F3.}\ \sum_{n=2}^{\infty}\left(\frac{1}{\ln n}\right)^n \qquad \textbf{F4.}\ \sum_{n=1}^{\infty}(-1)^n\frac{2n+1}{3n+2}$$

F5 (strategy drill, name the tool only, no computing). (i) $\sum \frac{n^2+3}{n^5+n}$ (ii) $\sum \frac{5^n}{n!}$ (iii) $\sum \left(\frac{3n}{4n+1}\right)^n$ (iv) $\sum_{n=2}^{\infty}\frac{(-1)^n}{n\ln n}$.

Nudges: **F1** alternating test, then strip the signs and check the shelf. **F2** who wins, factorial or exponential? The ratio knows. **F3** wall-to-wall nth power. **F4** list, step one. **F5** walk the list in order, out loud.

Next up (new unit): *Power Series*, infinite polynomials with an x riding inside, and the Ratio Test's finest hour.

➜ Want it on video?

Professor Leonard, *Calculus 2*, his lectures on the Alternating Series Test, absolute convergence, and the Ratio and Root Tests, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** sizes $\frac{1}{n^3}$ decrease to 0, converges; stripped, $\sum \frac{1}{n^3}$ converges ($p = 3$), so **absolutely convergent**. **P2:** $\left|\frac{a_{n+1}}{a_n}\right| = \frac{3}{n+1} \to 0 < 1$: converges (absolutely). **P3:** below the first dropped term, $b_{101} = \frac{1}{101} < 0.01$, you know $\ln 2$ to within a penny, so to speak, without summing anything further. **F1:** alternating with sizes $\frac{1}{\sqrt{n}} \downarrow 0$: converges; stripped it's $p = \frac{1}{2}$, divergent, so **conditionally convergent**. **F2:** ratio $= \frac{n+1}{2} \to \infty > 1$: **diverges** (factorial crushes exponential; the terms themselves blow up). **F3:** root $= \frac{1}{\ln n} \to 0 < 1$: converges absolutely. **F4:** sizes $\frac{2n+1}{3n+2}$ head to $\frac{2}{3}$, not 0: **diverges** by the Divergence Test, step one, no alternating analysis needed. **F5:** (i) dominant powers $\frac{1}{n^3}$, limit comparison with the shelf; (ii) Ratio Test; (iii) Root Test; (iv) Alternating Series Test (and for the absolute question, the Integral Test on $\frac{1}{n \ln n}$, which you know diverges, verdict: conditional).

Power Series

Calculus 2 • Power Series & Taylor, Chapter 1

Infinite polynomials, and the inputs they're allowed to swallow

❧ New unit. The summit is in sight.

Polynomials are the friendliest functions alive. Add them, differentiate them, integrate them, evaluate them with nothing but multiplication and addition, a machine can do it. Now picture a polynomial that *never ends*: a constant, plus a multiple of x, plus a multiple of x^2, plus a multiple of x^3, and so on forever. That's a **power series**, and here's the payoff waiting one chapter away: functions like e^x, $\sin x$, and $\ln(1+x)$, which you can't compute by hand, can be rewritten *exactly* as one of these. Hard-to-compute objects, reborn as endless arithmetic. That's the reason a chip with no notion of "sine" can still produce $\sin(1)$ to fifteen digits.

But an infinite polynomial comes with a catch, and the catch is this chapter. Feed it an x that's too big and the terms grow instead of shrink, the sum explodes, and the whole expression means nothing. So every power series carries a *range of legal inputs*, and finding that range is today's single skill.

The best part: you already own the tool. Last chapter's Ratio Test, pointed at terms with an x riding inside, hands you the answer almost mechanically. Everything in this unit has been quietly loading for four chapters. Time to fire it.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why infinite polynomials are worth building (your calculator is Exhibit A),
- read a power series in sigma notation and identify its center,
- find the *radius* of convergence with the Ratio Test,
- nail the full *interval*, including why the endpoints must be checked by hand,
- recognize the geometric series as the prototype power series,
- meet the three shapes an interval can take: a point, a stretch, or the whole line.

Why this exists, and what it's for

Ask the awkward question: *how does anything actually compute* $\sin(1)$*?* A processor can add, subtract, multiply, divide. It cannot "take a sine." Yet your calculator returns $0.8414709848\ldots$ instantly. The resolution is this unit: $\sin x$ equals an infinite polynomial, and a polynomial is pure arithmetic. Add a handful of terms, stop when the next one is below your precision, done. Most trig values, logarithms, and exponentials your machine has ever shown you came out of a polynomial approximation, a power series or a tuned cousin of one.

The uses run deeper than evaluation. Physics tames hard laws by keeping only the first term or two of a power series (that's where "for small angles, $\sin\theta \approx \theta$" comes from, it's the first term). Engineers replace brutal functions with polynomial stand-ins to make equations solvable. And integrals that have no antiderivative you can write with the functions you know, like $\int e^{-x^2}dx$, the integral under the bell curve that all of statistics leans on, get computed by expanding into a series and integrating term by term. Infinite polynomials are how hard functions get domesticated.

Which makes today's question the gatekeeper: the series only equals the function where it actually converges. Feed it an illegal x and you get nonsense with confidence. Try it yourself before any theory: add up $1 + x + x^2 + x^3 + \cdots$ with $x = \frac{1}{2}$, four or five

terms, and then again with $x = 2$. Which one is settling toward a number, and which is running away? Hold onto that feeling. So: which inputs are legal?

What a power series is, and the prototype you already know

A **power series centered at** a is

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

The constants c_n are the **coefficients**; a is the **center**, the point the whole thing is built around, and $|x-a|$, the distance from x to the center, is the quantity everything will hinge on. (One small convention: the $n = 0$ term $(x-a)^0$ is taken to be 1, so at $x = a$ the series is just c_0.) Most of our examples center at $a = 0$, giving plain $\sum c_n x^n$. The new feature compared to all of last unit: there's an x inside, so convergence is no longer a single verdict, it's a verdict *per input*.

And you've met one already. Take the geometric series with first term 1 and ratio x:

★ The prototype: geometric as a power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}, \qquad \text{valid exactly when } |x| < 1.$$

Read what this says: an infinite polynomial that *equals* a familiar function, but only on a limited range of x. Outside, at $x = 2$ say, the left side is $1 + 2 + 4 + 8 + \cdots$, nonsense, while the right side calmly reports -1. The function exists out there; the *series* doesn't. That mismatch is exactly why intervals of convergence matter, and the geometric condition $|x| < 1$ is your first interval.

The shape of legality: radius and interval

Convergence doesn't scatter randomly across the number line; it occupies a symmetric stretch around the center. There's a **radius** R such that inside you're safe, outside you're not, and the two edge points are genuinely up for grabs.

Before you look: suppose a series centered at 0 works at $x = 3$ and fails at $x = 7$. Where do you predict it works and fails on the negative side? Now check your guess against the shape:



Safe within R of the center, doomed beyond it, and the two edges decided individually.

Walk the line: at the center the series is trivially fine (every term after the first is zero). Move outward and the terms are still shrinking fast enough, until you cross distance R, where shrinkage loses to growth, the terms stop dying, and the sum can't settle. The two edge points sit exactly on the knife, which is why they're marked "check" rather than colored in, and the next section explains why they *must* be checked by hand.

★ The structure

For $\sum c_n(x-a)^n$ there's a radius R with: convergence for $|x-a| < R$, divergence for $|x-a| > R$, and **separate verdicts required** at $x = a \pm R$. Three possible shapes: $R = 0$ (legal only at the single point $x = a$), $R = \infty$ (legal everywhere), or a finite stretch whose edges may be in, out, or one of each.

Finding R, and why the edges are a blind spot

The method is mechanical: run last chapter's Ratio Test on the terms a_n (which now have x inside them), get the limit L of $\left|\frac{a_{n+1}}{a_n}\right|$, then demand $L < 1$ and solve for x.

➤ The full job: $\sum_{n=1}^{\infty} \frac{x^n}{n}$

Ratio-test it:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n}\right| = |x| \cdot \frac{n}{n+1} \longrightarrow |x| \cdot 1 = |x|,$$

since $\frac{n}{n+1}$ heads to 1 (divide top and bottom by n). Converges when $L = |x| < 1$, so $\mathbf{R = 1}$, centered at 0. Now the two edges, one at a time:

- $x = 1$: the series becomes $\sum \frac{1}{n}$, the harmonic series from two chapters back, *diverges*.
- $x = -1$: it becomes $\sum \frac{(-1)^n}{n}$, the alternating harmonic series from last chapter, *converges* (by the Alternating Series Test).

So the interval is $[-\mathbf{1}, \mathbf{1})$: in at the left edge, out at the right. (Notation: a square bracket means that endpoint is included, a round one means it's excluded.) The edges genuinely disagreed, which is the whole reason you can never assume.

☞ Watch out

Why must the endpoints be checked separately? Because whenever the Ratio Test is what handed you R, its limit at $x = a \pm R$ comes out exactly $L = 1$, and you know from last chapter that $L = 1$ is the test **shrugging**: no verdict. The very tool that found the radius goes deliberately silent at the edges. So you plug each endpoint in and judge the resulting *number* series with the old court system, alternating test, p-series, comparison. The edges are where the previous chapter earns its place in this one.

➤ **The other two shapes**

Legal everywhere: $\sum \frac{x^n}{n!}$.

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{|x|}{n+1} \longrightarrow 0$$

for *every* x, because the factorial ($n! = 1 \cdot 2 \cdots n$) crushes any fixed number. $L = 0 < 1$ always, so $R = \infty$ and the interval is $(-\infty, \infty)$, no endpoints to check. (That's e^x, and its infinite radius is exactly why e^x is so well-behaved everywhere.)

Legal nowhere but home: $\sum n!\, x^n$.

$$\left|\frac{a_{n+1}}{a_n}\right| = (n+1)|x| \longrightarrow \infty \quad \text{for any } x \neq 0.$$

$R = 0$: the series converges only at its center $x = 0$, where it's just c_0. With the factorial upstairs, the terms outrun any fixed x. A perfectly legal power series that's useless as a function, worth meeting once so the $R = 0$ case isn't a surprise.

☞ **Watch out**

Wrong ideas, named and retired. The ones that wreck power-series problems:

- *"Ratio Test says converges, done."* It says converges *inside* the radius. Without the two endpoint checks you have an open interval and a possibly-wrong answer; endpoints are graded.
- *"Both endpoints behave the same."* You just watched $[-1, 1)$ happen. Check each separately, always.
- *"Centered at a means plug in a."* The center is where the stretch is measured *from*: solve $|x - a| < R$, so the interval is $(a - R, a + R)$. For $\sum \frac{(x-3)^n}{n}$ the action is around 3, not 0.
- *"The series equals the function everywhere."* Only on the interval. Outside, the function may be perfectly happy while the series is gibberish, see $\frac{1}{1-x}$ at $x = 2$.

✎ **Quick practice, before you move on**

Practice 1. Radius of convergence of $\sum \frac{x^n}{2^n}$. (Shortcut available: is this secretly geometric? What's its ratio?)

Practice 2. Full interval, endpoints included, for $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$.

How to think about it: centered at 3. Ratio-test to get $|x - 3| < R$, write the base interval, then plug each endpoint in and recognize the two number series that come out, you've met both this unit.

Practice 3 (one sentence). Why can't the Ratio Test settle the endpoints for you?

Answers at the end of the chapter.

Carrying this forward

★ Leaving this chapter with this

A power series $\sum c_n(x-a)^n$ is an infinite polynomial that's only meaningful on an interval around its center. Find the *radius* by running the Ratio Test and solving $L<1$; find the *interval*, when R is finite, by then testing both endpoints by hand, because the Ratio Test returns exactly $L=1$ there and refuses to rule. Three shapes exist: a lone point ($R=0$), a stretch, or the whole line ($R=\infty$). Keep the prototype $\frac{1}{1-x}=\sum x^n$ on $|x|<1$ in your pocket; it's the ancestor of everything coming. And now the summit: next chapter we stop being handed power series and start *manufacturing* them, turning e^x, $\sin x$, and friends into infinite polynomials on demand.

✎ The chapter workout (get stuck on purpose)

F1. $\displaystyle\sum_{n=1}^{\infty}\frac{x^n}{n^2}$ (full interval) **F2.** $\displaystyle\sum_{n=0}^{\infty}2^n x^n$ (radius)

F3. $\displaystyle\sum_{n=1}^{\infty}\frac{x^n}{n\,3^n}$ (full interval) **F4.** $\displaystyle\sum_{n=0}^{\infty}\frac{(x+2)^n}{n!}$ (full interval)

F5 (one sentence). The prototype says $\sum x^n=\frac{1}{1-x}$ for $|x|<1$. At $x=\frac{1}{2}$ the sum is 2; check that by adding a few terms. Then say what goes wrong at $x=2$, where the formula would claim -1.

Nudges: **F1** $R=1$; at both edges you get $p=2$ series, one alternating. **F2** ratio $\to 2|x|$. **F3** ratio $\to \frac{|x|}{3}$; the edges are old friends. **F4** factorial downstairs, what does that always do to the radius? And where's the center? **F5** add $1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}$ and compare.

Next up: *Taylor & Maclaurin Series*, the recipe that manufactures a power series for almost any function you like, and the handful of expansions worth knowing by heart.

➜ Want it on video?

Professor Leonard, *Calculus 2*, his lectures on power series and intervals of convergence, at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** ratio $\to \frac{|x|}{2} < 1$ gives $\mathbf{R = 2}$; and yes, it's geometric with ratio $\frac{x}{2}$, so $|x/2| < 1$ gives the same thing in one line. (Habit check: at both endpoints the terms are $(\pm 1)^n$, which don't die, so the interval is $(-2, 2)$, both edges out.) **P2:** ratio $\to |x - 3|$, so $R = 1$ and the base interval is $(2, 4)$; at $x = 4$ you get $\sum \frac{1}{n}$ (diverges), at $x = 2$ you get $\sum \frac{(-1)^n}{n}$ (converges): interval $\mathbf{[2, 4)}$. **P3:** because at the endpoints the Ratio Test's limit is exactly 1, its no-verdict case, so the endpoints have to be judged as ordinary number series with the earlier tests. **F1:** $R = 1$; $x = 1$ gives $\sum \frac{1}{n^2}$ (converges, $p = 2$), $x = -1$ gives $\sum \frac{(-1)^n}{n^2}$ (converges, even with the signs stripped): interval $\mathbf{[-1, 1]}$, both edges in. **F2:** $2|x| < 1$, so $\mathbf{R = \frac{1}{2}}$. **F3:** $\frac{|x|}{3} < 1$ gives $R = 3$; $x = 3$ gives the harmonic (diverges), $x = -3$ gives alternating harmonic (converges): $\mathbf{[-3, 3)}$. **F4:** factorial downstairs crushes everything, ratio $\to 0$ for all x, so $R = \infty$ and the interval is $(-\infty, \infty)$, centered at -2 but legal everywhere anyway. **F5:** $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = 1.875$, climbing toward 2, correct. At $x = 2$ the series is $1 + 2 + 4 + 8 + \cdots$, which diverges; the formula $\frac{1}{1-x}$ still returns -1 because the *function* is defined there, but the series has no legal claim to it outside $|x| < 1$.

Taylor & Maclaurin Series

Calculus 2 • Power Series & Taylor, Chapter 2

Turning any smooth function into its own infinite polynomial

❧ Last chapter. The summit, and it's a real marvel.

Here's what you're about to be able to do. You will take a function like e^x or $\sin x$, things defined by limits and circles and growth, and rewrite them *exactly* as an endless polynomial: a constant, plus a multiple of x, plus a multiple of x^2, and on forever. On one side, a function you can't compute by hand. On the other, addition and powers, arithmetic a child could do. And they're *equal*. Last chapter you learned which inputs an infinite polynomial can legally swallow. This chapter you learn how to *manufacture* the polynomial in the first place.

The recipe is almost suspiciously simple, and I want you to guess it before I say it. If you wanted a polynomial to impersonate a function near one point, what would you make it copy? The value, sure. Then what? The slope. Then the bend. Then the rate the bend changes. Match enough of those and the impersonation gets good. Match *all* of them and, for the functions in this chapter, the impersonation becomes the real thing. That's the whole idea. Everything else is bookkeeping, and I'll show you where every piece of the bookkeeping comes from.

✓ In this chapter

Here's what you'll be able to do by the end, in plain words:

- say why anyone would bother rewriting a function as an infinite polynomial,
- explain the matching-derivatives idea and *derive* the Taylor formula, factorial and all,
- build a series from scratch for e^x and $\cos x$,
- recall the famous four expansions and know why they're the ones to know,
- build new series in one line by *substituting* into a known one,
- chop a series into a *Taylor polynomial* and use it to approximate a number.

Why this exists, and what it's for

Start with a question I asked last chapter and left hanging: how does a chip compute $\sin(1)$? It can add, subtract, multiply, divide. That's it. It has no circle inside it. And yet it hands you $0.8414709848\ldots$ before you lift your finger. Something has to translate "sine" into "arithmetic," and this chapter is the idea behind that translator. Once $\sin x$ is written as a polynomial, the machine just multiplies and adds a few terms and stops when the next term is smaller than it cares about.

Second use, and you've seen it even if nobody labeled it. In physics, "for small angles, $\sin\theta \approx \theta$" is what makes the pendulum solvable. Where does that come from? It's the first term of the sine series with the rest thrown away. Every "linear approximation" and every "to first order" you'll ever hear in a lab (both mean: keep the first term or two, drop the rest) is a Taylor series with the tail cut off.

Third use, and this one is almost magic. Some functions have no antiderivative you can write down. e^{-x^2}, the bell curve, is the famous one; the entire theory of statistics leans on its integral and there is no formula for it built from the functions you know. Dead end? No. Write e^{-x^2} as a series, integrate each term (they're just powers of x, you've been integrating those since Calc 1), and you have the integral as a series. Impossible integrals become routine ones.

So the question is: given a function, how do you find the polynomial that equals it? Think about that before reading on. If you could only make a polynomial match a function at *one* point, in *several* ways, which ways would you pick?

The idea: match every derivative

Here's a game. I pick a point, say $x = 0$, and a function f. You build a polynomial. You're allowed to look at f only at that one point, but you may ask for as much information there as you like. What do you ask for?

You'd ask for the value $f(0)$, so your polynomial at least starts in the right place. Then you'd ask for the slope $f'(0)$, so it heads off in the right direction. Then the second derivative, so it bends the right way. Then the third, so the bend changes at the right rate. Each derivative pins down one more piece of the function's *local personality*, and each one is a number you can bake into a coefficient.

Now watch what happens when you try to actually do it. Suppose the polynomial you're building is

$$p(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \cdots$$

and you want p to match f in every derivative at 0. Fill in this table before you read the next paragraph; it's the entire derivation in disguise.

want	**compute it from** p	**so** c_n **must be**
$p(0) = f(0)$	$p(0) = c_0$	$c_0 = f(0)$
$p'(0) = f'(0)$	$p'(x) = c_1 + 2c_2 x + 3c_3 x^2 + \cdots$, so $p'(0) = c_1$	$c_1 = f'(0)$
$p''(0) = f''(0)$	$p''(x) = 2c_2 + 6c_3 x + 12c_4 x^2 + \cdots$, so $p''(0) = 2c_2$	$c_2 = \dfrac{f''(0)}{2}$
$p'''(0) = f'''(0)$	$p'''(x) = 6c_3 + 24c_4 x + \cdots$, so $p'''(0) = 6c_3$	$c_3 = \dfrac{f'''(0)}{6}$
$p^{(4)}(0) = f^{(4)}(0)$	$p^{(4)}(0) = 24\,c_4$	$c_4 = \dfrac{f^{(4)}(0)}{24}$

Look at the numbers dividing in the last column: 1, 1, 2, 6, 24. Say them as products: 1, 1, $2 \cdot 1$, $3 \cdot 2 \cdot 1$, $4 \cdot 3 \cdot 2 \cdot 1$. Those are factorials. And now you can see *why*, not just *that*. Differentiating x^n once brings down an n. Differentiating again brings down $n - 1$. Do it n times and you've multiplied by $n \cdot (n-1) \cdots 2 \cdot 1 = n!$, leaving

a constant $n!\,c_n$. Every lower power died along the way (differentiated to zero), and every higher power still has an x on it and vanishes when you plug in 0. So the nth derivative at 0 sees exactly one survivor: $f^{(n)}(0) = n!\,c_n$. The factorial in the formula is just *undoing* the factorial that differentiation created. That's it. That's where the famous $n!$ comes from, and you'll never have to memorize it again because you can rebuild it from "differentiate x^n n times."

If you'd rather center somewhere other than 0, use powers of $(x-a)$ instead of powers of x and plug in $x = a$. Nothing else changes.

★ Taylor & Maclaurin series

The **Taylor series** of f centered at a is the power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Read it as instructions: *coefficient of $(x-a)^n$ equals the nth derivative at the center, divided by $n!$.* For every function you'll compute with in this book, this series *equals* $f(x)$ wherever it converges (the one honest exception is in the wrong-ideas box later). When the center is $a = 0$ it has its own name, the **Maclaurin series**: $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$. Same recipe, center at the origin.

Before the first example, predict: what's special about the derivatives of e^x that should make its series the easiest one in mathematics?

➤ Building e^x from scratch

Let $f(x) = e^x$. Every derivative of e^x is e^x again, and $e^0 = 1$, so $f^{(n)}(0) = 1$ for *every* n. The hardest part of the recipe, computing derivatives, is trivial. Drop the 1's into the Maclaurin formula:

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

That's the series from the pep talk, derived in three lines. **Why** it's legal everywhere: last chapter you ratio-tested exactly this series and found its radius of convergence R to be infinite. The polynomial is a valid impersonation of e^x at every x on the line.

Spot-check it. Plug $x = 1$ into the first five terms: $1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} = 2.7083\ldots$ The true e is $2.71828\ldots$ Five terms of arithmetic and you're already within 0.01 (just barely) of a number that has no finite decimal; a sixth term gets you within 0.002. That's the chip's trick, done by hand.

Now cosine. Before you compute: cosine is an *even* function (the graph is symmetric about the y-axis). If a polynomial is going to be symmetric that way, which powers of x can it contain? Odd powers flip sign when x flips sign; even powers don't. So bet on the answer before the derivation hands it to you.

➤ Building $\cos x$: the derivatives cycle

Let $f(x) = \cos x$. The derivatives run in a cycle of four: $\cos x \to -\sin x \to -\cos x \to \sin x \to \cos x \to \cdots$. Evaluate each at 0:

$$f(0) = 1, \quad f'(0) = 0, \quad f''(0) = -1, \quad f'''(0) = 0,$$

$$f^{(4)}(0) = 1, \quad f^{(5)}(0) = 0, \quad f^{(6)}(0) = -1, \ \ldots$$

Every odd derivative lands on $\pm \sin 0 = 0$ and dies. Every even derivative lands on $\pm \cos 0 = \pm 1$, alternating. So only even powers survive, with alternating signs:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

Your bet was right: an even function got an even polynomial, which is symmetry doing exactly what it must. (Sine, being odd, gets only odd powers. Try predicting its series now and check it against the table below.)

The famous four

A handful of series come up constantly, and the reason is not that teachers like them. It's that nearly every other function you meet is built from these by substitution, multiplication, differentiation, or integration. Know these four and you can build hundreds.

★ The ones to know

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots \qquad \text{(all } x\text{)}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \qquad \text{(all } x\text{)}$$

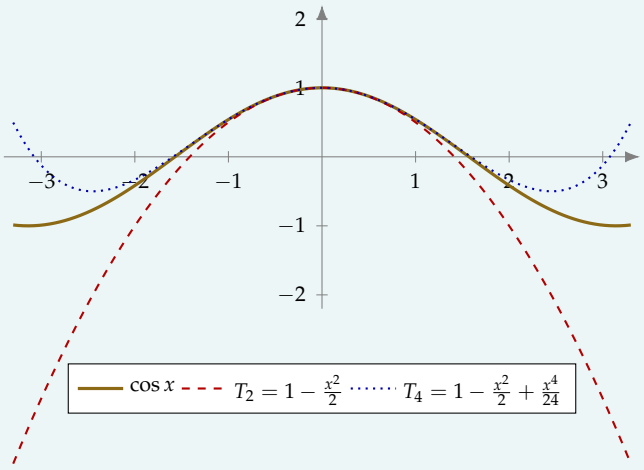
$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \qquad \text{(all } x\text{)}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots \qquad (|x| < 1)$$

Notice the last one is last chapter's prototype, the geometric series, now reread as a Maclaurin series. And notice the ranges: three of them work everywhere, the geometric one only on $|x| < 1$. The series carries its interval with it, always.

Here's the picture I want you to be able to redraw with your eyes closed. We'll chop the cosine series after the x^2 term, then after the x^4 term (call a series chopped after the x^N term the **Taylor polynomial** T_N; it gets its own section soon), and lay both on top of the real $\cos x$. Before you look: where should the short polynomials agree with cosine best, and where should they fall apart?

☆ Taylor polynomials closing in on $\cos x$



Each extra term hugs the true curve over a wider stretch before peeling away.

Walk through it. The solid curve is the real $\cos x$. The dashed one is $T_2 = 1 - \frac{x^2}{2}$, a plain downward parabola. At the center they're glued together: same height (1), same slope (0), same bend. Move away from 0 and the parabola keeps diving while cosine turns back up, so by $x \approx 2$ they've clearly split. Now the dotted one, T_4, with one more term. It tracks cosine further, through the first dip, and starts peeling off past about $x \approx 2.5$. Add T_6 and it would track further still. That's the pattern: *near the center, a few terms are indistinguishable from the real thing; the farther you go, the more terms you need.* Which is exactly why a program computing $\sin(1)$ from a polynomial like this needs only a handful of terms, and why "small angle" approximations say *small*.

The shortcut: substitute into a known series

In practice you almost never build a series from derivatives. You take one of the famous four and *plug something in*. It's instant, and it's legitimate because a series is an equation: whatever you're allowed to do to one side you're allowed to do to the other, as long as what you plug in lands inside the series' interval of convergence. Carry the interval through the substitution.

Predict first: what should the series for e^{x^2} look like, given the series for e^x? Don't compute anything. Just replace.

➤ Substitution: the series for e^{x^2}

Start from $e^u = \sum \frac{u^n}{n!}$, valid for every u, and substitute $u = x^2$:

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots$$

No derivatives of e^{x^2} needed, which is a mercy, because they're hideous (try the third one if you want to appreciate the shortcut). And since e^u converges for every u, the result is valid for every x. **Why this matters beyond saving time:** this is precisely how you integrate e^{-x^2}, the bell curve, a function with no antiderivative you can write in familiar functions. Substitute $u = -x^2$ instead, get $1 - x^2 + \frac{x^4}{2!} - \cdots$, and integrate term by term. The "impossible" integral becomes $x - \frac{x^3}{3} + \frac{x^5}{5 \cdot 2!} - \cdots$, a series you can evaluate to any precision you like. Statistics leans on this every day.

Taylor polynomials: using a few terms

Cut the series off after the degree-N term and what's left is the **Taylor polynomial** $T_N(x)$. It's not equal to the function anymore, it's an approximation, but near the center it's an excellent one, and it's a finite calculation. This is the version engineers actually use.

Before the example, guess: how many terms of e^x do you think it takes to get $e^{0.1}$ right to three decimal places? Two? Five? Twenty?

➤ **Approximating $e^{0.1}$**

Use the first three terms of e^x (that's T_2, degree 2) at $x = 0.1$:

$$e^{0.1} \approx 1 + 0.1 + \frac{(0.1)^2}{2} = 1 + 0.1 + 0.005 = 1.105.$$

The true value is $1.10517\ldots$, so three terms already agree to three decimals; the error is about 0.0002. **Why so few terms?** Because $x = 0.1$ is close to the center, and each new term carries another factor of 0.1 on top of a bigger factorial underneath. The next term is $\frac{(0.1)^3}{6} \approx 0.00017$, essentially the whole error. Add it and you're at 1.105167, within 0.000005 of the truth. Far from the center the terms shrink slower and you'd need more of them, which is what the cosine picture was telling you.

☞ Watch out

The wrong ideas that wreck Taylor-series problems, named and retired:

- *"The $n!$ is just a convention."* It's the exact undo of differentiating x^n n times. If you forget the formula, rebuild it from that one sentence.
- *"Any series with the right first term is the Taylor series."* Every coefficient must match the derivative recipe. $1 + x + x^2 + \cdots$ is *not* e^x (it's $\frac{1}{1-x}$); the factorials are what make the difference.
- *"Always compute derivatives."* Substitute into the famous four first. Derivatives are the backup plan, not the plan.
- *"A Taylor polynomial is good everywhere."* It's good *near the center*, and gets worse the farther you go. The T_2 parabola for cosine is wonderful at 0.2 and nonsense at 3.
- *"Matching every derivative always means equaling the function."* Almost always, and for every function you'll compute with in this book, yes. These well-behaved functions are called **analytic**. But one honest exception exists: $f(x) = e^{-1/x^2}$ (with $f(0) = 0$) has every derivative equal to 0 at the origin, so its Maclaurin series is just 0, yet the function isn't. Trust the famous four completely; just don't turn "matches derivatives" into a law of nature.

✎ Quick practice, before you move on

Practice 1. Write the first three nonzero terms of the Maclaurin series for $\sin x$. (You predicted which powers show up; now confirm with the derivative cycle $\sin \to \cos \to -\sin \to -\cos$.)

Practice 2. Find the Maclaurin series for $\dfrac{1}{1+x}$ *without* taking any derivatives. *How to think about it:* you already know $\frac{1}{1-u} = \sum u^n$. Rewrite $\frac{1}{1+x}$ as $\frac{1}{1-(-x)}$, substitute $u = -x$, and watch the signs. For which x is the result valid?

Practice 3 (one sentence). In your own words, why does the coefficient of x^n get divided by $n!$?

Answers at the end of the chapter.

Carrying this forward

★ Leaving this chapter, and Calculus 2

A smooth function (one with derivatives of every order) becomes a power series by matching its derivatives at a center: $f(x) = \sum \frac{f^{(n)}(a)}{n!}(x-a)^n$, Maclaurin when $a = 0$, with the $n!$ undoing what n differentiations did to x^n. Know e^x, $\sin x$, $\cos x$, and $\frac{1}{1-x}$ by heart, then build everything else by substituting into them. Chop to a Taylor polynomial when you just need a number, and remember it's trustworthy near the center. And that's Calculus 2: from integration tricks, through the geometry of area and volume, through the tests that judge infinite sums, all the way to rebuilding functions out of infinite arithmetic. Look back at the first page of this book. You earned this.

✎ The chapter workout (get stuck on purpose)

F1. first 3 nonzero terms of the Maclaurin series for $\cos x$

F2. use T_2 of e^x to approximate $e^{0.5}$

F3. Maclaurin series for $\frac{1}{1-2x}$ (substitute into the geometric series)

F4. Maclaurin series for $x \sin x$ (multiply a known series by x)

F5 (one sentence). The small-angle rule says $\sin\theta \approx \theta$. Use the sine series to say how big the first thing it ignores is, and why that justifies the word "small."

Nudges: **F1** even powers, alternating signs. **F2** $1 + x + \frac{x^2}{2}$ at $x = 0.5$; then compare with the true 1.6487 and say why it's worse than the $e^{0.1}$ case. **F3** put $u = 2x$ into $\frac{1}{1-u} = \sum u^n$, and carry the interval along. **F4** write out $\sin x$ and distribute the x. **F5** the first ignored term is $-\frac{\theta^3}{6}$.

That's the book. From "what is a limit" to rebuilding $\sin x$ out of arithmetic. If you want the map of how it all connects, the start-here page lays it out.

➜ Want it on video?

Professor Leonard, *Calculus 2*, his lectures on Taylor and Maclaurin series (the finale of his series unit), at https://www.youtube.com/@ProfessorLeonard.

Answers (wrestle first). **P1:** $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$; only odd powers, as the odd-function bet predicted. **P2:** $\frac{1}{1+x} = \sum(-x)^n = \sum(-1)^n x^n = 1 - x + x^2 - x^3 + \cdots$, valid for $|-x| < 1$, i.e. $|x| < 1$. **P3:** because differentiating x^n n times multiplies by $n!$, so the nth derivative at the center equals $n!\,c_n$, and dividing by $n!$ recovers c_n. **F1:** $1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$. **F2:** $1 + 0.5 + \frac{0.25}{2} = 1.625$; true value ≈ 1.6487, off by about 0.024, worse than at 0.1 because 0.5 is farther from the center so each term shrinks more slowly. **F3:** $\sum(2x)^n = \sum 2^n x^n = 1 + 2x + 4x^2 + 8x^3 + \cdots$, valid when $|2x| < 1$, so $|x| < \frac{1}{2}$. **F4:** $x \sin x = x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \cdots$, valid for all x. **F5:** the first ignored term is $-\frac{\theta^3}{6}$; for $\theta = 0.1$ that's about 0.00017, so dropping it costs almost nothing, but for $\theta = 1$ it's 0.167, a real error, which is why the rule only earns its keep for *small* angles.

ONE LAST THING, FROM ME

You sat through it. That's the whole thing.

If you're reading this page, you made it to the end of a calculus book that started from zero and did not let up. Take a second with that. Most people who open a book like this never get here.

Here's what I think you actually learned. Not the formulas. You'll forget half of those, and I told you on page two that I forgot them too. What you learned is how to get stuck. How to sit in front of something that makes no sense, stay there on purpose, go find what you need, and work it until it does make sense. You did that over and over for a few hundred pages. That's not a math skill. That's the skill. Once you have it, there is nothing you can't learn, because learning anything is just that loop, run enough times.

Things got hard in here. They were supposed to. And you kept showing up anyway. That is the thing you should be most proud of, and I mean that regardless of any grade. If you're sitting an exam, I hope it goes well, I really do. But whatever that number comes out to be, it doesn't measure what you just did. You sat through this. You finished it. That's the impressive part, and nobody gets to tell you otherwise.

So take this with you. Not the integrals, the method. Sit down. Get stuck. Stay stuck a while. Go find the right resource. Work it all the way through. Point that at whatever you want to do next, in school, in a career, in anything, and I'm telling you plainly: you're going to be great at it. I've got no doubt.

Thanks for reading.

Now go get stuck on something new.

Shriyash Ghimire

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